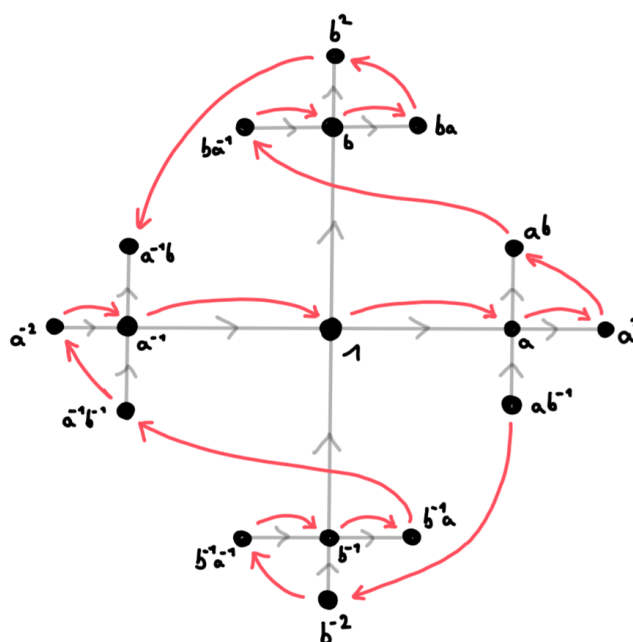


Orderability of Groups and Spaces of Orders



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Berlin, den 18.09.2025

Sandro Erras

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Abstract

This thesis serves as an introduction to the theory of ordered groups. A left-translation-invariant total order $<$ on a group G , or left-order for short, is a total order with the property that $g < g'$ implies $hg < hg'$ for all $g, g', h \in G$. Similarly, a bi-order is a left-order that is also invariant under right-translation. This has some immediate implications, the most prominent one being that left-orderability implies torsion-freeness. Moreover, the zero divisor conjecture for group rings is true for left-orderable groups. Aside from algebraic consequences, the topic regained interest in research of dynamical systems lately. This is mainly due to the fact that any countable group G is left-orderable if and only if G embeds as a subgroup into the group $\text{Homeo}_+(\mathbb{R})$ of order-preserving homeomorphisms on the real line. Furthermore, interesting connections between topology and orderability present themselves when one considers the space of all left-orders $\text{LO}(G)$ on a given group G . This space can be equipped with the Tychonoff topology in a natural way, making it into a compact and totally disconnected Hausdorff space. A well-aged result from Brouwer proven in 1910 then yields that for a countable group G , its space of left-orders $\text{LO}(G)$ is homeomorphic to the Cantor set if and only if it has no isolated points. This has motivated much of the recent interest in identifying isolated orders for specific groups.

After establishing the foundations for the theory of ordered groups, we introduce important notions such as the positive cone of an order, as well as explore some immediate algebraic consequences. The concepts are illustrated using well-known examples of left-ordered groups, such as $\mathbb{Z}^2$, the braid group B_n , as well as the fundamental group K of the Klein bottle, all of which prove to be recurring companions along our journey. Additionally, we discuss three important constructions for obtaining new left-orders from already existing ones, which will serve us well when studying spaces of orders at a later point in the thesis. Furthermore, they allow us to consider more examples of left-orderable groups, such as (semi-)direct products of ordered groups and more concretely the pure braid group on n strands PB_n . After that, we have a look at further instances of left-ordered groups, namely the aforementioned group $\text{Homeo}_+(\mathbb{R})$ of order-preserving homeomorphisms on the real line, two concrete left-orders on the free group on n generators F_n , as well as the famous Dehornoy-order on the braid group on n strands B_n .

The second half of the thesis focuses on the topology of the space of left-orders $\text{LO}(G)$ for a given group G , and in particular on the notion of isolated points. After a short detour on topological proofs for equivalent characterizations of orderability of a group, we turn to the study of spaces of orders of the braid groups B_n , the finitely generated free abelian groups $\mathbb{Z}^n$ as well as finitely generated free groups F_n , building upon more recent work. Here, it turns out that $\text{LO}(B_n)$ contains an isolated point in the form of the Dubrovina-Dubrovin order, which can be constructed from the Dehornoy order. On the contrary, $\text{LO}(\mathbb{Z}^n)$ and $\text{LO}(F_n)$ both contain no isolated orders, making them homeomorphic to the Cantor set.

Kurzzusammenfassung

Diese Arbeit stellt eine Einführung in die Theorie angeordneter Gruppen dar. Eine links-invariante totale Ordnung $<$ auf einer Gruppe G , kurz Links-Ordnung, ist eine totale Ordnung mit der Eigenschaft, dass $g < g'$ bereits $hg < hg'$ impliziert für alle $g, g', h \in G$. Ähnlich dazu ist eine Bi-Ordnung definiert als Links-Ordnung, welche zusätzlich rechts-invariant ist. Die wohl bekannteste Eigenschaft angeordneter Gruppen ist Torsionsfreiheit. Außerdem lässt sich die Nullteiler-Vermutung für Gruppenringe im Fall von angeordneten Gruppen zeigen. Abgesehen von algebraischen Konsequenzen hat das Thema in letzter Zeit wieder an Interesse in der Forschung zu dynamischen Systemen gewonnen. Das ist hauptsächlich auf ein Theorem zurückzuführen, welches besagt, dass man eine abzählbare Gruppe G links-anordnen kann genau dann wenn sich G als eine Untergruppe in die Gruppe $\text{Homeo}_+(\mathbb{R})$ von ordnungserhaltenen Homeomorphismen auf $\mathbb{R}$ einbetten lässt. Des Weiteren ergeben sich interessante Verbindungen zwischen Gruppentheorie und Topologie, wenn man die Menge aller Links-Ordnungen $\text{LO}(G)$ für eine gegebene Gruppe G betrachtet. Diese Menge lässt sich auf natürliche Weise mit der Tychonoff-Topologie ausstatten und wird dadurch zu einem kompakten, total unzusammenhängenden Hausdorff-Raum. Mit Hilfe eines älteren Resultats von Brouwer aus dem Jahr 1910 lässt sich dann für eine abzählbare Gruppe G zeigen, dass $\text{LO}(G)$ homeomorph zur Cantor-Menge ist genau dann wenn $\text{LO}(G)$ keine isolierten Punkte hat.

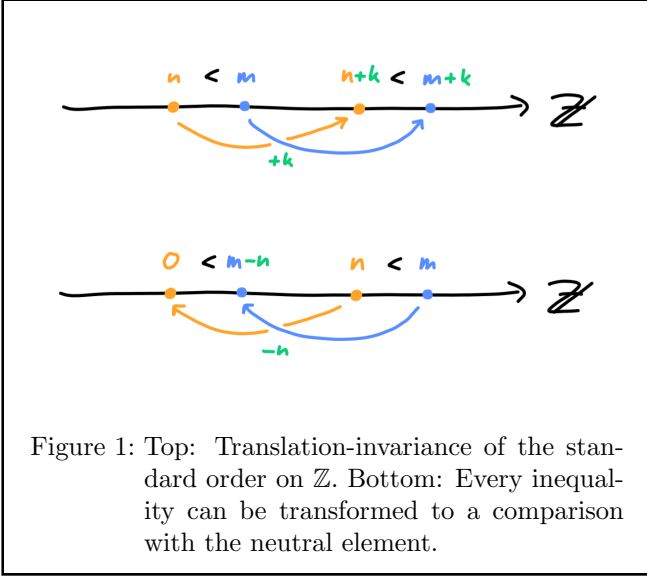
Nachdem die Grundlagen für die Theorie angeordneter Gruppen eingeführt wurden, widmen wir uns zentraler Begriffe wie dem positiven Kegel einer Ordnung, sowie direkten algebraischen Konsequenzen. Die Konzepte werden durch Beispiele dargestellt, die im Verlauf der Arbeit immer wieder von Bedeutung sind, wie zum Beispiel $\mathbb{Z}^2$, die Zopfgruppe B_n und die Fundamentalgruppe der Kleinschen Flasche K . Außerdem diskutieren wir drei wichtige Konstruktionen, um neue Links-Ordnungen aus bereits vorhandenen zu erzeugen, welche uns unter Anderem bei

der Analyse von Räumen von Links-Ordnungen von Nutzen sind. Darüber hinaus ergeben sich daraus weitere Beispiele von angeordneten Gruppen, wie (semi-)direkte Produkte von angeordneten Gruppen, zu denen auch die reine Zopfgruppe PB_n gehört. Danach untersuchen wir bekannte angeordnete Gruppen, nämlich die bereits erwähnte Gruppe $\text{Homeo}_+(\mathbb{R})$ von ordnungserhaltenden Homeomorphismen auf $\mathbb{R}$, zwei konkrete Links-Ordnungen der endlich generierten freien Gruppe F_n , sowie die berühmte Dehornoy-Ordnung auf der Zopfgruppe B_n .

Die zweite Hälfte der Arbeit beschäftigt sich mit der Topologie von Räumen von Ordnungen $\text{LO}(G)$ für eine gegebene Gruppe G , insbesondere mit isolierten Punkten. Nach einem kleinen Umweg, auf dem wir topologische Beweise für alternative Charakterisierungen von Anordbarkeit einer Gruppe sehen, wenden wir uns den Räumen von Ordnungen der Zopfgruppe B_n , der endlich generierten freien abelschen Gruppe $\mathbb{Z}^n$, sowie der endlich generierten freien Gruppe F_n zu. Es zeigt sich, dass $\text{LO}(B_n)$ die isolierte Dubrovina-Dubrovin-Ordnung enthält, welche ausgehend von der Dehornoy-Ordnung konstruiert werden kann. Im Gegensatz dazu enthalten sowohl $\text{LO}(\mathbb{Z}^n)$ als auch $\text{LO}(F_n)$ keine isolierten Punkte, was ihre Homeomorphie zur Cantor-Menge zeigt.

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1 Introduction

In school, we already learn about one of the simplest ordered groups, which is $(\mathbb{Z}, +)$ with the standard ordering $<$. Looking closely, what makes this order so special is that it is compatible with the group structure, in the sense that $n < m$ implies $n + k < m + k$ for all $n, m, k \in \mathbb{Z}$. More commonly, this property is called translation-invariance, see Fig. 1 on the top. Indeed, while there are numerous other total orders on $\mathbb{Z}$, only two of them exhibit this property, namely the standard ordering and its reversed order. This is due to the fact that for some translation-invariant order $\prec$ on $\mathbb{Z}$, every inequality can be translated to the origin, since

$$n \prec m \iff 0 \prec m - n$$

holds, see Fig. 1 on the bottom. Using the structure of $\mathbb{Z}$, in order to decide about the inequality on the right, we only need to know about positivity of the generating element 1, because we either have

$$m - n = \underbrace{1 + \dots + 1}_{m - n \text{ times}}$$

if $m - n \geq 0$ in the standard order, or

$$m - n = \underbrace{(-1) + \dots + (-1)}_{n - m \text{ times}}$$

otherwise. Say the former is true. Now either $0 \prec 1$ or $1 \prec 0$, hence translation-invariance yields either

$$0 \prec 1 \implies 1 \prec 2 \implies \dots \implies m - n - 1 \prec m - n$$

and thus $0 \prec m - n$ or

$$m - n \prec m - n + 1 \prec \dots \prec 2 \prec 1 \prec 0$$

in a similar fashion.

The existence of a translation-invariant order on $\mathbb{Z}$ has some immediate algebraic implications. For one, it implies torsion-freeness, since for $n > 0$ and $k > 0$ one obtains

$$0 < k < k + k < \dots < n \cdot k,$$

and similarly $n \cdot k < 0$ for $k < 0$, implying that $n \cdot k$ can never be equal to 0 for any $k \neq 0$. In fact, this holds for any group that admits a translation-invariant order, see Proposition 2.3. This automatically excludes all finite groups from catching our interest in this thesis.

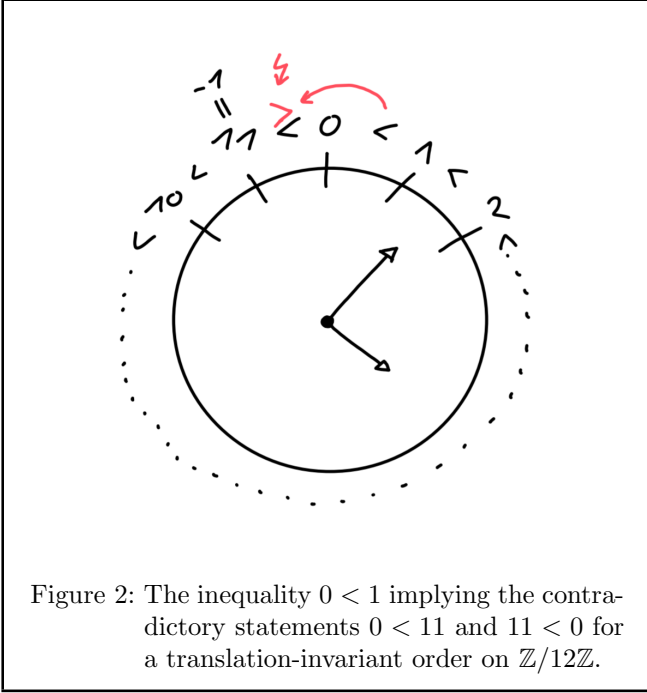
Secondly, we can use it to show that there are no non-trivial zero-divisors in $\mathbb{Z}$. Suppose we have $a \in \mathbb{Z}$ and $n \in \mathbb{Z} \setminus \{0\}$ with $n \cdot a = 0$. If either $a > 0$ or $a < 0$, then using translation-invariance we obtain $n \cdot a > 0$ or $n \cdot a < 0$, respectively. This means that a must be equal to 0. This is a special case of the more general statement that roots are unique in groups that admit an order which is invariant under left- as well as right-translation, see Lemma 2.9.

To further illustrate the incompatibility of translation-invariant orders with torsion groups, consider $\mathbb{Z}/12\mathbb{Z}$, which resemble the hours on a clock. This group is generated by the element 1, hence by choosing whether 1 should be negative or positive, we obtain an order on the whole clock. Say we choose $0 < 1$, then translation-invariance would yield $0 < 11$. But this is already a contradiction, since 11 is the inverse of 1 and thus $11 = -1 < 0$ follows from $0 < 1$. See Fig. 2 for an illustration.

In general, groups are non-abelian, so of course there is a difference between left- and right-translation-invariant orders, which we call left-orders and right-orders for short. Indeed, every left-order defines a unique right-order and vice versa, so it suffices to consider only one of those concepts. Although both left- and right-orderability are used in the literature, we stick to left-orders here, as most of our references do.

Another important algebraic consequence deals with zero divisors in group rings. The zero divisor conjecture claims that if G is torsion-free and R has no zero divisors, then the group ring RG also has no zero-divisors. While this is unproven in general, it holds when replacing torsion-freeness with the stronger notion of left-orderability, see Theorem 2.5.

Aside from the study of left-orders on groups in a purely algebraic context, they have regained interest in



the research of dynamical systems lately. This is mainly due to the remarkable result that left-orderability of a countable group G is equivalent to G being embeddable in the group of order-preserving homeomorphisms of the real line $\text{Homeo}_+(\mathbb{R})$, see Theorem 2.23. In other words, every countable group G acts faithfully on the real line by strictly increasing bijections. The embedding of G into $\text{Homeo}_+(\mathbb{R})$ depends (among other things) on the chosen left-order on G , hence the choice of order changes how G acts on $\mathbb{R}$. In general, due to a result from Linnell in [Lin11], the set of left-orders on a group G is either finite or uncountable. Furthermore, groups with only a finite number of left-orders have been completely characterized by Tararin, see Theorem 10.21 in [CR15].

In this dynamical context, one may be interested in studying the set $\text{LO}(G)$ of all left-orders on G . This naturally leads to questions about when two orders are “close to each other”. Luckily, $\text{LO}(G)$ can be equipped with a topology in a natural way by using the product topology as follows. We have seen before, that on $\mathbb{Z}$ every inequality may be translated to have 0 on either side. In general, this is also true for a left-order $<$ on some group G , since

$$g < h \iff 1 < g^{-1}h$$

holds for any $g, h \in G$, where 1 is the neutral element of G . This shows that the order $<$ is completely characterized by its positive elements, i.e. elements greater than 1. This leads to the important concept of the positive

cone $P_<$ of G , given by

$$P_< = \{g \in G \mid 1 < g\}.$$

It turns out that every left-order $<$ has a unique positive cone $P_<$, and since $P_<$ is nothing else than an element of the power set $\mathcal{P}(G)$ of G , this allows us to consider $\text{LO}(G)$ as a subset of $\mathcal{P}(G)$. Now we can use that there is a bijection

$$\varphi : \mathcal{P}(G) \xrightarrow{\cong} \prod_{g \in G} \{0, 1\},$$

where the space on the right naturally carries the product topology induced by the discrete space $\{0, 1\}$. This topology can then be transferred onto $\mathcal{P}(G)$ via φ , hence inducing a topology on $\text{LO}(G)$. This is called the Tychonoff topology.

This leads to an interesting connection between group theory and topology, for example giving rise to purely topological proofs of characterizations for left-orderability, see Sections 6.1 and 6.2. Also, due to a result from Brouwer, see Theorem 1 in [Fra12], it turns out that the space of left-orders $\text{LO}(G)$ of a countable group G is homeomorphic to the Cantor space if and only if it has no isolated points. This has led to the study of isolated left-orders for specific groups in recent years. Conveniently, isolated points in $\text{LO}(G)$ have a very explicit characterization. They consist of exactly those orders $<$, that admit a finite choice of elements $g_1, \dots, g_k \in G$ for which $<$ is the unique left-order on G such that all those elements g_i are positive, i.e. $1 < g_i$ for all $i = 1, \dots, k$.

Returning to the algebraic side, there are various constructions which are related to the topic of orderability. First, left-orderability is well-behaved with respect to group extensions, in the sense that given a short exact sequence

$$1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{p} H \longrightarrow 1$$

where the left and right factors N and H are left-ordered, the middle term G can also be left-ordered. See Proposition 3.4 for more details. This works via a “lexicographic construction”, where in order to determine if an element $g \in G \setminus \{1\}$ is positive or negative, it first gets mapped to H to see if $p(g)$ is positive or negative in H . If neither is true, exactness tells us that g must lie in the image of ι , hence the unique preimage $\iota^{-1}(g)$ can be compared to 1 in N . As a special case, this construction yields orderability of all direct and semidirect products of left-ordered groups.

Closely related to group extensions, we have the concept of convex subgroups. Intuitively, a subgroup $C \subseteq G$ is

<-convex, if it “behaves like an interval”, meaning

$$c < g < c' \implies g \in C$$

for all $c, c' \in C$ and $g \in G$. The usefulness of convexity is twofold: For one, it can be used to construct new orders via convex extensions, see Definition 3.9. On the other hand, if $C \subseteq G$ is <-convex, the order $<$ must be induced by the short exact sequence

$$1 \longrightarrow C \longrightarrow G \xrightarrow{p} G/C \longrightarrow 1$$

(see Corollary 3.15). This can prove useful when studying spaces of orders, for example in Corollary 3.16 it is used to see that $\text{LO}(K)$ for the Klein bottle fundamental group K consists of only four points.

The remaining construction has to do with automorphisms. In fact, given a group G , its automorphism group acts on $\text{LO}(G)$ by homeomorphisms via the map

$$\text{Aut}(G) \rightarrow \text{Homeo}(\text{LO}(G)), \quad \varphi \mapsto (<\mapsto<_{\varphi}),$$

where $<_{\varphi}$ is given by

$$g <_{\varphi} g' \iff \varphi(g) < \varphi(g').$$

As a special case, one obtains the right action by conjugation

$$G^{\text{op}} \rightarrow \text{Homeo}(\text{LO}(G)), \quad g \mapsto (<\mapsto<_g)$$

where G^{op} is the opposite group of G with multiplication reversed, and $<_g$ is given by

$$h <_g h' \iff ghg^{-1} < gh'g^{-1} \iff hg^{-1} < h'g^{-1}.$$

Again, these actions prove useful when studying spaces of orders. We will see some cases, where they are used to construct converging sequences in $\text{LO}(G)$ in order to show that specific orders are limit points and thus not isolated.

Let us return to some concrete examples of orderable groups and their respective spaces of orders. We have already seen that the free abelian group of rank one is left-orderable. Indeed, the same is true for free abelian groups $\mathbb{Z}^n$ of rank n . For $n \geq 2$, it turns out that $\text{LO}(\mathbb{Z}^n)$ has no isolated points and is thus homeomorphic to the Cantor set. This result is due to Sikora in [Sik04] and we present a proof different from the original one based on Theorem 10.5 and Problem 10.6 in [CR15], see Section 7.2.

Famous examples of orderable groups include the pure braid group on n strands PB_n , as well as more generally the braid group on n strands B_n . The former serves as an interesting application for constructing an order using group extensions, while the latter exhibits the

interesting property that its space of left-orders $\text{LO}(B_n)$ contains an isolated order $<_{DD}$ which is constructed from a non-isolated order $<_D$ in $n - 2$ steps using convex extensions, see Sections 4.3, 4.4 and 7.1. Indeed, there is a whole book written on the topic of ordering braid groups if the reader needs more confirmation that this is an interesting and complex field of study, see [Deh+08] for a reference. Astonishingly, there even are certain unprovability statements which can be shown using specific well-orderings on the braid monoid B_n^+ , see Section III.3.2 in [Deh+08] for details.

Lastly, finitely generated free groups (and in fact also non-finitely generated ones) are left-orderable. This requires a bit more work, because of cancellation problems occurring when multiplying elements. A famous example, albeit tedious to do explicit computations with, is the Magnus order, which we discuss in Section 4.2.2. Thankfully, using an action of the free group F_n on the real line $\mathbb{R}$ by homeomorphisms, a much more explicit order on the free group has been constructed recently, see [Š13] (Indeed, the picture on the title page of this thesis illustrates this order on F_2 for words of length 2). In the last two decades, some work has been done to also show that the space of left-orders $\text{LO}(F_n)$ has no isolated orders. There are different ways to do this, and in Section 7.3 we present a proof that is partly based on Chapter 2.2.2 from [DNR16] and partly uses a result that is borrowed from the theory of regular languages, see Theorem 1 and Corollary 2 from [Hv17].

As a rough overview, in Section 2, basic notions and examples regarding the theory of orderable groups are introduced. In Section 3, we discuss constructions of new orders via automorphisms, group extensions and convex subgroups. Section 4 showcases famous examples of concrete orders. Section 5 introduces a topology on the space of orders along with results about isolated points and cardinality. In Section 6 two equivalent characterizations for orderability are discussed. The thesis is concluded with Section 7, in which the existence or non-existence of isolated orders for braid groups, free abelian groups and free groups are discussed.

Throughout this thesis, basic knowledge about group theory and general topology is assumed. If more specific definitions and results occur, we will elaborate as needed. Finally, let us quickly introduce some notations that are frequently used throughout the thesis:

- i) The arrows $\hookrightarrow$ and $\twoheadrightarrow$ denote injective and surjective homomorphisms, respectively.
- ii) $\langle S \mid R \rangle$ denotes the group generated by some set S subject to the relations R .
- iii) Similarly, $\langle S \mid R \rangle^+$ denotes the semigroup generated by S with relations given by R .

- iv) If not stated otherwise, $\cong$ indicates a group isomorphism in the context of groups, and likewise a homeomorphism in the context of topological spaces. Sometimes we use $\cong_{\text{Set}}$ to clarify that there is only a bijective correspondence between underlying sets.
- v) In the context of non-abelian groups, we write $\cdot$ for the group operation, 1 for the neutral element and $\{1\}$ or just 1 for the trivial group. Likewise for the case of abelian groups, we denote the group operation as $+$, the neutral element as 0 and write $\{0\}$ or 0 for the trivial group.

2 Left-Invariant and Bi-Invariant Orders and their Positive Cones

In this section, we will establish the foundations for the theory of translation-invariant orders on groups and mainly follow the definitions, arguments and observations from Chapter 1 of [CR15], unless otherwise stated.

- Definition 2.1.** i) A *left-order* $<$ on a group G is a total order on G that is invariant under translation from the left, or *left-invariant* for short, i.e. $g < g'$ implies $fg < fg'$ for all $f, g, g' \in G$.
- ii) Similarly, a *right-order* on G is a total order that is right-invariant.
- iii) A left-order that is also a right-order is called *bi-order* or *bi-invariant*.

Before continuing, it should be noted that every left-order $<$ on G also defines a right-order $<'$ by saying

$$f <' g :\iff g^{-1} < f^{-1},$$

which yields

$$\begin{aligned} fh <' gh &\iff h^{-1}g^{-1} < h^{-1}f^{-1} \\ &\iff g^{-1} < f^{-1} \\ &\iff f <' g. \end{aligned}$$

In the same way, every right-order defines a left-order. Hence, there is a bijection between left-orders and right-orders on G , so it suffices to consider either one of them. We stick to left-orders in this thesis, but keep in mind that some references might use right- instead of left-invariance. Of course, for the case of abelian groups, all three notions from Definition 2.1 coincide. For the non-abelian case, however, there are left-orders that are not right-invariant, see Corollary 2.10.

Furthermore, all left-orders come in pairs, in the sense that for a given order $<$ one can define its *reversed order* $\overline{<}$ given by

$$g \overline{<} g' :\iff g' < g,$$

which is easily seen to inherit left- and/or bi-invariance from the original order.

Example 2.2. i) The standard order on $(\mathbb{Z}, +)$ is bi-invariant and so is its reversed order. In fact, these are the only two possible choices, since 1 is a generator for $\mathbb{Z}$ and we either have $0 < 1$ or $1 < 0$. The rest follows by translation.

ii) The standard lexicographic order on $(\mathbb{Z}^2, +)$ is left-invariant (hence also bi-invariant, since $\mathbb{Z}^2$ is abelian). This follows from the fact that the lexicographic order consists of component-wise comparison using the standard order on $\mathbb{Z}$, which is left-invariant. As we will see later in Example 3.5 i), this is a special case of a construction that works for any direct product, as well as more generally, any group extension of two left-orderable groups.

iii) The group $(\mathbb{Z}/2\mathbb{Z}, +)$ cannot be left-ordered. Indeed, suppose we are given a left-order on $\mathbb{Z}/2\mathbb{Z}$, then either $0 < 1$ or $1 < 0$. Assume the former, then by adding 1 on both sides we obtain $1 < 0$, contradicting the asymmetric property of a total order. The same argument works for the other case.

iv) Consider the fundamental group of the Klein bottle

$$K = \langle a, b \mid aba^{-1} = b^{-1} \rangle.$$

For reasons that will become clear throughout the thesis, this is an important example of a left-orderable group, which we will revisit many times. A possible left-order on K can be defined as follows. Since the defining relation for the group is equivalent to $ab = b^{-1}a$, given a word w in the generators a, b , it can be expressed in the form $w = a^n b^m$ for unique exponents $n, m \in \mathbb{Z}$. Then we order “lexicographically”, i.e. define

$$a^n b^m < a^{n'} b^{m'}$$

if and only if we either have $n < n'$ or $n = n'$ and $m < m'$. That this order is indeed left-invariant follows from a later result about left-orderability of group extensions of two left-ordered groups, see Proposition 3.4 and Example 3.5 iii). In fact, we will see that there are only four possible left-orders, which only differ in how we compare the exponents in the respective a - and b -factors. See Corollary 3.16 for details.

Part iii) of Example 2.2 is a special case of a general result, which is our first immediate algebraic consequence of left-orderability. This already excludes a big chunk of groups from being left-orderable, in particular finite groups.

Proposition 2.3. *Any left-orderable group G is torsion-free, i.e. if $g^n = 1$ for some $g \in G$ and $n > 1$, then $g = 1$.*

Proof. Let $<$ be a left-order on a group G . Take any element $g \in G \setminus \{1\}$ and assume that $g^n = 1$ for some positive integer n . Since $g \neq 1$, it is either strictly smaller or bigger than 1. Assume $1 < g$, the other case is analogous. By left-invariance we can repeatedly multiply the inequality with g from the left and obtain

$$1 < g < g^2 < \dots < g^{n-1} < g^n = 1,$$

leading to a contradiction. $\square$

Example 2.4. $\mathbb{R} \setminus \{0\}$ as a multiplicative group is not left-orderable by Proposition 2.3, since $(-1)^2 = 1$ implies that it has torsion.

As already mentioned in the introduction, orderability of groups has another very important application in the context of group rings. We include it here mainly for the sake of completeness and as an additional motivating factor, so we will not go into details.

Given a ring R and a group G , the group ring RG is the free R -module generated by G , i.e. the set of formal linear combinations of elements of G with coefficients in R . It can be shown that RG inherits a ring structure from R and G . The zero divisor conjecture then claims, that RG has no zero divisors if G is torsion-free and R has no zero divisors itself. In this generality, it still seems to be an open question. It is, however, true when replacing torsion-freeness with the stronger notion of left-orderability. In fact, the proof is quite simple, see Lemma III.1.3 in [Deh+08].

Theorem 2.5. *If G is left-orderable and R has no zero divisors, then the group ring RG has no zero divisors.*

At this point, let us establish some facts about “calculation” with inequalities in the context of left- and bi-orders.

Remark 2.6. Let $<$ be a left-order on a group G , and let $f, g, f', g' \in G$.

i) $1 < g$ always implies $g^{-1} < 1$.

ii) The equivalence

$$f < g \iff g^{-1} < f^{-1}$$

holds if f or g is in the center of G or if $<$ is a bi-order. To see the former, note that in this case we have

$$\begin{aligned} f < g &\iff 1 < f^{-1}g = gf^{-1} \\ &\iff g^{-1} < f^{-1}, \end{aligned}$$

and to see the latter, using bi-invariance one obtains

$$\begin{aligned} f < g &\iff 1 < f^{-1}g \\ &\iff g^{-1} < f^{-1}. \end{aligned}$$

To see that it fails in general, recall the fundamental group of the Klein bottle K with the left-order from Example 2.2 iv). Here we have

$$a^0b^1 = b < a = a^1b^0,$$

but

$$a^{-1}b^0 = a^{-1} > b^{-1} = a^0b^{-1}.$$

iii) If $<$ is a bi-order, then inequalities can be multiplied with each other: If we have $f < g$ and $f' < g'$, then bi-invariance yields

$$ff' < fg' < gg'.$$

This fails in general, again by considering K from Example 2.2 iv). We have seen before that $b < a$ and $b^{-1} < a^{-1}$, but multiplying these inequalities with each other yields $1 < 1$, a contradiction.

iv) The previous observation also holds for left-orders when one side of both inequalities consists only of the identity, i.e. if we have $1 < f$ and $1 < g$, then we get

$$1 < f < fg \quad \text{and} \quad 1 < g < gf.$$

Points ii) and iii) of Remark 2.6 show that we have to be careful with our usual intuitions regarding inequalities, when the order is only left- but not bi-invariant. Furthermore, it follows that the order on K we saw in Example 2.2 iv) serves as our first example of a left-order that is not right-invariant. Alternatively, this is also implied by the following stronger version of Proposition 2.3 for bi-orderable groups, which was posed as Problem 1.22 in [CR15]. For this we need the notion of generalized torsion.

Definition 2.7. Given a group G , an element $x \in G$ is called a *generalized torsion element*, if there exist $g_1, \dots, g_k \in G$, such that the product of the conjugates of x by g_i satisfies

$$g_1^{-1}xg_1 \cdots g_k^{-1}xg_k = 1.$$

Proposition 2.8. *Any bi-orderable group has no generalized torsion element other than the identity.*

Proof. Let $x \in G$ with $x \neq 1$. Without loss of generality assume $x > 1$. Using bi-invariance, this implies $g^{-1}xg > g^{-1}g = 1$ for every $g \in G$. Since a bi-order allows us to multiply inequalities by Remark 2.6 iii), for any choice of elements $g_1, \dots, g_k \in G$ we have

$$g_1^{-1}xg_1 \cdots g_k^{-1}xg_k > \underbrace{1 \cdots 1}_{k \text{ times}} = 1.$$

$\square$

As already mentioned, this is another way to show that the Klein bottle group K cannot be bi-ordered, since

$$1b1^{-1}aba^{-1} = baba^{-1} = 1$$

implies that it has generalized torsion.

We continue with an algebraic statement about the uniqueness of roots in bi-orderable groups, that allows us to consider another example of a group that is left-orderable, but not bi-orderable.

Lemma 2.9. *Let G be bi-orderable and take $x, y \in G$. If $x^n = y^n$ holds for some $n > 0$, then $x = y$.*

Proof. Assume $x^n = y^n$ and take a bi-order $<$ on G . Then either $x = y$, $x < y$ or $y < x$. If one of the latter two occurs, say $x < y$, then by Remark 2.6 iii), the inequality can be multiplied with itself n times, implying $x^n < y^n$. Hence $x = y$ must hold. $\square$

As remarked in [KT08] after Lemma 7.7, this leads to another important non-bi-orderable example. Here, let

$$B_n = \langle \sigma_1, \dots, \sigma_{n-1} \mid \sigma_i \sigma_j = \sigma_j \sigma_i \text{ for } |i - j| > 1, \\ \sigma_i \sigma_j \sigma_i = \sigma_j \sigma_i \sigma_j \text{ for } |i - j| = 1 \rangle.$$

denote the braid group on n strands. If we consider the symmetric group on n elements Σ_n with generators $s_1, \dots, s_{n-1}$, then there is a surjective homomorphism $\pi : B_n \rightarrow \Sigma_n$ which maps each braid to its underlying permutation. For more details, see Section 4.3.

Corollary 2.10. *For $n \geq 3$, the braid group B_n is not bi-orderable.*

Proof. Denote the generators of B_n as $\sigma_1, \dots, \sigma_{n-1}$. Then the braid relations imply

$$(\sigma_1 \sigma_2)^3 = \sigma_1 \sigma_2 \sigma_1 \sigma_2 \sigma_1 \sigma_2 = \sigma_2 \sigma_1 \sigma_2 \sigma_1 \sigma_2 \sigma_1 = (\sigma_2 \sigma_1)^3.$$

If B_n would be bi-orderable, then by Lemma 2.9 this would mean $\sigma_1 \sigma_2 = \sigma_2 \sigma_1$. But this is not true, since the projection $\pi : B_n \rightarrow \Sigma_n$ onto the symmetric group yields

$$\pi(\sigma_1 \sigma_2) = (312) \neq (231) = \pi(\sigma_2 \sigma_1),$$

hence $\sigma_1 \sigma_2 \neq \sigma_2 \sigma_1$, see Fig. 3. $\square$

Intuitively, an order $<$ on G being left-invariant just means that every inequality between elements $g, h \in G$ can be translated to the “origin”, or the identity element, in the sense that

$$g < h \iff 1 < g^{-1}h.$$

But this means that in order to completely describe a left-order, it suffices to know its positive elements, i.e. the ones that are greater than 1. This leads to one of the most important concepts when dealing with left-orders.

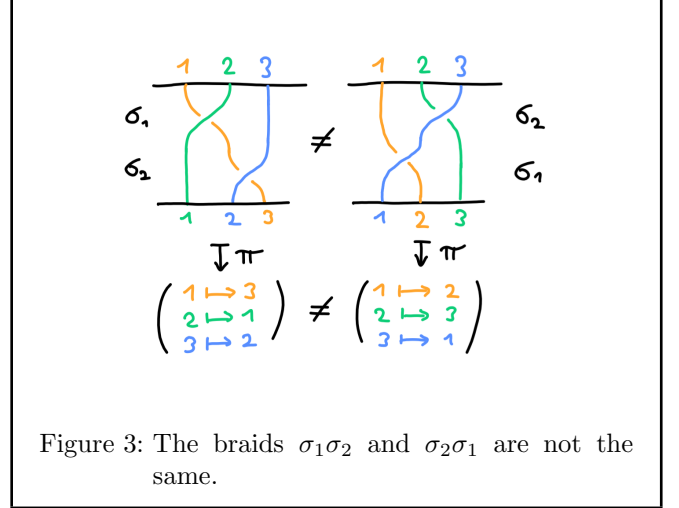


Figure 3: The braids $\sigma_1 \sigma_2$ and $\sigma_2 \sigma_1$ are not the same.

Definition 2.11. Given a left-order $<$ on G , denote its *positive cone* as

$$P_{<} := \{g \in G \mid 1 < g\}.$$

Furthermore, we call elements that are bigger than 1 *positive*, and likewise elements smaller than 1 *negative*.

Sometimes we omit the subscript and just write P if the corresponding order is clear from the context. Since a left-order $<$ on G is total, every element in G must be either positive, negative, or equal to 1. Also, an element $g \in G$ is positive if and only if its inverse g^{-1} is negative. This means, that G can be partitioned into disjoint subsets

$$G = P_{<} \sqcup \{1\} \sqcup P_{<}^{-1}.$$

Furthermore, in this context, Remark 2.6 iv) shows that the product of positive elements stays positive and likewise for negative elements, implying that $P_{<}$ is a semigroup.

Even better, we can also start the other way around: Given a subsemigroup $P \subseteq G$, such that G is partitioned into disjoint semigroups

$$G = P \sqcup \{1\} \sqcup P^{-1}, \quad (1)$$

which we just call a *positive cone*, we can define its corresponding left-order on G via

$$g <_P h :\iff g^{-1}h \in P.$$

Here, of course, the condition on the right wants to say that “ $1 <_P g^{-1}h$ ” but since we do not have the order $<_P$ defined yet, we need to define it as “ $g^{-1}h$ is positive”. Let us quickly check that this is indeed a left-invariant

total order. For left-invariance, note that

$$fg <_P fh \iff g^{-1}f^{-1}fh \in P \iff g <_P h$$

holds for all $f, g, h \in G$. Asymmetry follows from the fact that $P \cap P^{-1} = \emptyset$, transitivity is implied by the semigroup property of P and connectedness holds because G is partitioned by P, P^{-1} and $\{1\}$.

Summarising, the preceding arguments show that there is a bijection of sets

$$\{\text{left-orders on } G\} \xrightarrow{\cong} \{\text{positive cones on } G \text{ as in (1)}\}$$

$$< \longmapsto P_<$$

$$<_P \longleftarrow P$$

allowing us to freely move between left-orders and their respective positive cones.

We introduce the following related notation.

Definition 2.12. Given a subset $S \subseteq G$, write $\langle S \rangle^+$ for the semigroup generated by S inside G .

Now, what does the positive cone $P_<$ of a bi-order $<$ on G look like? Since the diagram

$$\begin{array}{ccc} 1 < g & \iff & g \in P_< \\ \updownarrow \text{right-invariance} & & \updownarrow P \text{ normal in } G \\ h < gh & \iff & h^{-1}gh \in P_< \end{array}$$

commutes, we have the following result.

Proposition 2.13. A left-order $<$ on G is a bi-order if and only if its positive cone $P_<$ is a normal subsemigroup of G .

Example 2.14. i) Consider $\mathbb{Z}^2$ with the standard lexicographic order. Its corresponding positive cone is

$$P = \{(n, m) \in \mathbb{Z}^2 \mid \text{Either } (n > 0) \\ \text{or } (n = 0 \text{ and } m > 0)\},$$

as visualized in Fig. 4.

ii) Consider the left-order $<$ on K as in Example 2.2 iv). Its positive cone is given as

$$P = \{a^n b^m \in K \mid \text{Either } (n > 0) \\ \text{or } (n = 0 \text{ and } m > 0)\},$$

as visualized in Fig. 5.

iii) Let us now construct a left-order from a positive cone. This is a fairly standard example, but important nonetheless. Consider $\mathbb{Z}^2$ embedded in $\mathbb{R}^2$ in

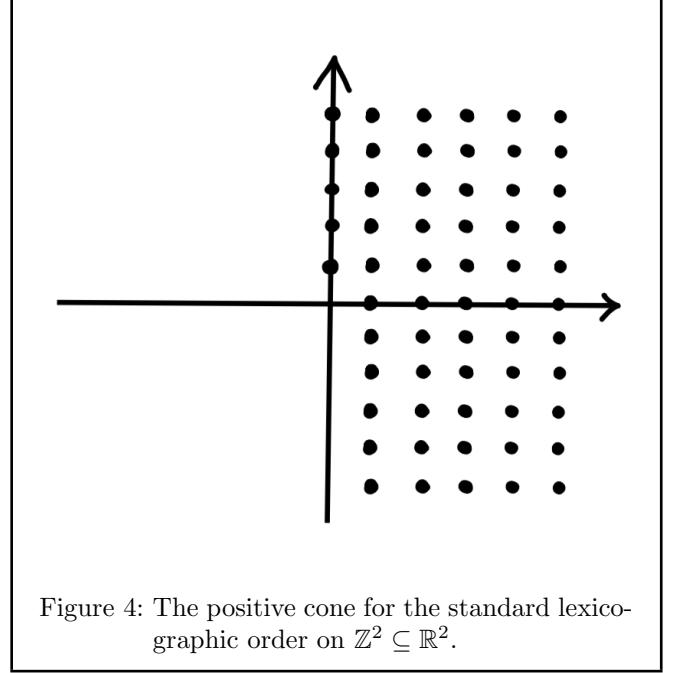


Figure 4: The positive cone for the standard lexicographic order on $\mathbb{Z}^2 \subseteq \mathbb{R}^2$.

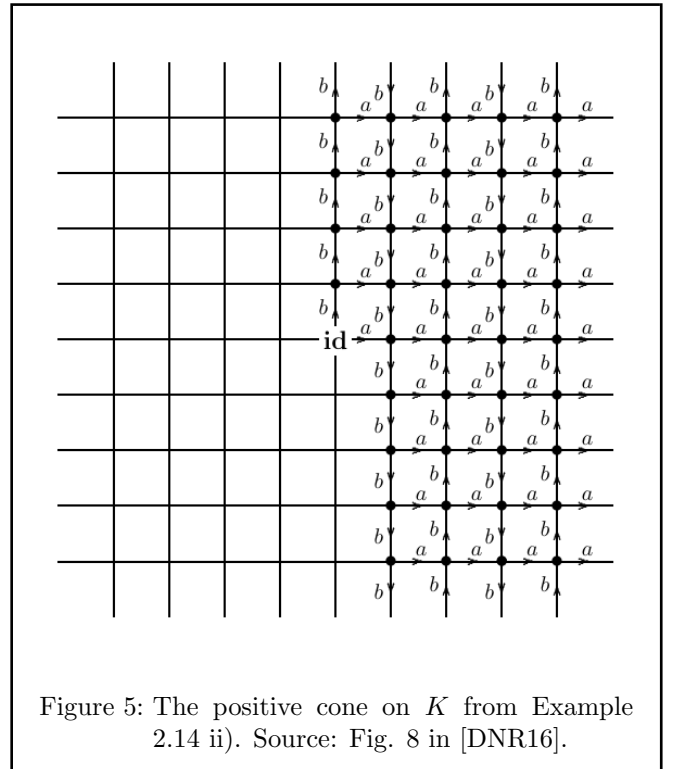


Figure 5: The positive cone on K from Example 2.14 ii). Source: Fig. 8 in [DNR16].

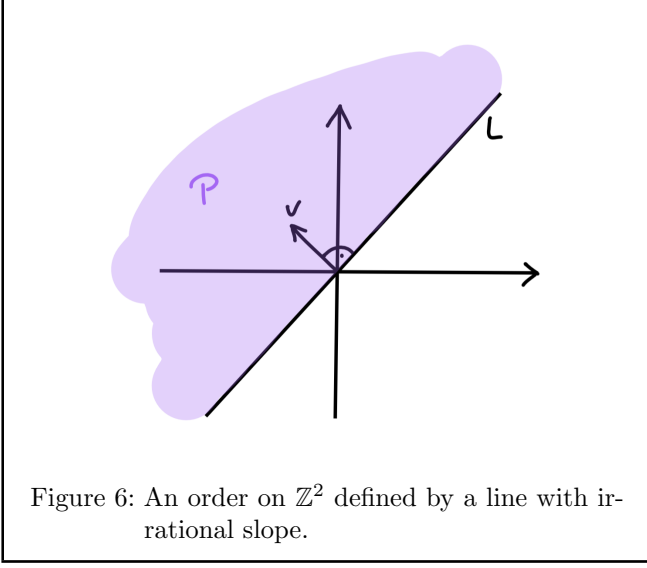


Figure 6: An order on $\mathbb{Z}^2$ defined by a line with irrational slope.

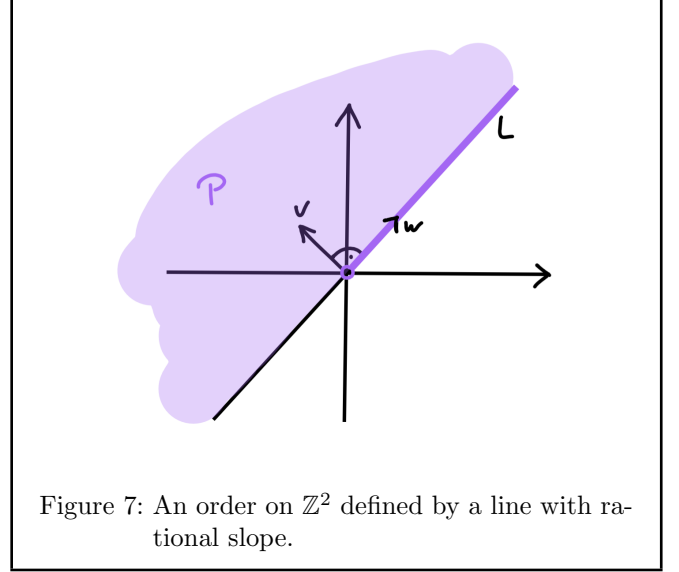


Figure 7: An order on $\mathbb{Z}^2$ defined by a line with rational slope.

the standard “grid” way. Next, choose some vector $v = (v_1, v_2) \in \mathbb{R}^2$ with irrational slope, i.e. $v_2/v_1 \notin \mathbb{Q}$. Taken as the normal vector for a line L in $\mathbb{R}^2$, this resembles a line through the origin with irrational slope.

Denoting the standard scalar product on $\mathbb{R}^2$ as $\langle \cdot, \cdot \rangle$, a point w lies on L if and only if $\langle v, w \rangle = 0$. Otherwise, it either lies on the same side as v if $\langle v, w \rangle > 0$, or on the opposite side if $\langle v, w \rangle < 0$.

Now define a positive cone $P \subseteq \mathbb{Z}^2$ as

$$P := \{w \in \mathbb{Z}^2 \mid \langle v, w \rangle > 0\},$$

see Fig. 6. Let us quickly check the defining properties of a positive cone. The semigroup property follows directly from the fact that the scalar product is bilinear. The negative cone is given by

$$P^{-1} = \{w \in \mathbb{Z}^2 \mid \langle v, w \rangle < 0\},$$

since $\langle v, w \rangle > 0$ implies $\langle v, -w \rangle < 0$, hence $P \cap P^{-1} = \emptyset$. It remains to check that $P \cup P^{-1} = \mathbb{Z}^2 \setminus \{0, 0\}$. Since the line L has irrational slope, the only vector with integer coefficients lying on the line is $(0, 0)$. Hence, either $w \in P$ or $w \in P^{-1}$ holds for every other vector $w \in \mathbb{Z}^2$.

The order given by this cone is easily described. One has $w < w'$ if and only if $(0, 0) < w' - w$, i.e. the vector going from w to w' points in the same direction as the chosen normal vector v . Note that the difference vector can never lie exactly on the line, since the slope was chosen to be irrational.

- iv) The previous construction also works for a line with rational slope, but then some extra steps are needed. As before, to define the line L , choose a normal vector $v = (v_1, v_2) \in \mathbb{R}^2$, but this time with rational slope $v_2/v_1 \in \mathbb{Q}$. Then choose another vector $w \in \mathbb{R}^2$

that is orthogonal to w , hence it lies on L .

The positive cone is then defined in a “lexicographic” way, namely

$$P := \{w' \in \mathbb{Z}^2 \mid \text{Either } (\langle v, w' \rangle > 0) \\ \text{or } (\langle v, w' \rangle = 0 \text{ and } \langle w, w' \rangle > 0)\},$$

see Fig. 7. The negative cone is then given as

$$P := \{w' \in \mathbb{Z}^2 \mid \text{Either } (\langle v, w' \rangle < 0) \\ \text{or } (\langle v, w' \rangle = 0 \text{ and } \langle w, w' \rangle < 0)\}.$$

Again, it is easily seen that P is a semigroup and that $P \cap P^{-1} = \emptyset$. For the fact that $P \cup P^{-1} = \mathbb{Z}^2 \setminus \{(0, 0)\}$, take some vector $w' \in \mathbb{Z}^2$ that is not the zero vector. Then either $\langle v, w' \rangle \neq 0$ and we are done or $\langle v, w' \rangle = 0$. In the second case, w' is orthogonal to v , meaning it is a multiple of w . Hence there is $\lambda \neq 0$ with $w' = \lambda \cdot w$. It then follows that

$$\langle w, w' \rangle = \lambda \underbrace{\langle w, w \rangle}_{>0}$$

is either positive or negative, depending on λ . In both cases we have $w' \in P \cup P^{-1}$.

Similar to before, the order here is described as $v' < w'$ if and only if the difference vector $w' - v'$ either points in the same direction as v or it lies on L and points in the same direction as w .

Interestingly, Example 2.14 iii) already shows that there are countable groups for which there are uncountably many different left-orders. As a matter of fact, a group admits either finitely many or uncountably many different left-orders, see Section 5.4.

Furthermore, looking at Example 2.14 iv), we see that the standard lexicographic order on $\mathbb{Z}^2$ is just a special

case of the order given by a line L with rational slope when one chooses $v = (1, 0)$ as a normal vector of L and $w = (0, 1)$ as a vector orthogonal to v . In fact, later in section 7.2 we will see that every left-order on $\mathbb{Z}^2$ (or more generally $\mathbb{Z}^n$ for $n \geq 2$) comes from such an order given by a line (respectively given by a hyperplane for higher dimensions).

Using their defining properties, one can show another useful result about positive cones.

Proposition 2.15. *Let P, P' be two positive cones in G . If $P \subseteq P'$, then $P = P'$.*

Proof. Assume that $P \subseteq P'$ for two positive cones. We know that

$$P \sqcup P^{-1} = G \setminus \{1\} = P' \sqcup (P')^{-1},$$

which together with the assumption implies $(P')^{-1} \subseteq P^{-1}$. Now suppose that $P' \neq P$, i.e. there exists

$$g \in P' \setminus P \subseteq G \setminus \{1\}.$$

But then g^{-1} must lie in $(P')^{-1} \setminus P^{-1} = \emptyset$, implying that g has no inverse in G , leading to a contradiction. $\square$

We conclude this section with results about the class of Archimedean orders. The prototypical example for this is $(\mathbb{R}, +)$ with the standard ordering, which is indeed universal for Archimedean ordered groups, in the sense that any such group embeds as a subgroup of $(\mathbb{R}, +)$, see Hölder's Theorem 2.22.

Definition 2.16. A left-ordered group $(G, <)$ is called *Archimedean* if for any two $g, h \in G$ with $1 < g, 1 < h$, there is some integer $n > 0$, such that $h < g^n$.

We start with a necessary condition for an order to be Archimedean, based on Lemma 2.2 from [CR15].

Proposition 2.17. *Every Archimedean left-order is a bi-order.*

Proof. Let $<$ denote a left-order on G and denote its positive cone as P . By Proposition 2.13, it suffices to show that P is normal in G , i.e. $g^{-1}Pg \subseteq P$ for all $g \in G$.

Let $h \in P$ and $g \in G$.

If $g = 1$ there is nothing to check.

For the case that $1 < g$, by the Archimedean property there is some integer $n > 0$ with $g < h^n$, which is equivalent to $1 < g^{-1}h^n$. This implies

$$1 < g^{-1}h^n < g^{-1}h^n g.$$

But then $1 < g^{-1}hg$ follows, since otherwise $1 \geq g^{-1}hg$ would imply $1 \geq g^{-1}h^n g$, and thus $g^{-1}hg \in P$.

For the case $g < 1$, assume that $g^{-1}hg \notin P$, i.e. $g^{-1}hg < 1$. This yields $1 < g^{-1}h^{-1}g$ and by the previous case then

$$1 < g(g^{-1}h^{-1}g)g^{-1} = h^{-1}$$

follows. But this means $h < 1$, which is a contradiction. $\square$

Example 2.18. $(\mathbb{Z}^2, +)$ with the lexicographic order is not Archimedean. Take for example $(0, 1)$ and $(1, 0)$ which are both positive, but $(0, 1)^n = (0, n)$ is smaller than $(1, 0)$ for every $n > 0$. This example shows that the converse of Proposition 2.17 does not hold.

The following result is based on Problem 2.3 in [CR15].

Proposition 2.19. *Let $(G, <)$ be an Archimedean ordered group and $g, h \in G$ with $g > 1$. Then there exists $n \in \mathbb{Z}$ such that*

$$g^n \leq h < g^{n+1}.$$

Proof. If $h > 1$, by the Archimedean property, there is some $n > 0$ with $h < g^n$. Let n be minimal with this property. Then $g^{n-1} \leq h < g^n$.

On the other hand, if $h < 1$, then $h^{-1} > 1$ and so by the first case, there is $k > 0$ with $g^{k-1} \leq h^{-1} < g^k$. Using that every Archimedean order is bi-invariant, this is equivalent to $g^{-k} < h \leq g^{-k+1}$. Now, if $h \neq g^{-k+1}$, we are done by choosing $n = -k$. If $h = g^{-k+1}$, one obtains $g^{-k+1} = h < g^{-k+2}$, where the last inequality follows by positivity of g and left-invariance. Here we are done by choosing $n = -k + 1$. $\square$

For proving Corollary 2.21, we need another lemma, which is based on Lemma 2.5 in [CR15].

Lemma 2.20. *Let $(G, <)$ be bi-ordered with no least positive element. Then for every $p > 1$ there exists $q > 1$ with $1 < q^2 < p$.*

Proof. By assumption, one can choose positive elements r, s with

$$1 < s < r < p.$$

Then $rs^{-1} > 1$. If $rs^{-1}rs^{-1} = (rs^{-1})^2 \geq r$, then $s^{-1}rs^{-1} \geq 1$, implying $r \geq s^2$ by bi-invariance. In this case we choose $q = s$. If on the other hand $(rs^{-1})^2 < r$ we can just choose $q = rs^{-1}$.

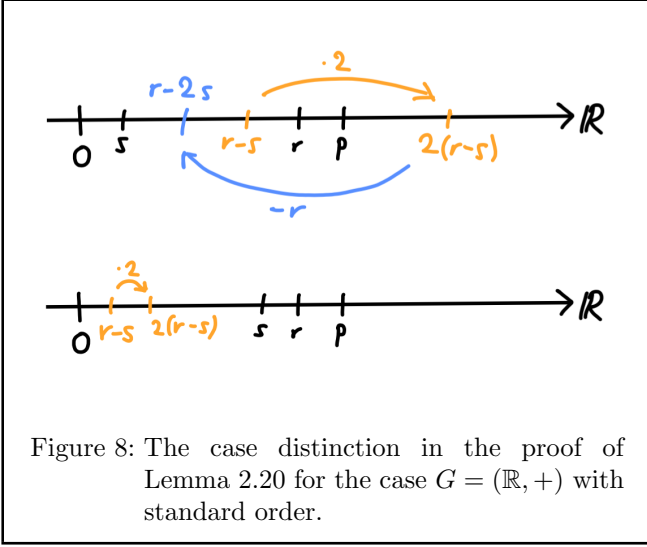
For visualization purposes, the situation is drawn in Fig. 8 for the case $G = (\mathbb{R}, +)$ with standard order. $\square$

We can now deduce that being abelian is a necessary condition for an Archimedean order to exist. The proof is based on Lemma 2.4 from [CR15].

Corollary 2.21. *If $(G, <)$ is Archimedean ordered, then it is abelian.*

Proof. Denote the positive cone of $<$ as P .

First assume that P has a least element p . In this case, one can show that G is already generated by p . Indeed, assume $G \setminus \langle p \rangle \neq \emptyset$, i.e. there exists some element $g \in G \setminus \langle p \rangle$. By Proposition 2.19, there is $n \in \mathbb{Z}$ with $p^n < g < p^{n+1}$, which implies $1 < p^{-n}g < p$. This contradicts the initial assumption.



On the other hand, suppose that P has no least element. Take some g, h that do not commute, i.e. their commutator is non-trivial. Without loss of generality we can assume that $g, h, ghg^{-1}h^{-1} > 1$. Since every Archimedean order is bi-invariant by Proposition 2.17, we can use Lemma 2.20 to obtain some $x > 1$ with $1 < x^2 < ghg^{-1}h^{-1}$. Additionally, by the Archimedean property, there are integers $n, m \in \mathbb{Z}$ with

$$x^m \leq g < x^{m+1}, \quad x^n \leq h < x^{n+1}.$$

In particular we have $g^{-1} \leq x^{-m}$ and $h^{-1} \leq x^{-n}$. Bi-invariance of the order allows us to combine all these inequalities into

$$ghg^{-1}h^{-1} < x^{m+1}x^{n+1}x^{-m}x^{-n} = x^2,$$

which leads to a contradiction. $\square$

As noted before, $(\mathbb{R}, +)$ is universal for Archimedean ordered groups. A proof of this Theorem can be found in Chapter 3.1 from [DNR16].

Theorem 2.22 (Hölder). *Every Archimedean ordered group $(G, <)$ is isomorphic to some subgroup of $(\mathbb{R}, +)$, such that the order on G corresponds to the standard ordering of $\mathbb{R}$ restricted to the embedding of G .*

As a conclusion to this Chapter, let us return to an important connection between the theories of dynamics and orderable groups that we already hinted at in the introduction. For countable groups, Hölder's Theorem is a special case of the fact that any countable left-orderable group embeds into the group $\text{Homeo}_+(\mathbb{R})$ of order-preserving homeomorphisms on the real line. For details, see Theorem 2.23 in [CR15] or Proposition 1.1.8 in [DNR16]. An example for a left-order on $\text{Homeo}_+(\mathbb{R})$ is given in Section 4.1.

Theorem 2.23. *A countable group G is left-orderable if and only if it is isomorphic to a subgroup of $\text{Homeo}_+(\mathbb{R})$.*

Although very important in general, Theorem 2.23 will not see any further use in this thesis and merely serves us as a motivational fact for the importance of the theory of orderable groups; the focus lies more on algebraic and topological aspects in the remaining Sections.

3 Producing New Orders from Old Ones

Left-orders rarely come alone, in the sense that there are various ways to obtain new left-orders from already given ones. In this section, we discuss some of these constructions.

3.1 Automorphisms

The first construction of new orders is via automorphisms on a group G and is based on the beginning of Chapter 10.1 in [CR15].

Assume we are given a left-order $<$ on some group G and an automorphism $\varphi \in \text{Aut}(G)$. Then for the positive cone $P_<$ of this order, we can consider its preimage

$$P_\varphi := \varphi^{-1}(P_<)$$

under the automorphism. This forms again a positive cone in G . First note that $P_\varphi^{-1} = \varphi^{-1}(P_<^{-1})$, since

$$\varphi(g) \in P_< \iff \varphi(g^{-1}) \in P_<^{-1} \iff g^{-1} \in \varphi^{-1}(P_<^{-1}).$$

The semigroup property of P_φ follows from linearity of φ , and the partition property from

$$\varphi^{-1}(P_<) \cap \varphi^{-1}(P_<^{-1}) = \varphi^{-1}(P_< \cap P_<^{-1}) = \emptyset$$

and

$$\begin{aligned} \varphi^{-1}(P_<) \cup \varphi^{-1}(P_<^{-1}) &= \varphi^{-1}(P_< \cup P_<^{-1}) \\ &= \varphi^{-1}(G \setminus \{1\}) \\ &= G \setminus \{1\}. \end{aligned}$$

The left-order $<_\varphi$ given by the cone P_φ is described by

$$\begin{aligned} g <_\varphi g' &\iff g^{-1}g' \in \varphi^{-1}(P_<) \\ &\iff 1 < \varphi(g^{-1}g') \\ &\iff 1 < \varphi(g)^{-1}\varphi(g') \\ &\iff \varphi(g) < \varphi(g'), \end{aligned}$$

leading to the following definition.

Definition 3.1. Given a left-order $<$ on G and an automorphism $\varphi \in \text{Aut}(G)$, define a new left-order $<_\varphi$ on

G by

$$g <_{\varphi} g' \iff \varphi(g) < \varphi(g').$$

As a special case, for some element $g \in G$, we have the *conjugated order* $<_g$ given by

$$h <_g h' \iff ghg^{-1} < gh'g^{-1} \iff hg^{-1} < h'g^{-1}.$$

Note that the set of all bi-orders consists exactly of those left-orders, which are fixed under all possible conjugations, i.e. $<_g = <$ for all $g \in G$.

Later we will see that this defines an action of $\text{Aut}(G)$ on the space of left-orders on G , see Proposition 5.23.

3.2 Group Extensions

The next construction, which is based upon Lemma 7.1 from [KT08], shows that left-orderability is well-behaved with respect to group extensions. For this, we need two additional definitions.

Definition 3.2. Given groups N, G and H , we say that G is an *extension* of H by N if they fit into a short exact sequence

$$1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{p} H \longrightarrow 1$$

of groups, i.e. ι is injective, p is surjective and $\ker p = \text{im } \iota$.

Definition 3.3. Let $(G, <), (G', <')$ be two left-ordered groups and $\varphi : G \rightarrow G'$ a homomorphism between them. Then we say that φ is *order-preserving*, if $g < h$ implies $\varphi(g) \leq' \varphi(h)$. We call it *strictly order-preserving*, if $g < h$ implies $\varphi(g) <' \varphi(h)$.

Note that because of left-invariance, for φ to be (strictly) order-preserving, it is enough that it is order-preserving with respect to 1, i.e. $1 < g$ implies $1 <' \varphi(g)$.

Proposition 3.4. Assume that G is an extension of H by N , i.e. there is a short exact sequence

$$1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{p} H \longrightarrow 1$$

of groups.

If N and H are left-ordered by $<_N$ and $<_H$ respectively, then there is a unique left-order $<$ on G , such that ι and p are order-preserving. The order is given by

$$g < g' \iff \text{Either } (p(g) <_H p(g')) \\ \text{or } (p(g) = p(g') \text{ and } 1 <_N g^{-1}g').$$

Furthermore, if the given orders $<_N$ and $<_H$ are bi-invariant and the image $\iota(P_N)$ of the positive cone P_N of $<_N$ is normal in G , then the resulting order on G is also bi-invariant.

Proof. Denote the positive cone of $<_N$ by P_N and the positive cone of $<_H$ by P_H . Construct a new order $<$ on

G by defining its positive cone as

$$P_{<} := p^{-1}(P_H) \sqcup \iota(P_N).$$

For the semigroup property, take $g, g' \in P_{<}$. If $p(g), p(g') \in P_H$, then $p(gg') \in P_H$, hence $gg' \in p^{-1}(P_H)$. If $p(g) \in P_H, g' \in \iota(P_N)$, then $p(gg') = p(g)$ by exactness, hence $gg' \in p^{-1}(P_H)$ and similarly for g and g' swapped. Lastly, if $g, g' \in \iota(P_N)$, then also $gg' \in P_N$, since P_N is a semigroup. In all cases, we have $gg' \in P_{<}$. For the partitioning property, first note that

$$p^{-1}(P_H \cap P_H^{-1}) = \emptyset = \iota(P_N \cap P_N^{-1})$$

by properties of P_H and P_N and that

$$p^{-1}(P_H) \cap \iota(P_N^{-1}) = \emptyset = \iota(P_N) \cap p^{-1}(P_H^{-1}),$$

since $\text{im } \iota = \ker p$ and $1 \notin P_H, 1 \notin P_H^{-1}$. Hence we obtain

$$P_{<} \cap P_{<}^{-1} = \emptyset.$$

Additionally, we have

$$\begin{aligned} P_{<} \cup P_{<}^{-1} &= p^{-1}(P_H \cup P_H^{-1}) \cup \iota(P_N \cup P_N^{-1}) \\ &= p^{-1}(H \setminus \{1\}) \cup \iota(N \setminus \{1\}) \\ &= G \setminus \ker p \cup (\iota(N) \setminus \{1\}) \\ &= G \setminus \{1\}, \end{aligned}$$

where we used that $\iota(N) = \ker p$.

The resulting order on G is then described by $g < g'$ if and only if either $p(g^{-1}g') \in P_H$ or $g^{-1}g' \in \iota(P_N)$, i.e. either

- i) $p(g) <_H p(g')$
- ii) or $p(g) = p(g')$, hence $g^{-1}g' \in \ker p = \text{im } \iota$, and $1 <_N g^{-1}g'$.

Regarding uniqueness, assume we have a left-order $\prec$ on G , such that ι and p are order-preserving. Furthermore assume $g \prec g'$. Then by order-preservation we have $p(g) \preceq p(g')$, i.e. either $p(g) <_H p(g')$ or $p(g) = p(g')$. In the second case, we have $g^{-1}g' \in \ker p = \text{im } \iota$, and thus also $1 <_N g^{-1}g'$, since we have $1 \prec g^{-1}g'$ and otherwise we would arrive at a contradiction because of order-preservation of ι . This shows that $g \prec g'$ implies $g < g'$, where $<$ is the order constructed above.

On the other hand, assume we have $g < g'$, then either $p(g) <_H p(g')$, i.e. $g \prec g'$ by order-preservation of p , or $p(g) = p(g')$ and $1 <_N g^{-1}g'$ and thus again $g \prec g'$ by order-preservation of ι . All in all, it was shown that $g < g'$ if and only if $g \prec g'$.

For the last part, assume that $<_N$ and $<_H$ are both bi-invariant and $\iota(P_N)$ is normal in G . Let $g, g', h \in G$ with $g < g'$.

If $p(g) < p(g')$, then $p(g)p(h) <_H p(g')p(h)$ by bi-invariance of $<_H$ and thus $p(gh) < p(g'h)$, hence $gh <$

$g'h$.

If $p(g) = p(g')$ and $1 <_N g^{-1}g'$, then

$$p(gh) = p(g)p(h) = p(g')p(h) = p(g'h)$$

and

$$(gh)^{-1}(g'h) = h^{-1}g^{-1}g'h >_N 1$$

by assumption, yielding again $gh < g'h$. $\square$

Note that for the last part about bi-invariance, the condition that $<_N$ is bi-invariant is actually not needed, since if $\iota(P_N)$ is normal in G , then in particular P_N is normal in N , hence $<_N$ is bi-invariant by Proposition 2.13.

Example 3.5. i) Lexicographic orders on direct products, in particular the one we saw in Example 2.2 ii), are a special case of the Proposition 3.4. A direct product comes with a short exact sequence

$$1 \longrightarrow H \xrightarrow{h \mapsto (1, h)} G \times H \xrightarrow{(g, h) \mapsto g} G \longrightarrow 1$$

so for the induced order on $G \times H$, we have $(g, h) < (g', h')$ if and only if either $g < g'$ in G or $g = g'$ and $h < h'$ in H .

ii) For the case of a semidirect product, we have a split exact sequence

$$1 \longrightarrow N \xrightarrow{\iota} N \rtimes H \xrightarrow[p]{p} H \longrightarrow 1$$

with $p \circ s = \text{id}_H$ and every element g in $N \rtimes H$ can be written as

$$g = \iota(n)s(h)$$

for unique $(n, h) \in N \times H$. Now suppose we have $g = \iota(n)s(h)$ and $g' = \iota(n')s(h')$, such that $g < g'$ with respect to the induced order on $N \rtimes H$. Then either $h = p(g) < p(g') = h'$ in H or $h = h'$ and

$$\begin{aligned} 1 < g^{-1}g' &\iff 1 < s(h)^{-1}\iota(n^{-1})\iota(n')s(h') \\ &\iff 1 < s(h)^{-1}\iota(n^{-1}n')s(h) \end{aligned}$$

with respect to the order on N . So in the second case we have to compare the conjugate of $\iota(n^{-1}n')$ by $s(h)$ with 1. So by identifying the underlying sets of $N \rtimes H$ and $N \times H$ via $g \cong_{\text{Set}} (n, h)$, the induced order on $N \rtimes H$ can be understood as a “twisted” lexicographic order, reading from the right. Of course, if the semidirect product here was indeed a direct product as considered above, then following the definitions of ι, p and s , we would see that elements of $\iota(N)$ and $s(H)$ commute and arrive again at the lexicographic order (but reading “from the right” instead of “from the left”).

iii) This example follows Example 1.9 and Problem 1.10 from [CR15]. As a special case of the previous point,

consider the fundamental group of the Klein Bottle

$$K = \langle x, y \mid xyx^{-1} = y^{-1} \rangle.$$

and the normal subgroup $\langle y \rangle$. Here we have a split exact sequence

$$1 \longrightarrow \langle y \rangle \xrightarrow{\text{inc}} K \xrightarrow[p]{p} K/\langle y \rangle \longrightarrow 1$$

where the right-split map j is given by $x^k \cdot \langle y \rangle \mapsto x^k$. In particular, K is the semidirect product of $\langle y \rangle$ with $K/\langle y \rangle$, hence every element $g \in K$ can be written as

$$g = y^m x^n = x^n y^{m \cdot (-1)^n},$$

for unique $n, m \in \mathbb{Z}$, where the last equation follows from the defining relations of K . Note that when identifying the underlying set of the semidirect product with $\langle y \rangle \times K/\langle y \rangle$, the group operation becomes

$$\begin{aligned} (y^m, x^n) \cdot (y^{m'}, x^{n'}) &= (y^m \cdot x^n y^{m'} x^{-n}, x^{n+n'}) \\ &= (y^{m+m' \cdot (-1)^n}, x^{n+n'}). \end{aligned}$$

Now $\langle y \rangle \cong \mathbb{Z}$ and indeed also $K/\langle y \rangle \cong \mathbb{Z}$. To see this, consider the surjective homomorphism

$$f : K \rightarrow \mathbb{Z}, \quad x \mapsto 1, y \mapsto 0,$$

i.e. the projection onto the x -component. Looking at the group operation on the semidirect product, it is easily seen that this projection is indeed linear.

Since $\langle y \rangle \subseteq \ker f$, this induces a homomorphism $\bar{f} : K/\langle y \rangle \rightarrow \mathbb{Z}$ that is also surjective. To see that it is injective, note that all elements in $K/\langle y \rangle$ are of the form $x^k \cdot \langle y \rangle$ for some $k \in \mathbb{Z}$. Then

$$\bar{f}(x^k \cdot \langle y \rangle) = f(x^k) = k,$$

thus a left-inverse of $\bar{f}$ is given by $k \mapsto x^k \cdot \langle y \rangle$, showing bijectivity of $\bar{f}$.

Under these identifications, we have

$$K \cong \mathbb{Z} \rtimes_{\varphi} \mathbb{Z},$$

where

$$\varphi : \mathbb{Z} \rightarrow \text{Aut}(\mathbb{Z}), \quad k \mapsto (n \mapsto (-1)^k \cdot n).$$

Using Proposition 3.4, combined with the fact that on $\mathbb{Z}$ there are only two possible left-orders (see Example 2.2 i)), yields 4 different left-orders on K . They only differ in how the exponents of elements of the form $y^m x^n$ are compared. One of these is described in Example 2.2 iv).

Furthermore, this construction serves as a counterexample for the second part of Proposition 3.4.

The orders on the $\mathbb{Z}$ -factors are bi-invariant, but no order on K can be bi-invariant, since K has generalized torsion, as already observed before.

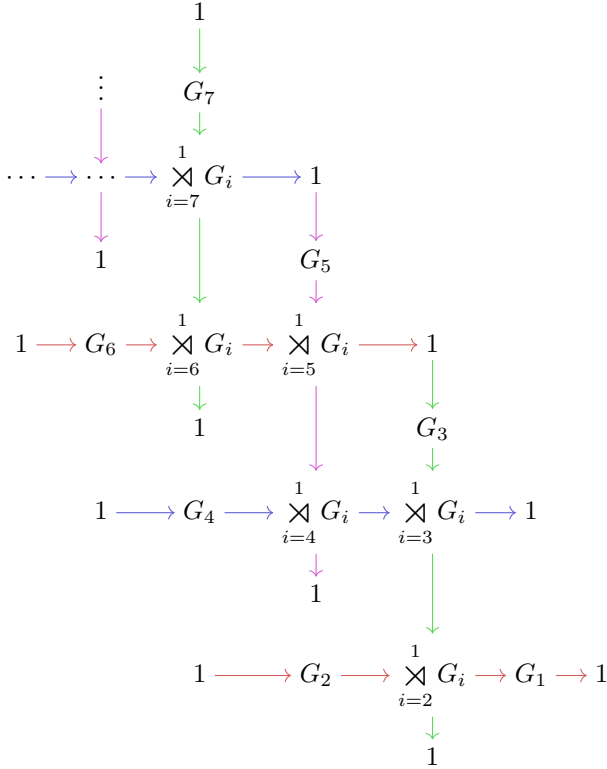
- iv) Lastly, as an extension to point ii), let us consider what happens when dealing with iterated semidirect products of the form

$$G = G_n \rtimes (G_{n-1} \rtimes (\cdots \rtimes (G_3 \rtimes (G_2 \rtimes G_1) \cdots)),$$

where each factor G_i is left-orderable. Writing

$$\bigtimes_{i=k}^1 G_i := G_k \rtimes \cdots \rtimes G_1,$$

this corresponds to a diagram of vertical and horizontal split exact sequences of the form



Then every element $g \in G$ is of the form

$$g = g_n \cdots g_1 \cong_{\text{Set}} (g_n, \dots, g_1)$$

with unique $g_i \in G_i$. By iteratively applying Proposition 3.4 to this diagram starting with $G_2 \rtimes G_1$, we get an induced order on G . Given $g = g_n \cdots g_1, h = h_n \cdots h_1$, in order to determine if $g < h$ we start “looking from the right” to spot the first index i where $g_i \neq h_i$ and then check if

$$1 < g_1^{-1} \cdots g_{i-1}^{-1} g_i^{-1} h_i h_{i-1} \cdots h_1$$

holds in G_i . If it is the case, then $g < h$, otherwise $g \geq h$. This will be relevant later when constructing

an order for the pure braid group PB_n in Section 4.3, as PB_n can be written as an iterated semidirect product.

3.3 Convex Subgroups

The constructions here are using the concept of convexity with respect to some order. Here, we follow Chapter 2.2. from [CR15] and Chapter 2.1.1 in [DNR16].

Definition 3.6. Given a left-ordered group $(G, <)$, a subset $S \subseteq G$ is *convex* with respect to $<$, also called *<-convex*, if $g < f < h$ implies $f \in S$ for all $g, h \in S$ and all $f \in G$. A subset $T \subseteq G$ is *relatively convex*, if there is a left-order $<'$ on G such that T is convex with respect to $<'$.

Intuitively, $<$ -convex subsets in a group behave like intervals in $\mathbb{R}$ (when considered with the usual ordering). As soon will become clear, our main interest lies in convex subsets that are also subgroups. For a $<$ -convex subgroup $C \subseteq G$, because of left-invariance, the condition

$$c < f < c' \implies f \in C$$

for $c, c' \in C, f \in G$ is equivalent to both

$$1 < c^{-1}f < c^{-1}c' \implies c^{-1}f \in C \implies f \in C$$

and

$$(c')^{-1}c < (c')^{-1}f < 1 \implies (c')^{-1}f \in C \implies f \in C,$$

so we can choose 1 to be on either side of the initial inequality. Intuitively, this means that convex subgroups are the subgroups “closest to 1”, in the sense that any element that lies between 1 and some element of C must already be contained in C . This is underlined by the fact that convex subgroups are totally ordered by inclusion, as stated as Problem 2.9 in [CR15], which we prove next.

Proposition 3.7. Given a left-order $<$ on a group G , the set

$$\{H \subseteq G \mid H \text{ is } <\text{-convex subgroup}\}$$

is totally ordered by inclusion.

Proof. Let C, D be subgroups of G which are convex with respect to $<$. We want to show that either $C \subseteq D$ or $D \subseteq C$. Assume the contrary, i.e. both $D \setminus C$ and $C \setminus D$ are non-empty. Then there exist $c \in C \setminus D$ and $d \in D \setminus C$. Both elements are different than 1, and we can assume $c, d > 1$, otherwise consider $\tilde{c} = c^{-1}$ and/or $\tilde{d} = d^{-1}$. Since the order on G is total, we either have $c < d$ or $d < c$. In the first case we obtain

$$1 < c < d$$

and thus $c \in D$ by convexity of D and in the second case it follows that

$$1 < d < c$$

and thus $d \in C$ by convexity of C . Both lead to a contradiction. $\square$

Next, we have a look at some examples for convex subgroups.

Example 3.8. i) The subgroup

$$\Delta = \{(n, n) \mid n \in \mathbb{Z}\}$$

of $(\mathbb{Z}^2, +)$ is not convex with respect to the standard lexicographic order $<_{\text{lex}}$, since for example $(0, 0) < (1, 0) < (1, 1)$ but $(1, 0) \notin \Delta$.

However,

$$H := \{(0, n) \mid n \in \mathbb{Z}\}$$

is a $<_{\text{lex}}$ -convex subgroup. Indeed, assume

$$(0, n) <_{\text{lex}} (a, b) <_{\text{lex}} (0, n')$$

for some $n, a, b, n' \in \mathbb{Z}$ and suppose $a \neq 0$. Then the first inequality implies $0 < a$, whereas the second inequality dictates $a < 0$, leading to a contradiction. This forms a special case of the next point.

ii) As noted in [CR15] Problem 2.12, the orderings of $\mathbb{Z}^2$ as in Example 2.14 iv) have a non-trivial convex subgroup when the line used for the ordering has rational slope. Indeed, let $v \in \mathbb{R}^2$ be a vector with rational slope and choose another vector w that is orthogonal to v . Consider

$$A = L \cap \mathbb{Z}^2 = \{p \in \mathbb{Z}^2 \mid \langle v, p \rangle = 0\},$$

i.e. exactly those vectors with integer coordinates that lie on the line L which is spanned by w .

Then A is a subgroup of $\mathbb{Z}^2$ by properties of the inner product. For convexity, let $p, p' \in A$ with $p < p'$ and $q \in \mathbb{Z}^2$ with $p < q < p'$. This means we have $\langle v, p \rangle = \langle v, p' \rangle = 0$ and either

a) $\langle v, q - p \rangle > 0$ or

b) $\langle v, q - p \rangle = 0$ and $\langle w, q - p \rangle > 0$

and either

A) $\langle v, p' - q \rangle > 0$ or

B) $\langle v, p' - q \rangle = 0$ and $\langle w, p' - q \rangle > 0$.

Assume that a) or A) holds, then

$$0 = \langle v, p' - p \rangle = \langle v, p' - q \rangle + \langle v, q - p \rangle > 0,$$

which is not possible. Hence, b) and B) must be true. In particular, we have

$$\langle v, q \rangle = \langle v, p + q - p \rangle = \langle v, p \rangle + \langle v, q - p \rangle = 0$$

and thus $q \in A$.

Lastly, note that A is non-trivial, since v was chosen to have rational slope and thus the line spanned by w intersects $\mathbb{Z}^2$ in more than one point.

iii) This example is based on Problem 2.13 and the paragraphs before in [CR15].

Recall from Proposition 3.4 that we can construct a left-order $<$ on G induced via the short exact sequence

$$1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{p} Q \longrightarrow 1$$

whenever N and Q are left-ordered.

Then $\text{im } \iota$ is $<$ -convex in G . Indeed, assume that $n < g < n'$ for $n, n' \in \text{im } \iota$ and $g \in G$. If

$$\underbrace{p(n)}_{=1} = p(g) \quad \text{or} \quad p(g) = \underbrace{p(n')}_{=1},$$

then $g \in \ker p = \text{im } \iota$. On the other hand, if $p(g) \neq 1$, then $1 < p(g) < 1$ in Q , a contradiction.

In particular, given a (not necessarily order-preserving) homomorphism

$$\varphi : (G, <) \rightarrow (H, <'),$$

we can construct new left-order $\prec$ on G for which $\ker \varphi$ is convex and $\varphi : (G, \prec) \rightarrow (H, <')$ is order-preserving. Indeed, restrict $<$ to $\ker \varphi$ and consider the short exact sequence

$$1 \longrightarrow \ker \varphi \hookrightarrow G \xrightarrow{\varphi} H \longrightarrow 1$$

Then take $\prec$ to be the left-order induced by this sequence. It is given by

$$g < g' \iff \text{Either } \varphi(g) <' \varphi(g') \\ \text{or } \varphi(g) = \varphi(g') \text{ and } g < g'.$$

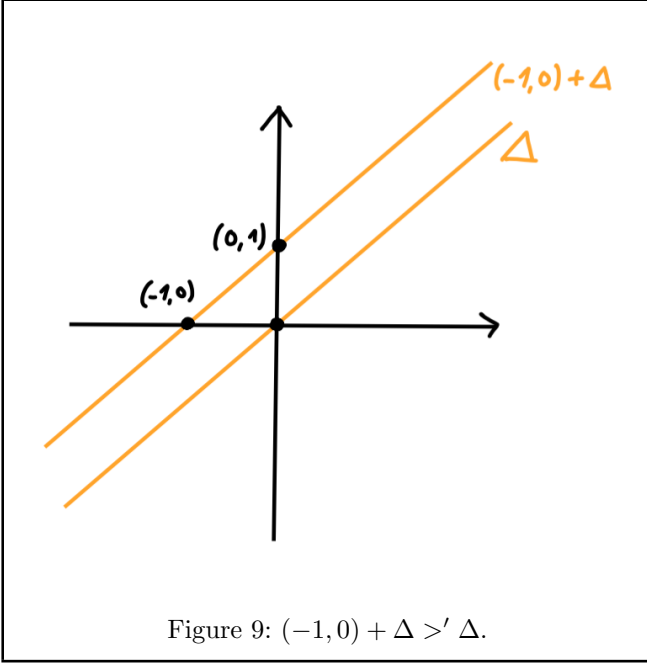
iv) For a concrete realization of the previous point, consider the short exact sequence

$$1 \longrightarrow \Delta \hookrightarrow (\mathbb{Z}^2, <_{\text{lex}}) \xrightarrow{p} (\mathbb{Z}^2/\Delta, <') \longrightarrow 1$$

where $\Delta \subseteq \mathbb{Z}^2$ is the diagonal consisting of elements of the form (n, n) for $n \in \mathbb{Z}$ and $p : (n, m) \mapsto (n, m) + \Delta$. Here, $<_{\text{lex}}$ is the standard lexicographic order on $\mathbb{Z}^2$ and define $<'$ on $\mathbb{Z}^2/\Delta$ by “ordering by y -coordinate”, i.e. we consider the point in $\mathbb{Z}^2$ where the line spanned by an element $(n, m) + \Delta$ meets the y -axis. Thus we have to define

$$\Delta <' (n, m) + \Delta = (0, m - n) + \Delta :\iff m - n > 0.$$

Note that p is not order-preserving in this case, since for example we have $(-1, 0) <_{\text{lex}} (0, 0)$ but $(-1, 0) +$



$\Delta = (0, 1) + \Delta >' \Delta$, see Fig. 9. Then the induced order $\prec$ from Proposition 3.4 on $\mathbb{Z}^2$ is

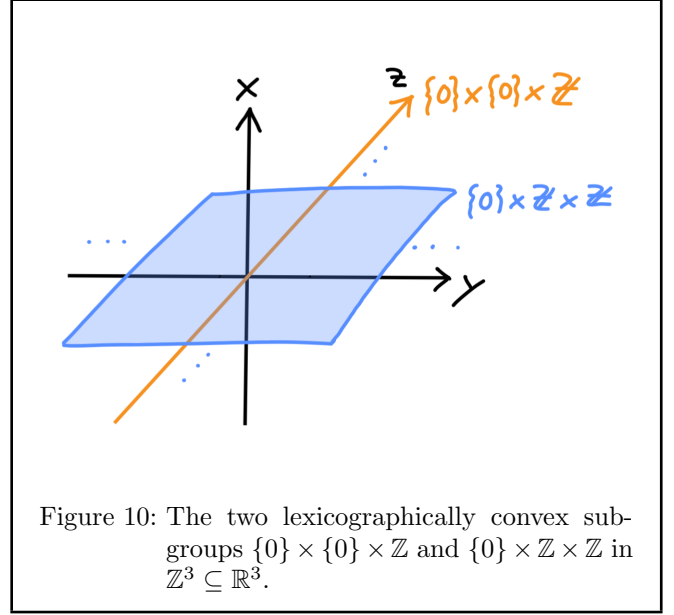
$$(0, 0) \prec (n, m) \iff \text{either } n = m \text{ and } n > 0 \\ \text{or } m > n.$$

Here we can see that the positive cone $P_{\prec}$ consists of those elements in $\mathbb{Z}^2$ that either lie “above” the diagonal Δ or lie on the diagonal with positive y -coordinate. Hence, this order is the same as the order defined in Example 2.14 iv) when choosing $v = (-1, 1)$ and $w = (1, 1)$.

- v) Similar to point i) in this Example, the lexicographic order on $\mathbb{Z}^3 = \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ induced by the two short exact sequences

$$\begin{array}{ccccccc} & & 1 & & & & \\ & & \downarrow & & & & \\ & & \mathbb{Z} & & & & \\ & & \downarrow z \mapsto (0, 0, z) & & & & \\ & & \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z} & & & & \\ & & \downarrow (x, y, z) \mapsto (x, y) & & & & \\ 1 & \longrightarrow & \mathbb{Z} & \xhookrightarrow{y \mapsto (0, y)} & \mathbb{Z} \times \mathbb{Z} & \twoheadrightarrow_{(x, y) \mapsto x} & \mathbb{Z} \longrightarrow 1 \\ & & & & \downarrow & & \\ & & & & 1 & & \end{array}$$

has $\{0\} \times \{0\} \times \mathbb{Z}$ and $\{0\} \times \mathbb{Z} \times \mathbb{Z}$ as its only proper, non-trivial convex subgroups, see Fig. 10. Visually, this underlines the intuition of convex subgroups being “closest to the identity”. That these are indeed



the only convex subgroups besides $\{0\}^3$ and $\mathbb{Z}^3$ itself follows from Proposition 3.17.

The next Definition is based on a remark in Chapter 2.1.1 in [DNR16].

Definition 3.9. Let C be a $<$ -convex subgroup of G . Choose a left-order $<'$ on C . Denote the positive cones of $<$ and $<'$ as $P_{<}$ and $P_{<'}$ respectively. Then the *convex extension* of $<'$ by $<$ is the left-order on G given by the positive cone $P := (P_{<} \setminus C) \sqcup P_{<'}$.

Intuitively, a convex extension changes the order on a subgroup C , while keeping the original order on the complement $G \setminus C$. The proof that a convex extension indeed yields a well-defined left-order is done later in Proposition 3.14, as we will need some additional preparation first.

An important example for a convex extension is the following Example 3.10, which is based on Example 2.1.4 in [DNR16]. This construction will become relevant later when examining the space of orders on the braid group B_n see Section 7.1.

Example 3.10. Suppose we have a group G with some left-order $<$ and a $<$ -convex subgroup C . Then the order on C may be “flipped”, while keeping the order the same on $G \setminus C$. More formally, consider the order $<' = <|_C$ on C and then extend its reversed order $\overline{<'}$ by $<$.

There is also the possibility to “push” an order $<$ via some homomorphism $G \rightarrow H$ if its kernel is $<$ -convex. This construction is based on Problem 2.13 in [CR15] and the discussion before.

Suppose we have a left-orderable G and a homomorphism $\varphi : G \rightarrow H$. Then $\varphi(G)$ is not left-orderable in general, as can be seen in the following example.

Example 3.11. Consider $\mathbb{Z}$ with the standard ordering and the projection $\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$. Its image is not left-orderable as it is not torsion-free.

However, if $\ker \varphi$ is relatively convex, i.e. if there is a left-order $<$ on G such that $\ker \varphi$ is $<$ -convex, then $\varphi(G)$ is left-orderable as follows. Denote the positive cone of $<$ in G as P_G , then

$$P_{\varphi(G)} := \varphi(P_G) \setminus \{1\} = \{\varphi(g) \mid 1 < g, \varphi(g) \neq 1\}$$

is a positive cone in $\varphi(G)$:

For the semisubgroup-property, let $\varphi(g), \varphi(h) \in P_{\varphi(G)}$. Then $1 < g < gh$. Suppose $\varphi(gh) = 1$, i.e. $gh \in \ker \varphi$, then by $<$ -convexity we would have $g \in \ker \varphi$, which is a contradiction. Hence $\varphi(g)\varphi(h) = \varphi(gh) \in P_{\varphi(G)}$.

Since φ is a homomorphism, the corresponding negative cone is

$$\begin{aligned} P_{\varphi(G)}^{-1} &= \varphi(P_G)^{-1} \setminus \{1\} \\ &= \{\varphi(g)^{-1} \mid 1 < g, \varphi(g) \neq 1\} \\ &= \{\varphi(g^{-1}) \mid g^{-1} < 1, \varphi(g^{-1}) \neq 1\} \\ &= \varphi(P_G^{-1}) \setminus \{1\}. \end{aligned}$$

This implies

$$\begin{aligned} \varphi(G) \setminus \{1\} &= \varphi(G \setminus \{1\}) \setminus \{1\} \\ &= \varphi(P_G \cup P_G^{-1}) \setminus \{1\} \\ &= P_{\varphi(G)} \cup P_{\varphi(G)}^{-1}. \end{aligned}$$

Lastly, suppose $P_{\varphi(G)} \cap P_{\varphi(G)}^{-1} \neq \emptyset$, i.e. there exists $x \in P_{\varphi(G)} \cap P_{\varphi(G)}^{-1}$. Then there exist $g, h \in G$ with $h < 1 < g$, $\varphi(g) = x = \varphi(h)$ and $g, h \notin \ker \varphi$. The first inequality implies $1 < h^{-1} < h^{-1}g$ and the second equality means $\varphi(h^{-1}g) = x^{-1}x = 1$, hence $h^{-1}g \in \ker \varphi$ and thus $h^{-1} \in \ker \varphi$ by convexity, which leads to a contradiction.

The order on $\varphi(G)$ defined by $P_{\varphi(G)}$ is characterized by

$$\begin{aligned} \varphi(g) < \varphi(h) &\iff g < h \text{ and } g^{-1}h \notin \ker \varphi \\ &\iff g < h \text{ and } \varphi(g) \neq \varphi(h). \end{aligned}$$

We summarize this discussion as follows.

Proposition 3.12. *If the kernel of some homomorphism $\varphi : G \rightarrow H$ is $<$ -convex, then its image $\varphi(G)$ can be left-ordered by*

$$\varphi(g) < \varphi(h) :\iff g < h$$

for $\varphi(g) \neq \varphi(h)$.

Note that Proposition 3.12 implies that a subgroup $H \subseteq G$ for which the quotient G/H has no left-order (for example if it has torsion), is not relatively convex.

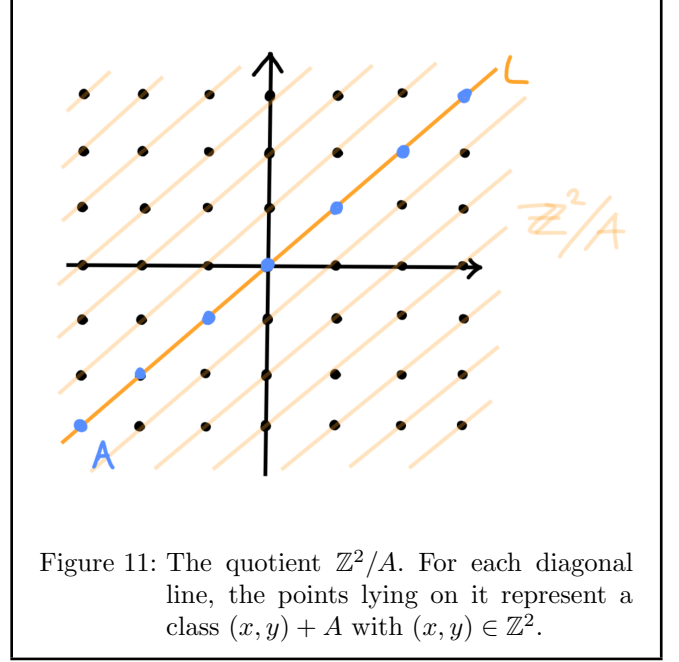


Figure 11: The quotient $\mathbb{Z}^2/A$. For each diagonal line, the points lying on it represent a class $(x, y) + A$ with $(x, y) \in \mathbb{Z}^2$.

Example 3.13. Recall Example 3.8 ii), using the same notation for the vectors v and w and for the subgroup A . Since $(\mathbb{Z}^2, +)$ is abelian, A is a normal subgroup and we can consider the short exact sequence

$$1 \longrightarrow A \hookrightarrow \mathbb{Z}^2 \xrightarrow{p} \mathbb{Z}^2/A \longrightarrow 1$$

We have seen that A is convex with the order $<$ defined by v and w . Hence, $\mathbb{Z}^2/A$ can be left-ordered by

$$(n, m) + A < (n', m') + A \iff (n, m) < (n', m')$$

for $(n, m) - (n', m') \notin A$.

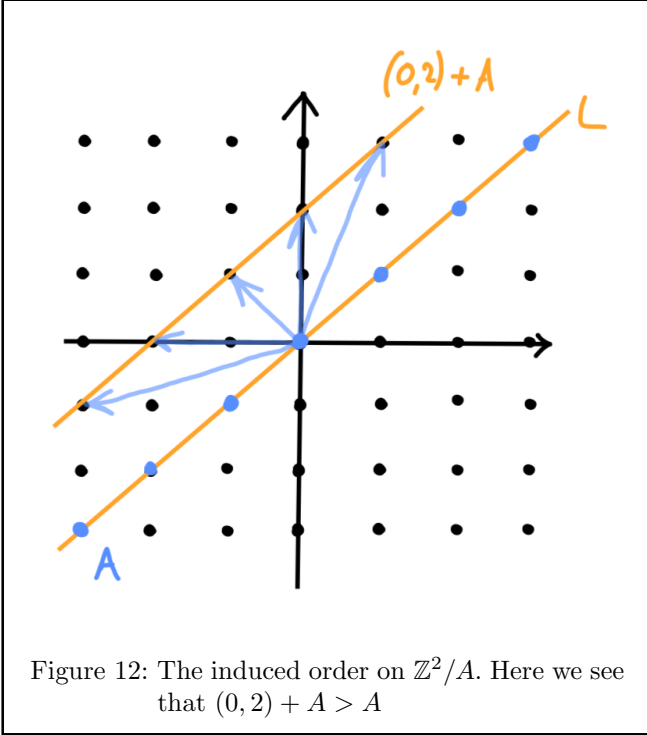
Visually, the quotient $\mathbb{Z}^2/A$ can be understood as the set of lines parallel to L , see Fig. 11. Here, one can see that the induced order on $\mathbb{Z}^2/A$ is independent of the chosen representatives, since for every pair of representatives for two parallel lines, their difference will always be on one side of L which depends only on the chosen lines. This fact is visualized in Fig. 12

Also based on Problem 2.13 from [CR15], we can now show that a convex extension as defined in Definition 3.9 yields a well-defined left-order.

Proposition 3.14. *Let C be a $<$ -convex subgroup of G and choose a left-order $<'$ (that might possibly be different from $<$) on C . Then the convex extension of $<'$ by $<$ given by the positive cone $P = (P_{<} \setminus C) \sqcup P_{<'}$ is a left-order on G .*

Proof. By Proposition 3.12, the quotient G/C can be left-ordered by

$$gC < hC \iff g < h$$



for $gC \neq hC$. Now consider the short exact sequence

$$1 \longrightarrow C \hookrightarrow G \xrightarrow{p} G/C \longrightarrow 1$$

where C is ordered by $<'$ and G/C by $<$, with positive cones $P_{<'}$ and $P_{<}$ respectively. Then by Proposition 3.4 we get an induced order on G with positive cone

$$p^{-1}(P_{<}) \sqcup P_{<' } = (P_{<} \setminus C) \sqcup P_{<' }.$$

Regarding convexity of C , denote the order obtained by the extension process as $<_{\text{ext}}$. Assume we have

$$c <_{\text{ext}} g <_{\text{ext}} c'$$

with $c, c' \in C$. This implies $1 <_{\text{ext}} c^{-1}g$ and $1 <_{\text{ext}} g^{-1}c'$. Assume $g \notin C$, then also $c^{-1}g, g^{-1}c \notin C$, leading to $1 < c^{-1}g$ and $1 < g^{-1}c'$, which combines to

$$c < g < c'.$$

Hence $g \in C$ by $<$ -convexity of C , contradicting the assumption. $\square$

As a special case, for a $<$ -convex subgroup $C \subseteq G$, we can consider the convex extension of the restricted order $<|_C$ by the induced quotient order on G/C and obtain the following result.

Corollary 3.15. *Given a left-order $<$ on G and a $<$ -convex subgroup $C \subseteq G$, the order $<$ is induced by the*

short exact sequence

$$1 \longrightarrow C \hookrightarrow G \xrightarrow{p} G/C \longrightarrow 1$$

where C is left-ordered by the restriction $<|_C$ of $<$ to C and G/C is given the order $<$ from Proposition 3.12.

In particular, there are only 4 left-orders on the fundamental group of the Klein bottle. This result is based on Example 9.2 from [CR15].

Corollary 3.16. *The fundamental group of the Klein Bottle*

$$K = \langle x, y \mid xyx^{-1} = y^{-1} \rangle$$

admits only 4 left-orders, all of which are induced by the short exact sequence

$$1 \longrightarrow \langle y \rangle \hookrightarrow K \xrightarrow{p} K/\langle y \rangle \longrightarrow 1$$

Proof. As stated in Problem 9.1 in [CR15], the subgroup $\langle y \rangle$ is convex with respect to all left-orders on K , hence by Corollary 3.15 any left-order on K is induced by the short exact sequence.

We have seen earlier that $K/\langle y \rangle \cong \mathbb{Z}$, leaving only two possibilities for the order on the right factor. The same is true for $\langle y \rangle \cong \mathbb{Z}$. As a result, any left-order on K must already be induced by choices of left-orders on the two $\mathbb{Z}$ -factors, of which there are 4 possible combinations in total. Recalling that any element in K can be uniquely written as $y^k x^n$ for some $k, n \in \mathbb{Z}$, those orders only differ in how one compares first the exponents from the x -component and then the exponents from the y -component. $\square$

With Proposition 3.12 at hand, one can deduce an upper bound for the number of proper convex subgroups of $\mathbb{Z}^n$, which is Problem 2.15 in [CR15].

Proposition 3.17. *For a given left-order $<$ on $\mathbb{Z}^n$, there are at most $n - 1$ proper, non-trivial $<$ -convex subgroups of $\mathbb{Z}^n$.*

Proof. Assume that $C \subseteq \mathbb{Z}^n$ is a $<$ -convex subgroup. Then $\mathbb{Z}^n/C$ is finitely generated abelian and torsion-free, since it can be left-ordered as in Proposition 3.12. Hence $\mathbb{Z}^n/C \cong \mathbb{Z}^k$ for some $k \leq n$. If C is non-trivial, this inequality must be strict.

Suppose there are n pairwise distinct, proper, non-trivial $<$ -convex subgroups $C_1, \dots, C_n$ of $\mathbb{Z}^n$. We know that $<$ -convex subgroups are totally ordered, so we can assume

$$1 \neq C_1 \subsetneq \dots \subsetneq C_n \neq \mathbb{Z}^n.$$

By the isomorphism theorem, we have

$$\underbrace{\mathbb{Z}^n/C_{i+1}}_{\cong \mathbb{Z}^{k_{i+1}}} \cong \underbrace{(\mathbb{Z}^n/C_i)}_{\cong \mathbb{Z}^{k_i}} / (C_{i+1}/C_i)$$

for some $k_1, \dots, k_n < n$. Since C_{i+1}/C_i is non-trivial, this implies $k_{i+1} > k_i$ for all i . Inductively, we then obtain $k_n \geq n$, contradicting $k_n < n$ as observed initially. $\square$

We conclude this section with a class of orders that have no convex subgroups other than the trivial group and the group itself. This result is needed later for the proof of Sikora's Theorem in section 7.2.

Corollary 3.18. *Archimedean ordered groups have no non-trivial proper convex subgroups.*

Proof. Assume there is a convex subgroup C for some Archimedean ordered group $(G, <)$ such that $\{1\} \neq C \neq G$. In particular, there exist $z \in G \setminus C$ and $g \in C \setminus \{1\}$. By Proposition 2.19, there is an integer $n \in \mathbb{Z}$ with

$$g^n \leq z < g^{n+1}.$$

But then convexity of C implies $z \in C$, which contradicts the assumption. $\square$

In Example 3.8 i) we have seen that there is a proper non-trivial $<_{\text{lex}}$ -convex subgroup of $\mathbb{Z}^2$. Combined with Corollary 3.18, this provides another way to show that the standard lexicographic order $<_{\text{lex}}$ on $\mathbb{Z}^2$ is not Archimedean, as observed in Example 2.18.

4 More Elaborate Examples

In this section, we explore some more involved examples. All of these are fairly standard in the literature and they serve as interesting examples in addition to the ones we have already seen.

4.1 Ordering $\text{Homeo}_+(\mathbb{R})$

As we have seen before in Theorem 2.23, the group $\text{Homeo}_+(\mathbb{R})$ of order-preserving homeomorphisms on the real line serves as an important example in the class of left-orderable groups.

The following construction is based on Example 1.11 from [CR15].

Example 4.1. Consider $(\text{Homeo}_+(\mathbb{R}), \circ)$, the group of all strictly increasing homeomorphisms $f : \mathbb{R} \rightarrow \mathbb{R}$. By choosing some countable and dense set $\{x_1, x_2, \dots\} \subseteq \mathbb{R}$, we can do the following: Let $f \in \text{Homeo}_+(\mathbb{R}) \setminus \{\text{id}\}$, then there exists $y \in \mathbb{R}$ for which $f(y) \neq y$. Since f is continuous and the chosen set is dense in $\mathbb{R}$, there is at least one x_i such that $f(x_i) \neq x_i$ and so the set

$$\{i \in \mathbb{N} \mid f(x_i) \neq x_i\}$$

is non-empty. Thus the minimum

$$m(f) := \min\{i \in \mathbb{N} \mid f(x_i) \neq x_i\}$$

exists. With this we are able to specify a positive cone for a left-order on $\text{Homeo}_+(\mathbb{R})$ as

$$P := \{f \in \text{Homeo}_+(\mathbb{R}) \setminus \{\text{id}\} \mid x_{m(f)} < f(x_{m(f)})\}.$$

Let us check the defining properties of a positive cone.

For $f, g \in P$ we have

$$x_{m(f)} < f(x_{m(f)}), \quad x_{m(g)} < g(x_{m(g)}).$$

There are some cases to consider. If $m(g) \leq m(f)$, then $m(f \circ g) = m(g)$ and $x_{m(g)} \leq f(x_{m(g)})$. Thus

$$x_{m(g)} \leq f(x_{m(g)}) < f \circ g(x_{m(g)}),$$

since $x_{m(g)} < g(x_{m(g)})$ and f is order-preserving. On the other hand, if $m(f) < m(g)$, then $m(f \circ g) = m(f)$ and $g(x_{m(f)}) = x_{m(f)}$, yielding

$$x_{m(f)} < f(x_{m(f)}) = f \circ g(x_{m(f)}).$$

In both cases we obtain $f \circ g \in P$.

Now what is P^{-1} ? For $f \in P$ we have $x_{m(f)} < f(x_{m(f)})$, so by order-preservation we get $f^{-1}(x_{m(f)}) < x_{m(f)}$. Also $m(f) = m(f^{-1})$, so this means

$$P^{-1} = \{g \in \text{Homeo}_+(\mathbb{R}) \setminus \{\text{id}\} \mid g(x_{m(g)}) < x_{m(g)}\}.$$

Hence it is clear that $P \cap P^{-1} = \emptyset$.

Finally, $P \cup P^{-1} = \text{Homeo}_+(\mathbb{R}) \setminus \{\text{id}\}$ follows from the initial discussion, since for every $f \in \text{Homeo}_+(\mathbb{R})$ different from the identity we either have $f(x_{m(f)}) < x_{m(f)}$ or $f(x_{m(f)}) > x_{m(f)}$. Concluding this example, we have a look at the left-order that is derived from positive cone we defined. For $f \neq g$ We have

$$\begin{aligned} f < g &: \iff \text{id} < f^{-1} \circ g \\ &: \iff f^{-1} \circ g \in P \\ &\iff x_{m(f^{-1} \circ g)} < f^{-1} \circ g(x_{m(f^{-1} \circ g)}) \\ &\iff f(x_{m(f^{-1} \circ g)}) < g(x_{m(f^{-1} \circ g)}), \end{aligned}$$

and

$$\begin{aligned} m(f^{-1} \circ g) &= \min\{i \in \mathbb{N} \mid f^{-1} \circ g(x_i) \neq x_i\} \\ &= \min\{i \in \mathbb{N} \mid g(x_i) \neq f(x_i)\}, \end{aligned}$$

so $f < g$ if and only if $f(x_i) < g(x_i)$ at the smallest index i where they differ.

4.2 Two Orders on the Finitely Generated Free Group F_n

Throughout this section, let $F_n = \langle x_1, \dots, x_n \rangle$ be the free group on n generators.

Note that while simple approaches like lexicographic orders work for free monoids $\langle 1, x_1, \dots, x_n \rangle^+$, construct-

ing left-orders for free groups requires more work. The reason for this are cancellations that might occur when passing from the free monoid to the free group, as seen in the next Example.

Example 4.2. Consider $F_2 = \langle a, b \rangle$. Say we order the generators as $b^{-1} < a^{-1} < 1 < a < b$. Trying a simple lexicographic order based on this ordering of generators, we would have

$$1 < ab^{-1}a^{-1} \quad \text{and} \quad 1 < ba^{-1},$$

but $ba^{-1} \cdot ab^{-1}a^{-1} = a^{-1} < 1$, violating left-invariance. Similarly, a shortlex-type order, where one first compares word-length and then orders lexicographically, would yield $ab < bb$ but $b^{-1} \cdot ab > b$, again not satisfying left-invariance.

If one would do something similar to the order on the fundamental group K of the Klein Bottle as in Example 3.5 iii), i.e. comparing via sums of exponents, then for example $bab^{-1}a^{-1}$ would be neither greater or smaller than 1, violating totality of the order. Note that the order on K was constructed via a short exact sequence. The same approach does not work here for F_2 , since $\langle b \rangle$ is not normal in F_2 as opposed to K .

Before we come to the examples for left-orders on F_n , we introduce the concept of a N -ball in the Cayley graph Γ_A for some generating set A of a group G . We need this notion at various points throughout this thesis, see Section 6.1 and in particular Section 7.3.

Definition 4.3. Let G be a group and fix a generating set A of G , for which we consider the corresponding Cayley graph Γ_A .

- i) Define the *length* $|g|$ of an element $g \in G$ as

$$|g| := \min\{k \in \mathbb{Z}_{\geq 0} \mid \text{There exist } g_i \in A, \varepsilon_i = \pm 1, \text{ such that } g = g_1^{\varepsilon_1} \cdots g_k^{\varepsilon_k}\}.$$

- ii) Denote the N -ball around 1 in Γ_A as

$$B_N := \{g \in G \mid |g| \leq N\}.$$

Let us have a look at some examples.

Example 4.4. i) Consider the group

$$(\mathbb{Z}^2, +) \xrightarrow[(n,m) \mapsto a^n b^m]{\cong} \langle a, b \mid ab = ba \rangle.$$

Under this identification we have

$$B_k = \{(n, m) \in \mathbb{Z}^2 \mid \underbrace{|n| + |m|}_{=|| (n, m) ||_1} \leq k\},$$

see Fig. 13 for the case $k = 2$.

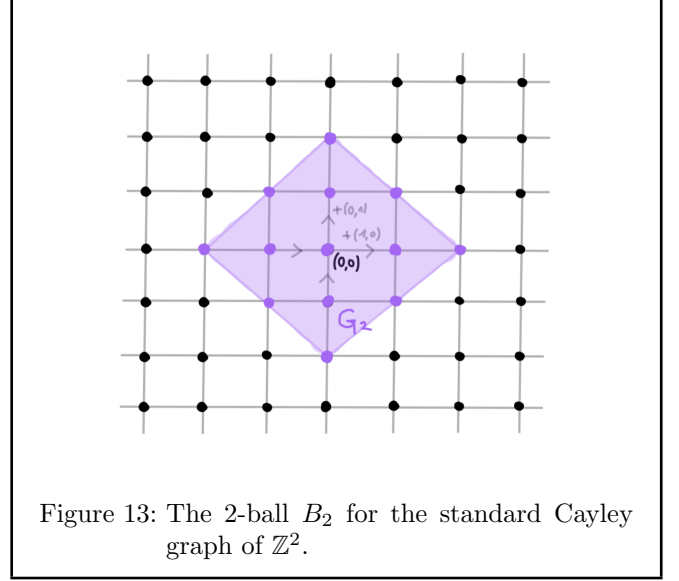


Figure 13: The 2-ball B_2 for the standard Cayley graph of $\mathbb{Z}^2$.

- ii) Consider a different generating set for $\mathbb{Z}^2$, for example take

$$\mathbb{Z} \cong \langle a^2, a^3 \rangle,$$

then

$$\mathbb{Z}^2 \cong \langle a^2, a^3, b^2, b^3 \mid ab = ba \rangle.$$

Note that then we obtain

$$B_1^{\mathbb{Z}} = \{1, a^2, a^3, a^{-2}, a^{-3}\}$$

as the 1-ball for $\mathbb{Z}$. Since $a = a^{-2}a^3, a^4 = a^2a^2, a^5 = a^2a^3$ and $a^6 = a^3a^3$, we have

$$B_2^{\mathbb{Z}} = B_1^{\mathbb{Z}} \cup \{a, a^4, a^5, a^6, a^{-1}, a^{-4}, a^{-5}, a^{-6}\}.$$

For $\mathbb{Z}^2$, we have

$$B_2^{\mathbb{Z}^2} = \{1, a^{\pm 1}, \dots, a^{\pm 6}, b^{\pm 1}, \dots, b^{\pm 6}, a^{\pm 2}b^{\pm 2}, a^{\pm 3}, b^{\pm 2}, a^{\pm 2}b^{\pm 3}, a^{\pm 3}b^{\pm 3}\},$$

see Fig. 14

- iii) Consider the free group on 2 generators

$$F_2 = \langle a, b \rangle.$$

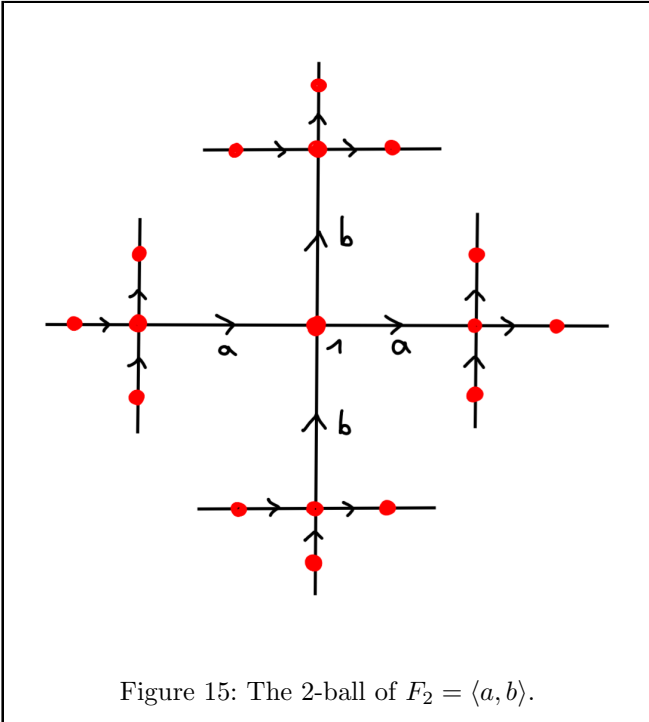
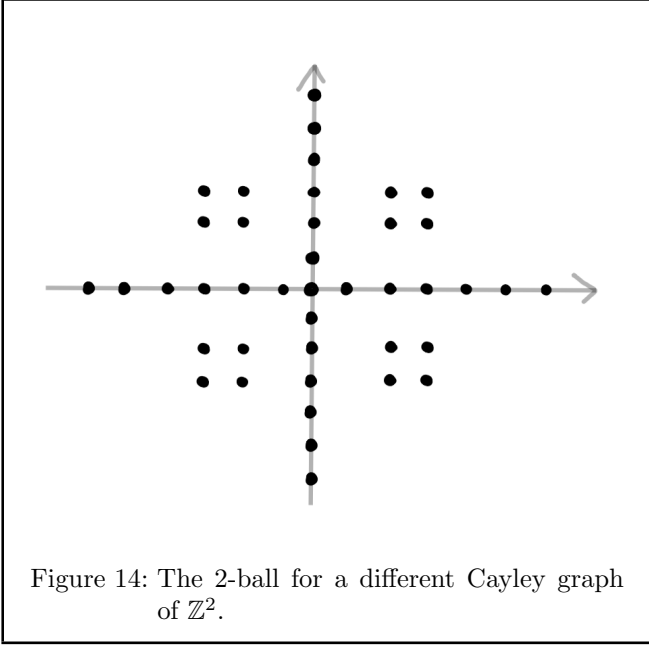
Here the 2-ball is given as

$$B_2 = \{1, a, b, a^{-1}, b^{-1}, a^2, b^2, a^{-2}, b^{-2}, ab, ba, ab^{-1}, ba^{-1}, b^{-1}a, b^{-1}a^{-1}, a^{-1}b, a^{-1}b^{-1}\}$$

see Fig. 15.

4.2.1 Explicit Left-order on F_n

In [Š13], an explicit left-order on $F_n = \langle x_1, \dots, x_n \rangle$ is constructed, which extends the lexicographic order on



the free monoid $\langle 1, x_1, \dots, x_n \rangle^+$ given by $x_1 < \dots < x_n$.

First, consider the weight function $w : F_n \rightarrow \mathbb{R}$ defined as

$$w(g) := \#\{\text{subwords of } g \text{ of the form } x_j x_i^{-1} \text{ for } j > i\} \\ - \#\{\text{subwords of } g \text{ of the form } x_j^{-1} x_i \text{ for } j > i\} \\ + \frac{1}{2} \cdot \begin{cases} 1, & \text{if } g \text{ ends in any positive letter,} \\ -1, & \text{if } g \text{ ends in any negative letter,} \\ 0, & \text{if } g \text{ is trivial.} \end{cases}$$

Then define $<_E$ to be the order given by the positive cone

$$P_E := \{g \in F_n \mid w(g) > 0\}.$$

Next we have a look at some exemplary inequalities with respect to this order.

Example 4.5. Let $F_2 = \langle a, b \rangle$ with $a < b$.

i) We have

$$\begin{aligned} w(a) &= 1/2, \\ w(a^{-1}b) &= 1/2, \\ w(ba^{-1}) &= 1 - 1/2 = 1/2, \\ w(a^{-1}) &= -1/2 \end{aligned}$$

and thus

$$b^{-1} <_E a^{-1} <_E 1 <_E a <_E b.$$

ii) More generally, the intersection of P_E with the 2-ball B_2 is

$$P_E \cap B_2 = \{a, b, ab, a^2, ba, ba^{-1}, b^2, a^{-1}b\},$$

see Fig. 16. See Fig. 17 for a visualization of $P_E \cap B_3$.

iii) Considering only elements in the 2-ball B_2 , the elements are in ascending order

$$ab^{-1} <_E b^{-2} <_E \dots <_E b^2 <_E a^{-1}b$$

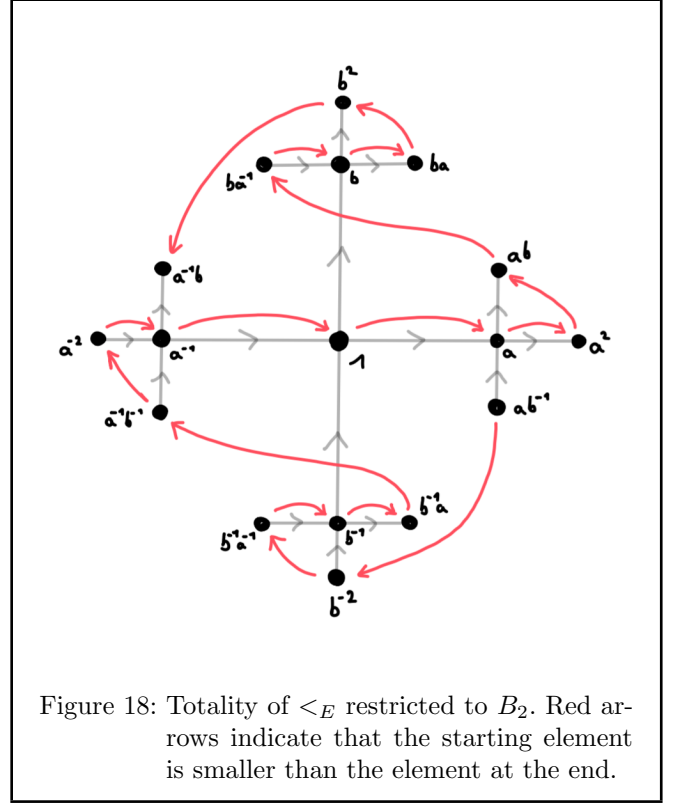
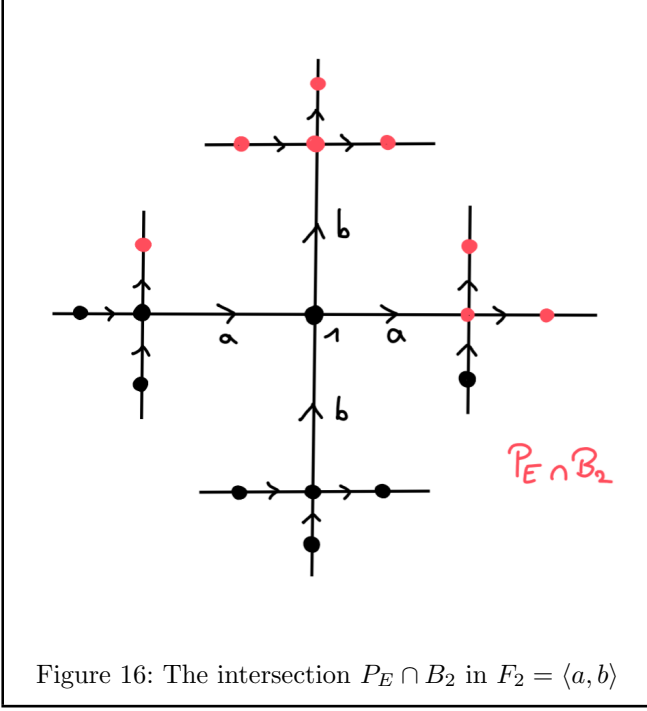
as shown in Fig. 18.

As already noted in [Š13] in the very last paragraph, this left-order is not a bi-order. Since the lexicographic order on $\langle 1, x_1, \dots, x_n \rangle^+$ is given by

$$x_{i_1} \dots x_{i_k} < x_{j_1} \dots x_{j_k}$$

if and only if $x_{i_\ell} < x_{j_\ell}$ for the first index ℓ where the two words differ, in particular we have

$$x_1 = x_1 \cdot 1 < x_1 x_1$$



and

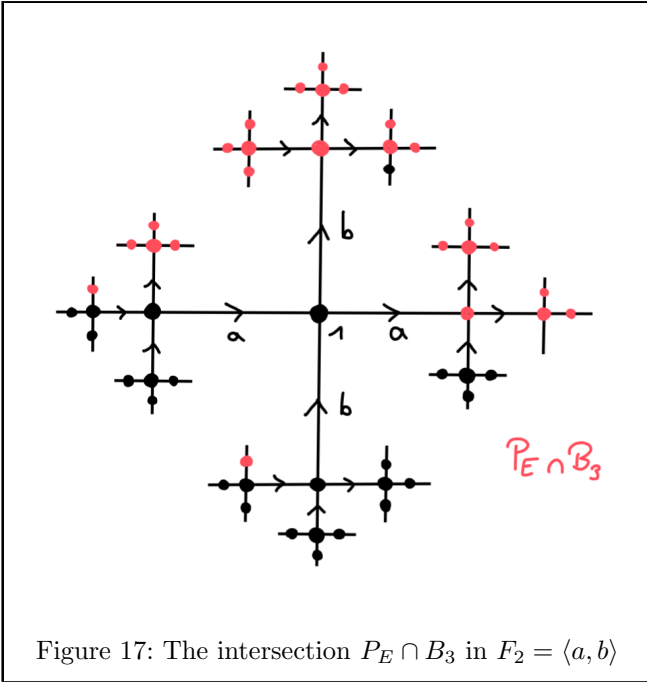
$$x_1 x_1 x_2 < x_1 x_2 \cdot 1 = x_1 x_2.$$

As a consequence, for any left-order that extends the lexicographic order above, the inequality $x_1 < x_1^2$ cannot imply $x_1 x_2 < x_1^2 x_2$, hence such a left-order cannot be bi-invariant.

4.2.2 Magnus Order on F_n

As already mentioned before, the difficulty in constructing a left-order on F_n comes mainly from cancellations. The idea for the construction shown in this Section is to avoid this problem by first embedding F_n as a multiplicative subgroup in a ring of formal power series, where no negative exponents can occur, and then defining an order on this ring, which in turn restricts to the embedding of F_n . Historically, this order has been discovered earlier than the order $<_E$ from Section 4.2.1, but the downside is that it is very “implicit”: Computing the order of two words is often tedious, even though the computation might be straightforward. This is in contrast to $<_E$, where the order is revealed by counting certain subwords.

During this construction we mainly follow the first part from Chapter 7.2 in [KT08] as well as Chapter 5 in [KR03]. Note that in the latter reference, right-orders instead of left-orders are used.



Definition 4.6. Denote as $\mathbb{Z}[[X_1, \dots, X_n]]$ the *formal power series* in the non-commuting variables $X_1, \dots, X_n$. For a multi-index $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$, write

$$X^\alpha := X_1^{\alpha_1} \cdots X_n^{\alpha_n}.$$

The elements of $\mathbb{Z}[[X_1, \dots, X_n]]$ are formal sums of the form

$$\sum_{\alpha \in \mathbb{N}^n} c_\alpha X^\alpha, \quad c_\alpha \in \mathbb{Z}$$

with addition given by

$$\left(\sum_{\alpha \in \mathbb{N}^n} c_\alpha X^\alpha \right) + \left(\sum_{\alpha \in \mathbb{N}^n} d_\alpha X^\alpha \right) = \sum_{\alpha \in \mathbb{N}^n} (c_\alpha + d_\alpha) X^\alpha$$

and multiplication given by

$$\left(\sum_{\alpha \in \mathbb{N}^n} c_\alpha X^\alpha \right) \cdot \left(\sum_{\alpha \in \mathbb{N}^n} d_\alpha X^\alpha \right) = \sum_{\alpha, \beta \in \mathbb{N}^n} c_\alpha d_\beta X^{\alpha+\beta}.$$

Elements of the form X^α are called *monomials*. For a monomial $X_1^{\alpha_1} \cdots X_n^{\alpha_n}$ in $\mathbb{Z}[[X_1, \dots, X_n]]$ define its *degree* as the sum of its exponents, i.e.

$$\deg(X_1^{\alpha_1} \cdots X_n^{\alpha_n}) = \alpha_1 + \cdots + \alpha_n.$$

Given an element in $\mathbb{Z}[[X_1, \dots, X_n]]$, write $\mathcal{O}(d)$ to denote the sum of all of its terms of degree $\geq d$.

Furthermore, define as

$$\mathcal{G} := \{1 + \mathcal{O}(1) \in \mathbb{Z}[[X_1, \dots, X_n]]\}$$

the subset of all formal power series with constant term $1 = X^0$.

The following result can be found as Lemma 5.3 in Chapter 5.5 from [MKS04] and we state it here without proof.

Lemma 4.7. $\mathcal{G}$ is a subgroup of $\mathbb{Z}[[X_1, \dots, X_n]]$ under multiplication. The inverse of $1 + W$ with $W \in \mathbb{Z}[[X_1, \dots, X_n]]$ having no constant term is given by

$$(1 + W)^{-1} = 1 - W + W^2 - W^3 + \dots$$

Definition 4.8. Define the *Magnus embedding* as

$$\mu : \{x_1, \dots, x_n\} \rightarrow \mathcal{G} \subseteq \mathbb{Z}[[X_1, \dots, X_n]], \quad x_i \mapsto 1 + X_i.$$

By the universal property of free groups, it follows that the Magnus embedding extends uniquely to a group homomorphism

$$\mu : F_n = \langle x_1, \dots, x_n \rangle \rightarrow \mathcal{G}, \quad x_i \mapsto 1 + X_i.$$

Note that the equality

$$\mu(x_i^{-1}) = (1 + X_i)^{-1} = 1 - X_i + X_i^2 - X_i^3 + \dots$$

can be understood as eliminating negative exponents which might be present in words from F_n .

The next step is showing that the Magnus embedding is indeed an embedding, i.e. injective. For this we need the following Lemma, which is stated as Lemma 7.9 in [KT08].

Lemma 4.9. For any X_i and any integer $k \geq 2$ there is a formal power series $H_k(X_i) \in \mathbb{Z}[[X_1, \dots, X_n]]$ in the variable X_i , such that

$$(1 + X_i)^k = 1 + kX_i + X_i^2 H_k(X_i).$$

Proof. This can be proven inductively. For $k = 2$ one has

$$(1 + X_i)^2 = 1 + 2X_i + X_i^2.$$

Assuming that the statement is true for some $k \geq 2$, we obtain

$$\begin{aligned} (1 + X_i)^{k+1} &= (1 + X_i)^k (1 + X_i) \\ &= (1 + kX_i + X_i^2 H_k(X_i))(1 + X_i) \\ &= 1 + X_i + kX_i + kX_i^2 + X_i^2 H_k(X_i) + X_i^2 H_k(X_i) X_i \\ &= 1 + (k+1)X_i + X_i^2 \underbrace{(k + H_k(X_i) + H_k(X_i)X_i)}_{=: H_{k+1}(X_i)}. \end{aligned}$$

□

The proof for injectivity of μ is based on [KT08] Proposition 7.10.

Proposition 4.10. The Magnus embedding

$$\mu : F_n \rightarrow \mathcal{G}, \quad x_i \mapsto 1 + X_i$$

is injective.

Proof. Let $w \in F_n \setminus \{1\}$. Then

$$w = x_{i_1}^{k_1} \cdots x_{i_r}^{k_r}$$

for some choice of $i_1, \dots, i_r \in \{1, \dots, n\}$, such that $x_{i_j} \neq x_{i_{j+1}}$ for $j = 1, \dots, r-1$ and all the exponents $k_j \neq 0$. Then by Lemma 4.9 we have

$$\begin{aligned} \mu(w) &= (1 + X_{i_1})^{k_1} \cdots (1 + X_{i_r})^{k_r} \\ &= (1 + k_1 X_{i_1} + \mathcal{O}(2)) \cdots (1 + k_r X_{i_r} + \mathcal{O}(2)). \end{aligned}$$

Calculating the term on the right, the monomial $X_{i_1} \cdots X_{i_r}$ has coefficient $k_1 \cdots k_r \neq 0$ and thus $\mu(w) \neq 1$. □

After having successfully embedded F_n as a multiplicative group into $\mathbb{Z}[[X_1, \dots, X_n]]$, one needs to define an appropriate order on the latter.

Definition 4.11. The *Magnus order* on $\mathbb{Z}[[X_1, \dots, X_n]]$ is defined as follows. First order the variables in ascending order depending on their index, i.e.

$$X_1 < X_2 < \dots < X_n.$$

Monomials in $\mathbb{Z}[[X_1, \dots, X_n]]$ can then be ordered via “shortlex” with respect to $<$, i.e.

$$\underbrace{X_1^{k_1} \dots X_n^{k_n}}_{=:M} <_{\text{slex}} \underbrace{X_1^{\ell_1} \dots X_n^{\ell_n}}_{=:N}$$

if and only if $\deg(M) < \deg(N)$ or $\deg(M) = \deg(N)$ and there exists an index $i \in \{1, \dots, n\}$, such that $k_j = \ell_j$ for all $j < i$ and $k_i < \ell_i$. Note that $<_{\text{slex}}$ is a well-order in this case.

Given two distinct power series $V \neq W \in \mathbb{Z}[[X_1, \dots, X_n]]$, first sort their respective monomials using the described lexicographic order. Then consider the smallest monomial in which they differ and define $V < W$ if and only if the coefficient of this monomial in V is smaller than the coefficient of this monomial in W .

Note that for two power series $V \neq W$ one can write

$$W - V = c_M \cdot M + (\text{terms } >_{\text{slex}} M)$$

for some monomial $M \in \mathbb{Z}[[X_1, \dots, X_n]]$ with integer coefficient c_M , then the condition $V < W$ is equivalent to $c_M > 0$. This, in turn is equivalent to $W - V > 0$.

Example 4.12. i) We have $1 < 1 + X_i$ for all $i = 1, \dots, n$.

ii) We obtain

$$1 + X_i > 1 + 0 \cdot X_i + X_{i+1} = 1 + X_{i+1}$$

for any $i \in \{1, \dots, n-1\}$. it then follows that

$$1 + X_1 > 1 + X_2 > \dots > 1 + X_n.$$

Example 4.13. We have

$$\begin{aligned} & (1 + X_1)(1 + X_2)^{-1} \\ &= (1 + X_1) \cdot \sum_{k \geq 0} (-1)^k X_2^k \\ &= 1 + X_1 + \underbrace{\sum_{k \geq 1} (-1)^k (1 + X_1) X_2^k}_{\text{terms } >_{\text{slex}} X_1} \end{aligned}$$

and

$$\begin{aligned} & (1 + X_1)^{-1}(1 + X_2)^{-1} \\ &= \left(\sum_{k \geq 0} (-1)^k X_1^k \right) \cdot \left(\sum_{s \geq 0} (-1)^s X_2^s \right) \\ &= \sum_{k \geq 0} \sum_{s=0}^k (-1)^s X_1^s (-1)^{k-s} X_2^{k-s} \\ &= \sum_{k \geq 0} \sum_{s=0}^k (-1)^k X_1^s X_2^{k-s} \\ &= 1 - X_1 - X_2 + \underbrace{\sum_{k \geq 2} \sum_{s=0}^k (-1)^k X_1^s X_2^{k-s}}_{\text{terms } >_{\text{slex}} X_1}. \end{aligned}$$

Let us quickly check, that the Magnus order is total on $\mathbb{Z}[[X_1, \dots, X_n]]$.

Antisymmetry follows from the fact that for $V \neq W$ they must differ in some monomial and then either $V < W$ or $W < V$ follows.

For transitivity assume $V < W$ and $W < U$, i.e.

$$W - V = c_M \cdot M + (\text{terms } >_{\text{slex}} M)$$

and

$$U - W = c_N \cdot N + (\text{terms } >_{\text{slex}} N)$$

where N, M are monomials and $c_M, c_N > 0$. Then

$$\begin{aligned} U - V &= (U - W) + (W - V) \\ &= c_M \cdot M + c_N \cdot N + (\text{terms } >_{\text{slex}} M) + (\text{terms } >_{\text{slex}} N) \end{aligned}$$

Now if $M = N$, then the smallest monomial in this expression has coefficient $c_M + c_N > 0$. If $M <_{\text{slex}} N$, then the smallest monomial is M with coefficient $c_M > 0$ and similarly if $N <_{\text{slex}} M$, we get $c_N > 0$ as coefficient for the smallest monomial. In all three cases this means that $V < U$.

The Magnus order is already invariant under addition, but not under multiplication, since for example multiplying an inequality with -1 reverses the order. However, it becomes bi-invariant under multiplication when restricted to $\mathcal{G}$. The proof for this is based on the first part of Theorem 4 from [KR03].

Theorem 4.14. The Magnus order on $\mathbb{Z}[[X_1, \dots, X_n]]$ restricts to a bi-order on $\mathcal{G}$ and thus on F_n .

Proof. Note that in order to show left-invariance of the order restricted to $\mathcal{G}$ it suffices to verify that

$$g < h \iff 1 < g^{-1}h$$

for all $g, h \in \mathcal{G}$. Let $g, h \in G$, i.e. $g = 1 + G, h = 1 + H$ with $G, H \in \mathcal{O}(1)$ having no constant terms. Then

$1 + G = g < h = 1 + H$ is equivalent to $H - G > 0$. We have

$$\begin{aligned} g^{-1}h &= (1 + G)^{-1}(1 + H) \\ &= (1 - G + G^2 - G^3 + \dots)(1 + H) \\ &= 1 + H - G - \underbrace{GH + G^2 + G^2H + \dots}_{=:R} \end{aligned}$$

where each term in R is bigger (with respect to $<_{\text{slex}}$) than the smallest term of $H - G$. Hence $g^{-1}h > 1$ if and only if $H - G > 1$, which in turn is equivalent to $H - G > 0$, since both H and G have no constant terms. In the same way, one can show that $g < h \Leftrightarrow 1 < hg^{-1}$ also holds, giving right-invariance of the Magnus order. Since F_n can be embedded as a subgroup of $\mathcal{G}$, the Magnus order thus induces a bi-invariant order on F_n . $\square$

Let us revisit Examples 4.12 and 4.13 in order to see what the Magnus order looks like for $F_n = \langle x_1, \dots, x_n \rangle$.

Example 4.15.

We obtain $x_i > 1$ for all i and

$$x_1 > x_2 > \dots > x_n.$$

From

$$\begin{aligned} \mu(x_1 x_2^{-1}) &= (1 + X_1)(1 + X_2)^{-1} \\ &= 1 + X_1 + (\text{terms } >_{\text{slex}} X_1) \\ &> 1 \end{aligned}$$

and

$$\begin{aligned} \mu(x_1^{-1} x_2^{-1}) &= (1 + X_1)^{-1}(1 + X_2)^{-1} \\ &= 1 - X_1 + (\text{terms } >_{\text{slex}} X_1) \\ &< 1 \end{aligned}$$

we can deduce $x_1 x_2^{-1} > 1$ and $x_1^{-1} x_2^{-1} < 1$. Of course, we already know these from i) by bi-invariance.

Finally, it should be noted that the same construction works for arbitrary free groups $F(X)$ with any set X as its generating set. Then one has to choose a total order on X , which then induces an order on the free monoid X^* , which in turn induces an order on $\mathbb{Z}[[X]]$ similar to above. This means that any free group (even on uncountably many generators) is bi-orderable. For details, see Chapter 7.2 in [KT08].

4.3 Artin-Magnus Order on the Pure Braid Group PB_n

The Magnus order from Section 4.2.2 can be used to define a bi-invariant order on the pure braid group PB_n , which is called Artin-Magnus order. This serves as an excellent example of Proposition 3.4. The construction of the order mainly follows arguments from Chapter

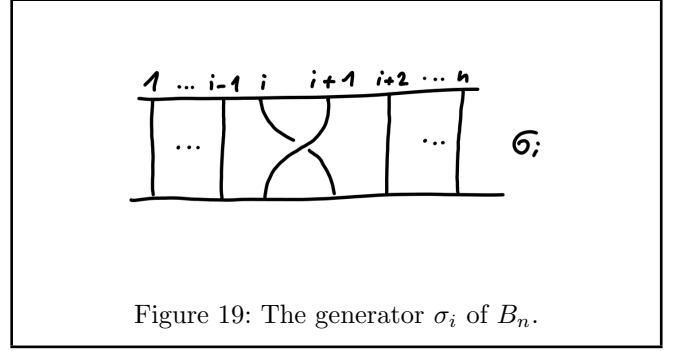


Figure 19: The generator σ_i of B_n .

6 of [KR03]. Throughout this Section we need some knowledge about braid groups, all of which can be found for example in [KT08].

For what follows, let

$$\begin{aligned} B_n &= \langle \sigma_1, \dots, \sigma_{n-1} \mid \sigma_i \sigma_j = \sigma_j \sigma_i \text{ for } |i - j| > 1, \\ &\quad \sigma_i \sigma_j \sigma_i = \sigma_j \sigma_i \sigma_j \text{ for } |i - j| = 1 \rangle. \end{aligned}$$

denote the braid group on n strands. A generator σ_i can be visualized via a *braid diagram* as in Fig. 19. The group operation then corresponds to concatenation of these braid diagrams, where continuous deformations of the individual strands are allowed, as long as no two strands intersect in this process.

Consider the symmetric group on n elements Σ_n generated by the transpositions $s_1, \dots, s_{n-1}$. Since these transpositions also satisfy the braid relations, there is a surjective homomorphism

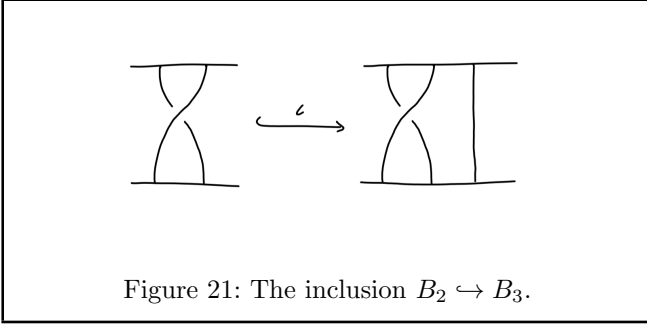
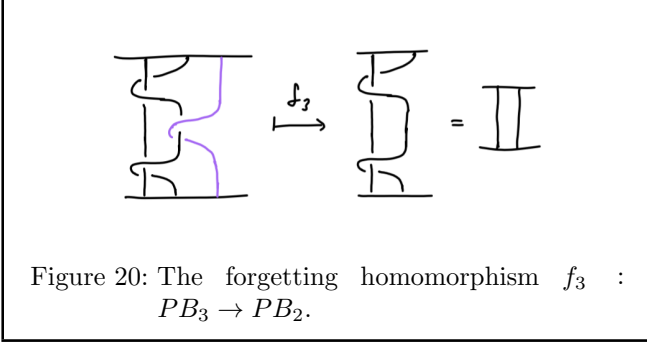
$$\pi : B_n \rightarrow \Sigma_n, \quad \sigma_i \mapsto s_i.$$

This projection can be regarded as mapping a braid to its underlying permutation, via “forgetting” the twisting of the strands in B_n and only looking at how starting- and end-points of the individual strands are permuted. A braid $\beta \in B_n$ is then considered *pure*, if it each strand ends at the same point where it started, i.e. the underlying permutation $\pi(\beta)$ is trivial.

Definition 4.16. The *pure braid group on n strands* is

$$PB_n := \ker(\pi : B_n \rightarrow \Sigma_n).$$

Given a pure braid β on n strings, we can consider what happens when we “delete” the n -th string, obtaining a pure braid on $n - 1$ strands. This works, since the underlying permutation of a pure braid is trivial, so the first $n - 1$ strands can be considered on their own. It can happen, however, that some strands become “disentangled”, in the extreme case the whole resulting braid becomes trivial, as depicted in Fig. 20. This yields the



forgetting homomorphism

$$f_n : PB_n \rightarrow PB_{n-1}.$$

Of course, we can also go the other way around, taking a pure braid on $n-1$ strands and add an additional n -th strand that is unlinked to any of the other strands, see Fig. 21. This can be considered as an inclusion homomorphism

$$\iota : PB_{n-1} \rightarrow PB_n.$$

With this we obtain $f_n \circ \iota = \text{id}_{PB_{n-1}}$, showing injectivity of ι and surjectivity of f_n .

Furthermore, if we denote as

$$U_n := \ker f_n$$

those pure braids where removing the n -th strands “dis-entangles” the braid as in Fig. 20, then we have a split exact sequence

$$1 \longrightarrow U_n \hookrightarrow PB_n \xrightarrow{f_n} PB_{n-1} \longrightarrow 1$$

$\swarrow \quad \searrow$
 $\quad \quad \quad \iota$

In other words, PB_n is the semidirect product $U_n \rtimes PB_{n-1}$. Furthermore, it can be shown that by defining

$$A_{i,n} := \sigma_{n-1} \cdots \sigma_{i+1} \sigma_i^2 \sigma_{i+1}^{-1} \cdots \sigma_{n-1}^{-1},$$

the kernel U_n is free on the $n-1$ generators

$A_{1,n}, \dots, A_{n-1,n}$, i.e.

$$U_n = \langle A_{i,n} \rangle_{i=1, \dots, n-1} \cong F_{n-1}.$$

More explicitly, write

$$F_{n-1} = \langle x_1^{(n)}, \dots, x_{n-1}^{(n)} \rangle,$$

then there is an isomorphism

$$U_n \xrightarrow{\cong} F_{n-1}, \quad A_{i,n} \mapsto x_i^{(n)}.$$

This tells us that each U_n in itself can be bi-ordered using the Magnus embedding

$$U_n \xrightarrow{\cong} F_{n-1} \xrightarrow{\mu} \mathcal{G}, \quad A_{i,n} \mapsto 1 + X_i.$$

Thus for example we obtain directly that $A_{i,n} > 1$ for all i and

$$A_{1,n} > A_{2,n} > \cdots > A_{n-1,n},$$

but $A_{i,j}$ and $A_{r,s}$ for $j \neq s$ cannot be compared without further work. To overcome this problem, we use the split exact sequence from above. Inductively, we obtain

$$\begin{aligned} P_n &\cong U_n \rtimes P_{n-1} \\ &\cong U_n \rtimes (U_{n-1} \rtimes P_{n-2}) \\ &\cong \dots \\ &\cong U_n \rtimes (U_{n-1} \rtimes (\dots \rtimes (U_4 \rtimes (U_3 \rtimes (\underbrace{U_2 \rtimes P_1}_{\cong U_2}) \dots) \dots) \dots) \\ &\cong F_{n-1} \rtimes (F_{n-2} \rtimes (\dots \rtimes (F_3 \rtimes (F_2 \rtimes F_1) \dots) \dots) \end{aligned}$$

So any pure braid $\beta \in P_n$ can be written uniquely in its *normal form* as

$$\beta = \beta_n \cdots \beta_3 \beta_2 \cong_{\text{Set}} (\beta_n, \dots, \beta_3, \beta_2) \quad (2)$$

with $\beta_j \in U_j$. Repeated application of Proposition 3.4 then induces a lexicographic-type order on P_n , which is called the *Artin-Magnus order*.

Recalling Example 3.5 iv), for two pure braids

$$\beta = \beta_n \cdots \beta_2 \neq \alpha_n \cdots \alpha_2 = \alpha$$

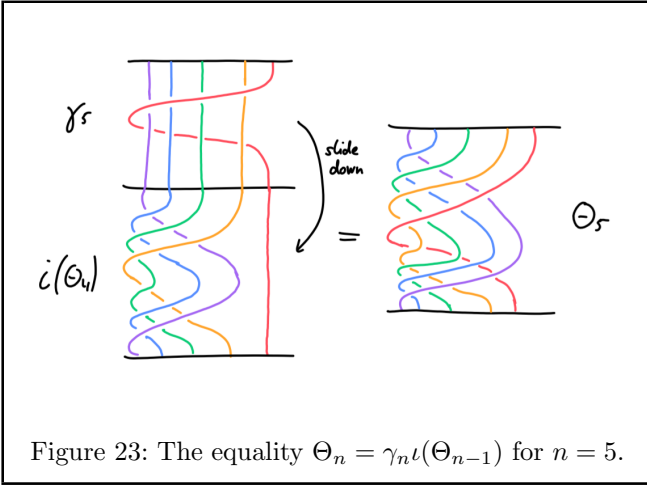
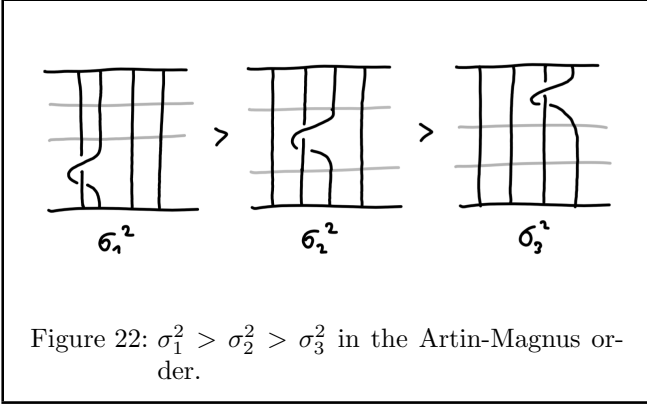
we have $\beta < \alpha$ if and only if

$$1 < \beta_2^{-1} \cdots \beta_i^{-1} \alpha_i \cdots \alpha_2$$

for the first index i where $\beta_i \neq \alpha_i$.

The following examples are based on Example 2 and 3 from [KR03].

Example 4.17. Consider $P_4 \cong U_4 \rtimes (U_3 \rtimes U_2)$.



i) Take the pure braids

$$\begin{aligned}\sigma_1^2 &= 1 \cdot 1 \cdot A_{1,2}, \\ \sigma_2^2 &= 1 \cdot A_{2,3} \cdot 1, \\ \sigma_3^2 &= A_{3,4} \cdot 1 \cdot 1.\end{aligned}$$

Comparing σ_1^2 and σ_2^2 , we see that starting from the right, they differ in the U_2 -factor, and since $1 < A_{1,2}$ we have $\sigma_2^2 < \sigma_1^2$.

Likewise, for σ_2^2 and σ_3^2 , they first differ in the U_3 -factor, and we have

$$1 < 1 \cdot 1 \cdot A_{2,3} \cdot 1,$$

hence $\sigma_3^2 < \sigma_2^2$. See Fig. 22 for a visualization.

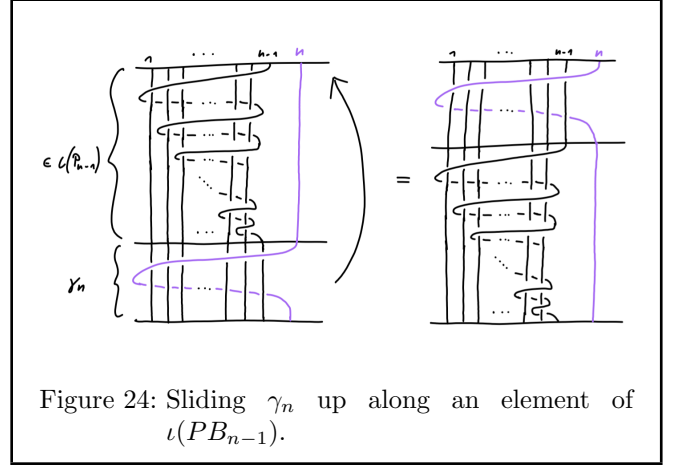
ii) Consider the full-twist $\theta_n = \Delta_n^2$, where

$$\Delta_n = (\sigma_1 \cdots \sigma_{n-1})(\sigma_1 \cdots \sigma_{n-2}) \cdots (\sigma_1 \sigma_2) \sigma_1$$

is the half-twist. It can be built from the pure braids

$$\gamma_n = A_{1,n} A_{2,n} \cdots A_{n-1,n}$$

via the formula $\theta_n = \gamma_n \iota(\theta_n)$, see Fig. 23. So for



example in P_4 the full-twist is

$$\theta_4 = \gamma_4 \gamma_3 \gamma_2$$

Let us compare it to

$$\beta := A_{3,4} \gamma_3 \gamma_2$$

and see if $\theta_4 < \beta$ is true. Starting from the right, we see that the factors corresponding to U_2 and U_3 coincide but they differ in the third factor. Hence we have to look at the term

$$1 \stackrel{?}{<} \gamma_2^{-1} \gamma_3^{-1} \gamma_4^{-1} A_{3,4} \gamma_3 \gamma_2$$

and decide if it is greater than 1 with respect to the order on U_4 . Now $\gamma_n \in Z(\iota(PB_{n-1}))$, as can be seen by drawing the corresponding braid diagram and “sliding” γ up or down, see Fig. 24. Thus

$$\gamma_2^{-1} \gamma_3^{-1} \gamma_4^{-1} A_{3,4} \gamma_3 \gamma_2 = \gamma_3^{-1} \gamma_4^{-1} \gamma_2^{-1} A_{3,4} \gamma_2 \gamma_3.$$

Furthermore, $A_{3,4}$ commutes with γ_2 , so the term in question becomes $\gamma_3^{-1} \gamma_4^{-1} A_{3,4} \gamma_3$. Drawing the corresponding braid diagrams, one arrives at

$$\gamma_3^{-1} \gamma_4^{-1} A_{3,4} \gamma_3 = A_{2,4}^{-1} A_{1,4}^{-1},$$

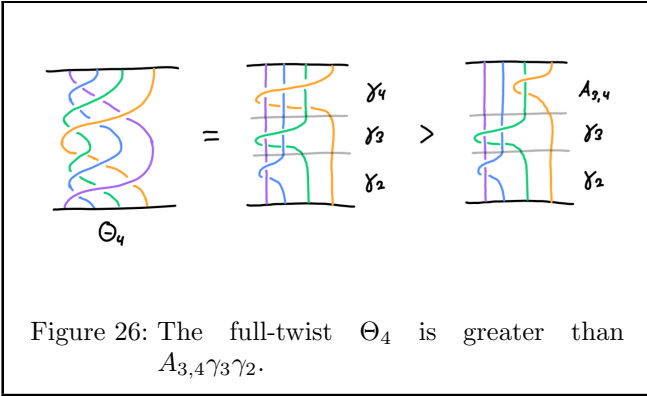
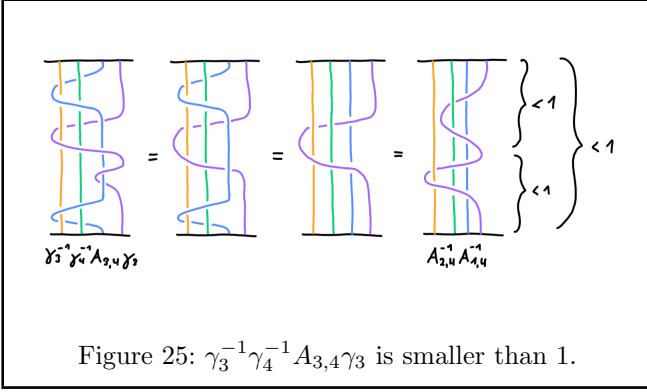
see Fig. 25. We have already seen that $A_{1,4}^{-1} < 1$ and $A_{2,4}^{-1} < 1$ hold in U_4 , since they correspond to $(1 + X_1)^{-1}$ and $(1 + X_2)^{-1}$ under the Magnus embedding. Thus, using bi-invariance of the order on U_4 , we have

$$A_{2,4}^{-1} A_{1,4}^{-1} < 1$$

and so it follows that

$$1 > \gamma_2^{-1} \gamma_3^{-1} \gamma_4^{-1} A_{3,4} \gamma_3 \gamma_2,$$

hence $\gamma_2 \gamma_3 \gamma_4 = \theta_4 > \beta = \gamma_2 \gamma_3 A_{3,4}$, as depicted in Fig. 26.



The tricky thing with the Artin-Magnus order is, that for comparing two braids, one first has to compute their respective normal forms as in (2). Depending on the specific braids in question, this might be a very tedious exercise if done by hand.

We conclude this section with the fact that the Artin-Magnus order is not only left-invariant, but also bi-invariant. This builds upon arguments from Chapter 7.2.3 in [KT08]. Before, we need two auxiliary results, the first of which is posed as Exercise 7.2.1 in the same reference.

Lemma 4.18. *Given generators x_i, x_j of $F_n = \langle x_1, \dots, x_n \rangle$, the Magnus expansion of their commutator is*

$$\mu([x_i, x_j]) = 1 + (X_i X_j - X_j X_i) + \mathcal{O}(3).$$

Proof. In $\mathbb{Z}[[X_1, \dots, X_n]]$ we have

$$\begin{aligned} & (1 + X_i)^{-1} \cdot (1 + X_j)^{-1} \\ &= (1 - X_i + X_i^2 - X_i^3 + \dots) \cdot (1 - X_j + X_j^2 - X_j^3 + \dots) \\ &= 1 - X_i - X_j + X_i X_j + X_i^2 + X_j^2 + \mathcal{O}(3). \end{aligned}$$

and thus

$$\begin{aligned} \mu([x_i, x_j]) &= \mu(x_i^{-1} x_j^{-1} x_i x_j) \\ &= (1 + X_i)^{-1} (1 + X_j)^{-1} (1 + X_i) (1 + X_j) \\ &= (1 - X_i - X_j + X_i X_j + X_i^2 + X_j^2 + \mathcal{O}(3)) \\ &\quad \cdot (1 + X_i + X_j + X_i X_j) \\ &= 1 + X_i + X_j + X_i X_j - X_i - X_j + X_i X_j + X_i^2 + X_j^2 \\ &\quad - X_i^2 - X_i X_j - X_j X_i - X_j^2 + \mathcal{O}(3) \\ &= 1 + X_i X_j - X_j X_i + \mathcal{O}(3). \end{aligned}$$

□

The next result claims that given $\beta \in P_n$, we always have $\beta A_{i,n} \beta^{-1} = A_{i,n} u_i$, where $u_i \in [U_n, U_n]$ lies in the commutator subgroup of U_n . The reasoning for this lies in the defining relations for the elements $A_{i,j}$ and how they behave when being conjugated with each other. For details, look into the third paragraph at the beginning of Chapter 7.2.3 in [KT08].

Proposition 4.19. *For every $i = 1, \dots, n-1$ and every $\beta \in P_n$, there is $u_i \in [U_n, U_n]$, such that*

$$\beta A_{i,n} \beta^{-1} = A_{i,n} u_i.$$

Proposition 4.20. *The Artin-Magnus order is bi-invariant on P_n .*

Proof. We show that for any $\beta \in P_n$ and any $u \in U_n$ with $u > 1$ we also have $\beta u \beta^{-1} > 1$. The statement then follows by induction from the second part of Proposition 3.4.

Considering a generator $A_{i,n}$ of U_n and $\beta \in P_n$, with Proposition 4.19 and Lemma 4.18 we obtain

$$\begin{aligned} \mu(\beta A_{i,n} \beta^{-1}) &= \mu(A_{i,n} u_i) \\ &= (1 + X_i) \cdot (1 + \mathcal{O}(2)) \\ &= 1 + X_i + \mathcal{O}(2), \end{aligned}$$

for some $u_i \in [U_n, U_n]$. In other words the diagram

$$\begin{array}{ccc} A_{i,n} & \xrightarrow{\quad} & \beta A_{i,n} \beta^{-1} \\ \mu \downarrow & & \downarrow \mu \\ 1 + X_i & \xrightarrow{\quad} & 1 + X_i + \mathcal{O}(2) \end{array}$$

commutes. Using that $u \in U_n$ is a (reduced) word in $A_{i,n}$, we see that passing from u to $\beta u \beta^{-1}$, this corresponds to replacing each X_i with $X_i + \mathcal{O}(2)$ in the respective images under the Magnus embedding, hence $\mu(u)$ and $\mu(\beta u \beta^{-1})$ have the same smallest non-constant term. In particular this implies $u > 1$ if and only if $\beta u \beta^{-1} > 1$. □

4.4 Dehornoy Order on the Braid Group B_n

This Section is mainly based on Chapter II from [Deh+08] and is about the famous Dehornoy order. As before, let

$$B_n = \langle \sigma_1, \dots, \sigma_{n-1} \mid \sigma_i \sigma_j = \sigma_j \sigma_i \text{ for } |i-j| > 1, \\ \sigma_i \sigma_j \sigma_i = \sigma_j \sigma_i \sigma_j \text{ for } |i-j| = 1 \rangle.$$

denote the braid group on n strands.

Definition 4.21. A word w in $\sigma_1, \dots, \sigma_{n-1}$ is called a *braid word*. It is called σ_i -positive, if

- i) The word has at least one occurrence of σ_i ,
- ii) There are no occurrences of $\sigma_1, \dots, \sigma_{i-1}$,
- iii) Every occurrence of σ_i has positive exponent.

A braid $\beta \in B_n$ is called σ_i -positive, if it can be represented by an σ_i -positive braid word. A braid is called σ -positive, if it is σ_i -positive for some $i \in \{1, \dots, n-1\}$. The definitions for σ_i -negative are the same but with negative exponent instead of positive.

Example 4.22. i) The braid $\sigma_1 \sigma_2 \sigma_1^{-1} = \sigma_2^{-1} \sigma_1 \sigma_2$ is σ_1 -positive, even though the braid word $\sigma_1 \sigma_2 \sigma_1^{-1}$ is neither σ_1 -positive nor σ_1 -negative.

- ii) If $\beta \in B_n$ is σ_{n-1} -positive, then it is of the form $\beta = \sigma_{n-1}^k$ with positive exponent $k > 0$.

Definition 4.23. The *Dehornoy order* $<_D$ on B_n is given by the positive cone P_D containing all σ -positive braids. Furthermore, define

$$P_i := \{\beta \in B_n \mid \beta \text{ is } \sigma_i\text{-positive}\}$$

for all $i = 1, \dots, n-1$. Then

$$P_D = P_1 \cup \dots \cup P_{n-1}.$$

That this indeed defines a left-order on B_n follows from the following three properties, proofs of which can be found in [Deh+08].

- Property A (Acyclicity): A σ -positive braid is non-trivial.
- Property C (Comparison): Every non-trivial braid of B_n can be represented by an n -strand braid word that is σ -positive or σ -negative.
- Property S (Subword): Every braid of the form $\beta^{-1} \sigma_i \beta$ for some braid β is σ_i -positive.

While it is left-invariant, the Dehornoy order is not bi-invariant, since there are no bi-invariant orders on B_n by Corollary 2.10.

Moreover, it is a discrete order, as shown by Proposition

II.3.9 in [Deh+08]. This property will become relevant later when studying the space of left-orders on B_n in section 7.1.

Definition 4.24. A left-order $<$ on a group G is called *discrete*, if its positive cone $P_<$ has a smallest element. Otherwise it is called *dense*.

Proposition 4.25. The Dehornoy order $<_D$ on B_n is discrete with smallest positive element σ_{n-1} .

Proof. Let $1 <_D \beta$. If β is σ_i -positive for some $i \leq n-2$, then $\sigma_{n-1}^{-1} \beta$ is also σ_i -positive, so $\sigma_{n-1} <_D \beta$. If β is σ_{n-1} -positive, then $\beta = \sigma_{n-1}^p$ for some $p > 0$. Then $\sigma_{n-1}^{-1} \beta = \sigma_{n-1}^{p-1} \geq_D 1$, implying $\sigma_{n-1} \leq_D \beta$. $\square$

We will mainly use the Dehornoy order as a starting point for constructing the Dubrovina-Dubrovin order in Chapter 7.1. It should be noted, however, that the Dehornoy order also has various other applications, for example see Chapter III in [Deh+08].

5 Spaces of Orderings

We saw before that there are various ways to produce new orders from already existing ones, either “internally” via automorphisms and convex subgroups or “externally” via group extensions. This naturally leads to the following question: Given a group G , what does its set of left-orders $\text{LO}(G)$ look like? When are two different left-orders “similar” or “near” to each other? Of course, to answer the latter questions, one would need to equip $\text{LO}(G)$ with a metric, or at least a topology, which is exactly what has been done already in the literature, and what we will explore in this section. We mainly follow Chapter 1.5 from [CR15].

Definition 5.1. For a group G , denote its set of left-orders as $\text{LO}(G)$. Likewise, denote its set of bi-orders as $\text{BO}(G)$.

Luckily for us, there is a natural way to equip $\text{LO}(G)$ with a topology. Recall that the set of left-orders on G is in bijection with the set of all positive cones on G , and since a positive cone is nothing else than a subset of G with special properties, we can consider $\text{LO}(G)$ as a subset of the power set $\mathcal{P}(G)$ of G . Hence, if we find a reasonable topology on $\mathcal{P}(G)$, then $\text{LO}(G)$ is successfully topologized as well as a subspace of the power set. The good news is that there is a natural choice for a topology on $\mathcal{P}(G)$.

5.1 A Topology on the Power Set

First recall another characterisation for the power set $\mathcal{P}(X)$ of an arbitrary set X , namely the bijection

$$\psi : \mathcal{P}(X) \xrightarrow{\cong} \prod_{x \in X} \{0, 1\}, \quad A \mapsto (a_x)_{x \in X},$$

where

$$a_x = \begin{cases} 0, & x \notin A, \\ 1, & x \in A. \end{cases}$$

The product on the right-hand side can be equipped with the product topology, where $\{0, 1\}$ is considered as a discrete space. A subbase for this topology is given sets of the form $p_y^{-1}(U)$, where

$$p_y : \prod_{x \in X} \{0, 1\} \rightarrow \{0, 1\}, \quad (a_x)_{x \in X} \mapsto a_y$$

is the projection onto the y -factor, where y is some element of X and U is any subset of $\{0, 1\}$.

This subbase can then be translated via ψ to obtain a topology on $\mathcal{P}(X)$. More concretely, if either $U = \emptyset$ or $U = \{0, 1\}$, then either $\psi^{-1}(p_y^{-1}(U)) = \emptyset$ or $\psi^{-1}(p_y^{-1}(U)) = \mathcal{P}(X)$, so these cases are trivial. On the other hand, if $U = \{0\}$, we have

$$\begin{aligned} \psi^{-1}(p_y^{-1}(U)) &= \psi^{-1}(\{(a_x)_{x \in X} \mid a_y = 0\}) \\ &= \{A \subseteq X \mid y \notin A\} \end{aligned}$$

and likewise if $U = \{1\}$ then

$$\psi^{-1}(p_y^{-1}(U)) = \{A \subseteq X \mid y \in A\}.$$

We give this topology a name in the following definition.

Definition 5.2. Equip $\mathcal{P}(X)$ with the *Tychonoff topology*, that is the topology which has as a subbase all subsets of $\mathcal{P}(X)$ of the form

$$U_x := \{A \subseteq X \mid x \in A\}, \quad U_x^c := \{A \subseteq X \mid x \notin A\}.$$

Thus, the open sets are exactly arbitrary unions of finite intersections of subsets of this form.

Remark 5.3. If X is finite, then any singleton $\{A\}$ in $\mathcal{P}(X)$ can be written as a finite intersection of open sets

$$\{A\} = \bigcap_{a \in A} U_a \cap \bigcap_{x \in X \setminus A} U_x^c$$

and is thus itself open. So in this case the Tychonoff topology is just the discrete topology.

The discrete 2-space is compact, hence by Tychonoff's Theorem the product $\prod_{x \in X} \{0, 1\}$ is also compact. Since the Tychonoff topology is a result of making ψ into a homeomorphism, we also get compactness of $\mathcal{P}(X)$.

Proposition 5.4. *The power set $\mathcal{P}(X)$ is compact for any set X .*

5.2 A Topology on the Space of Left-Orders

As promised, the space of left-orders $\text{LO}(G) \subseteq \mathcal{P}(G)$ for some group G now inherits a topology as a subspace of $\mathcal{P}(G)$ equipped with the Tychonoff topology. For $g \in G$,

let U_g and U_g^c be the generating sets from Definition 5.2, then the topology on $\text{LO}(G)$ is generated by sets of the form

$$\begin{aligned} V_g &:= U_g \cap \text{LO}(G) = \{P_< \in \text{LO}(G) \mid g \in P_<\} \\ &\cong \{< \in \text{LO}(G) \mid 1 < g\} \end{aligned}$$

and

$$\begin{aligned} V_g^c &= U_g^c \cap \text{LO}(G) = \{P_< \in \text{LO}(G) \mid g \notin P_<\} \\ &\cong \{< \in \text{LO}(G) \mid g < 1\} \\ &= \{< \in \text{LO}(G) \mid 1 < g^{-1}\} \\ &\cong V_{g^{-1}}, \end{aligned}$$

for $g \in G \setminus \{1\}$ and $V_1 = \emptyset$ as well as $V_1^c = \text{LO}(G)$.

It turns out that $\text{LO}(G)$ is a compact subspace of $\mathcal{P}(G)$, as will follow from Example 5.5, which is based on Example 1.36 in [CR15].

Example 5.5. Let G be a group and $\mathcal{P}(G)$ its power set with the Tychonoff topology.

i) Consider

$$\begin{aligned} \mathcal{S}(G) &:= \{S \subseteq G \mid S \text{ is a subsemigroup}\} \\ &= \{S \subseteq G \mid g, h \in S \implies gh \in S\}. \end{aligned}$$

Note that $A \in \mathcal{P}(G) \setminus \mathcal{S}(G)$ if and only if there exist $g, h \in A$ with $gh \notin A$. Hence we have

$$\mathcal{P}(G) \setminus \mathcal{S}(G) = \bigcup_{g, h \in G} \underbrace{U_g \cap U_h \cap U_{gh}^c}_{\text{open in } \mathcal{P}(G)}.$$

Thus $\mathcal{S}(G)$ is closed as the complement of an open set.

ii) Consider

$$\mathcal{D}(G) := \{D \in G \mid D \cap D^{-1} = \emptyset\}.$$

Then $B \in \mathcal{P}(G) \setminus \mathcal{D}(G)$ if and only if there exists $g \in B$ with $g^{-1} \in B$. This implies

$$\mathcal{P}(G) \setminus \mathcal{D}(G) = \bigcup_{g \in G} \underbrace{U_g \cap U_{g^{-1}}}_{\text{open in } \mathcal{P}(G)}.$$

Thus $\mathcal{D}(G)$ is closed.

iii) Finally, consider

$$\mathcal{U}(G) := \{V \in G \mid V \cup V^{-1} = G \setminus \{1\}\}.$$

Then $C \in \mathcal{P}(G) \setminus \mathcal{U}(G)$ if and only if there is $g \in G \setminus \{1\}$, such that $g \notin V$ and $g \notin V^{-1}$. The latter condition is equivalent to $g^{-1} \notin V$. Thus we can

write

$$\mathcal{P}(G) \setminus \mathcal{U}(G) = \bigcup_{g \in G \setminus \{1\}} \underbrace{U_g^c \cap U_{g^{-1}}^c}_{\text{open in } \mathcal{P}(G)}$$

and so $\mathcal{U}(G)$ is closed.

Recalling the defining properties for positive cones, we have that

$$\text{LO}(G) = \mathcal{S}(G) \cap \mathcal{D}(G) \cap \mathcal{U}(G)$$

is closed as a finite intersection of closed sets. Hence we have the following result.

Proposition 5.6. *$\text{LO}(G)$ is a closed subspace of $\mathcal{P}(G)$, hence compact.*

In addition to compactness, the space of left-orders $\text{LO}(G)$ has additional interesting properties. It is totally disconnected and Hausdorff for any group G , as well as metrizable in the case of a countable group G . For a proof of the latter, see [Sik04], as we will not do it here. We will show, however, the two former properties, which are inherited from the power set $\mathcal{P}(G)$. Recall that a space is connected if it cannot be divided into two disjoint non-empty open sets.

Definition 5.7. A topological space X is called *totally disconnected* if the only connected subsets of X are its singleton sets.

Example 5.8. The rational numbers $\mathbb{Q}$ considered as a subspace of $\mathbb{R}$, with the usual topology, is totally disconnected. In order to see this, we show that any subset $U \subseteq \mathbb{Q}$ that contains two distinct points $p \neq q$ must be disconnected. Assume $p < q$, otherwise swap the roles of p and q . Take any irrational number $m \in \mathbb{R} \setminus \mathbb{Q}$ with $p < m < q$. Define

$$U_1 := (-\infty, m) \cap U, \quad U_2 := (m, \infty) \cap U.$$

Then $p \in U_1, q \in U_2$, so both sets are non-empty. Furthermore they are both open in U and we have $U = U_1 \cup U_2$ as well as $U_1 \cap U_2 = \emptyset$. Thus U is disconnected.

The next example for a totally disconnected space is the Cantor set, considered as a subspace of $\mathbb{R}$ with the standard topology. The reason for it being so important to us is the following result, which is due to Brouwer. A (modern) proof can be found for example in [Fra12], Chapter 4. Until now, we did not define what an “isolated point” actually is in the context of topology, but we will make up for that soon in Definition 5.12.

Theorem 5.9 (Brouwer, 1910). *Every non-empty, totally disconnected, compact and metrizable space without isolated points is homeomorphic to the Cantor set.*

As already hinted at before, the space of left-orders $\text{LO}(G)$ for some countable group G exhibits all these

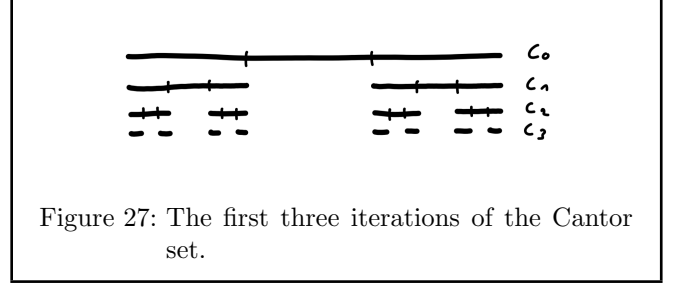


Figure 27: The first three iterations of the Cantor set.

properties, except for the fact that the existence of isolated points depends on the group in question.

While totally disconnectedness of the Cantor set of course is implied by the result of Brouwer, we still include here a direct way to see it as an illustrative example. The idea for how to do this comes from the answer by user642796 in [use12].

Example 5.10. The Cantor set $\mathcal{C}$ is totally disconnected. It consists of all those real numbers in $[0, 1]$ which can be expressed with only zeros and twos in base-3 decimal representation and is defined recursively via

$$C_0 := [0, 1], \quad C_n := \frac{1}{3}(C_{n-1} \cup (2 + C_{n-1}))$$

as then taking the intersection $\mathcal{C} := \bigcap_{n=0}^{\infty} C_n$. Looking at the first few iterations

$$\begin{aligned} C_0 &= [0, 1] \\ C_1 &= \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right], \\ C_2 &= \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right], \end{aligned}$$

we can see that in each step the “middle third” of each interval is deleted. In particular, C_n consists of a disjoint union of closed intervals of length $\frac{1}{3^n}$, see Fig. 27.

Take $x, y \in \mathcal{C}$ with $x \neq y$. Since $|x - y| > 0$, there is some $N \in \mathbb{N}$ with $\frac{1}{3^N} < |x - y|$, showing that x and y belong to different intervals in C_N . Denote the interval containing x as $I_x \subseteq C_N$, which is closed in C_N . Then $y \notin I_x$ and $C_N \setminus I_x$ is a finite union of closed intervals, thus it is closed and so I_x is open in C_N . Hence, the Cantor set can be partitioned into two disjoint open sets

$$\mathcal{C} = (\mathcal{C} \cap I_x) \sqcup (\mathcal{C} \cap (C_N \setminus I_x)).$$

This argument shows that any subset of $\mathcal{C}$ containing more than one point is disconnected.

Proposition 5.11. *For any set X , its power set $\mathcal{P}(X)$ is totally disconnected and Hausdorff.*

Proof. Take two distinct subsets $A, B \subseteq X$. Assume that $B \setminus A$ is non-empty, otherwise swap the roles of B and

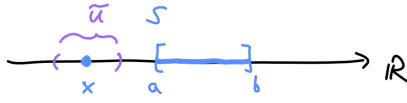


Figure 28: Example of an isolated point x with an open neighborhood $\tilde{U}$ that does not intersect with $S \setminus \{x\}$.

A. Take some $x \in B \setminus A$, then $B \in U_x$ and $A \in U_x^c$. Hence $\mathcal{P}(X)$ is partitioned into the two open sets $\mathcal{P}(X) = U_x \sqcup U_x^c$. This shows that $\mathcal{P}(X)$ is Hausdorff and that any subset of $\mathcal{P}(X)$ containing more than one point can be partitioned into two disjoint open sets. $\square$

Since these two properties carry over to subspaces, also $\text{LO}(G)$ is totally disconnected and Hausdorff for any group G .

5.3 Isolated Points and Easy Examples

As promised, we now give a precise definition for isolated points.

Definition 5.12. Let X be a topological space, $S \subseteq X$ some subset and $x \in X$. Then x is called an *isolated point* of S if there is a neighborhood of x in X that does not contain any point of S other than x .

Example 5.13. Take some closed interval $[a, b] \subseteq \mathbb{R}$ and $x \in \mathbb{R} \setminus [a, b]$ and define $S := \{x\} \cup [a, b]$ as in Fig. 28. Then $\{x\}$ is an isolated point of S .

It is easy to see that the following alternative characterizations hold for isolated points.

Proposition 5.14. In a topological space X with some subspace $S \subseteq X$ and a point $x \in S$, the following are equivalent:

- i) x is an isolated point in S ;
- ii) $\{x\}$ is open in S ;
- iii) x is not a limit point of $S \setminus \{x\} \subseteq S$.

Example 5.15. $\mathbb{Q} \subseteq \mathbb{R}$ has no isolated points, since for any $x \in \mathbb{Q}$, $(x - \frac{1}{n})_{n \in \mathbb{N}}$ is a sequence in $\mathbb{Q} \setminus \{x\}$ that has x as its limit.

Example 5.16. The space of left-orders $\text{LO}(K)$ of the fundamental group K of the Klein bottle has only isolated points, since it is a discrete space consisting of only 4 points, see Corollary 3.16.

Proposition 5.17. If a set X is countably infinite, then its power set $\mathcal{P}(X)$ has no isolated points.

Proof. A point $A \in \mathcal{P}(X)$ not being isolated is equivalent to it being a limit point of $\mathcal{P}(X)$ by Proposition 5.14. Furthermore we know that

$$\mathcal{P}(X) \cong \prod_{x \in X} \{0, 1\} \cong \prod_{i=1}^{\infty} \{0, 1\},$$

since X is countably infinite. Hence, A can be identified with some sequence $(a_1, a_2, a_3, \dots)$ where $a_i \in \{0, 1\}$ for all i . We then define for each $n \in \mathbb{N}$ a new sequence via

$$A_i^{(n)} := \begin{cases} a_i, & i \leq n, \\ 1 - a_i, & i > n, \end{cases}$$

i.e.

$$A^{(n)} = (a_1, \dots, a_n, 1 - a_{n+1}, 1 - a_{n+2}, \dots).$$

So $A^{(n)}$ represents the same sequence as A but with every entry with index greater than n “flipped”. This defines a sequence

$$(A^{(n)})_{n \in \mathbb{N}} \subseteq \prod_{i=1}^{\infty} \{0, 1\}$$

and since the i -th component converges to a_i in the i -th factor of the product, this sequence converges to A . Additionally we have $A^{(n)} \neq A$ for all $n \in \mathbb{N}$, showing that A is a limit point of $\mathcal{P}(X)$. $\square$

Note that this does not hold if X is finite, because we have seen that in this case $\mathcal{P}(X)$ is discrete and thus any point is isolated.

For what comes next, we need some knowledge about topological separation axioms. Recall that a space X is *second countable* if it has a countable base for its topology, *normal* if any two disjoint closed subsets of X can be separated by disjoint open neighborhoods, and *regular* if for any choice of a point $x \in X$ and a closed subset $S \subseteq X \setminus \{x\}$ one can find disjoint open neighborhoods separating x and S .

By Urysohn’s Metrization Theorem, every Hausdorff second-countable regular space is metrizable. Additionally, we have the implications

$$\text{Compact \& Hausdorff} \implies \text{Normal}$$

and

$$\text{Totally disconnected} \implies \text{Singletons are closed,}$$

hence every space that is compact, Hausdorff and totally disconnected is regular. Thus, Brouwer’s Theorem 5.9 can be reformulated as follows.

Theorem 5.18. Every non-empty, totally disconnected, compact, Hausdorff, second-countable space without isolated points is homeomorphic to the Cantor set.

Since X being countable means that the Tychonoff topology on $\mathcal{P}(X)$ has a countable base, this yields the following result as remarked in [CR15] after Problem 1.33.

Corollary 5.19. *If a set X is countably infinite, then its power set $\mathcal{P}(X)$ is homeomorphic to the Cantor set.*

Note that while $\mathcal{P}(G)$ is homeomorphic to the Cantor set for a countable group G , this property can fail when passing to its subspace $\text{LO}(G)$, since points may become isolated in the process. Lemma 5.20 is based on the introduction of Chapter 10 in [CR15] and is an important characterization of isolated orders in $\text{LO}(G)$.

Lemma 5.20. *Given a group G , a left-order $<$ is an isolated point in $\text{LO}(G)$ if and only if there are elements $g_1, \dots, g_n$ such that $<$ is the unique left-order on G with $1 < g_i$ for all $i = 1, \dots, n$.*

Proof. Let $< \in \text{LO}(G)$ be isolated, which is equivalent to saying that $\{<\}$ is open in $\text{LO}(G)$. Recalling the generating sets for the topology on $\text{LO}(G)$, this is the case if and only if there is a finite set of elements $g_1, \dots, g_n \in G \setminus \{1\}$, such that

$$\{<\} = \bigcap_{i=1}^n V_{g_i} = \bigcap_{i=1}^n \{<' \in \text{LO}(G) \mid 1 < g_i\}.$$

In other words, $<$ is the unique left-order satisfying $1 < g_i$ for all $i = 1, \dots, n$. $\square$

By contraposition, we obtain that $<$ is a limit point in $\text{LO}(G)$ if and only if for every finite choice of $<$ -positive elements $g_1, \dots, g_n \in G$, there is another left-order in $\text{LO}(G) \setminus \{<\}$ for which these elements are also positive.

This leads to a sufficient condition for an order being isolated, which is based on the second part of Problem 10.26 in [CR15].

Corollary 5.21. *If a positive cone $P \in \text{LO}(G)$ is finitely generated as a subsemigroup, then P is isolated in $\text{LO}(G)$.*

Proof. Assume that P is finitely generated as a semisubgroup, meaning there are $g_1, \dots, g_n \in G$, such that

$$P = \langle g_1, \dots, g_n \rangle^+.$$

Suppose there is another positive cone P' with $g_1, \dots, g_n \in P'$, then we must have $P \subseteq P'$ by assumption. But this already implies $P = P'$ by Proposition 2.15. Hence, P is the unique positive cone for which the elements $g_1, \dots, g_n$ are positive, so it is isolated in $\text{LO}(G)$ by Lemma 5.20. $\square$

A left-order $<$ with a positive cone P that is finitely generated as a semigroup

$$P = \langle g_1, \dots, g_n \rangle^+$$

is in a sense explicit, since any $<$ -positive element can be expressed as a word in $g_1, \dots, g_n$ with only positive exponents.

The following example comes from [DNR16], before Theorem 2.2.44.

Example 5.22. Recall the fundamental group of the Klein bottle

$$K = \langle x, y \mid xyx^{-1} = y^{-1} \rangle.$$

As seen in Corollary 3.16, the 4 points in $\text{LO}(K)$ correspond to the positive cones $\langle x, y \rangle^+, \langle x^{-1}, y \rangle^+, \langle x, y^{-1} \rangle^+$ and $\langle x^{-1}, y^{-1} \rangle^+$.

Contrary to Example 5.22, the opposite direction of Corollary 5.21 does not hold in general, as there are cases where a positive cone is isolated but not finitely generated as a semigroup, see Example 2.2.43 in [DNR16]. However, it turns out that the condition of being finitely generated is not only sufficient but also necessary for isolated positive cones in finitely generated free groups. This is a vital ingredient in showing that the space of left-orders on a free group on n generators has no isolated points, as we will see in section 7.3.

We conclude this section with an application of Lemma 5.20 to show that the spaces of left-orders for both the free group on countably infinite generators $\text{LO}(F_\infty)$ as well as for the braid group on countably infinite strands $\text{LO}(B_\infty)$ are homeomorphic to the Cantor set. In both cases, we show that each left-order $<$ is a limit point in the respective space of left-orders, which amounts to showing that for each choice of $<$ -positive elements $g_1, \dots, g_n$ there is a different left-order $<'$ for which these elements are also positive.

Before coming to that, we revisit an old friend. Recall from Definition 3.1, that given an automorphism $\varphi \in \text{Aut}(G)$ and some element $< \in \text{LO}(G)$, we obtain a new element $<_\varphi \in \text{LO}(G)$ via $g <_\varphi h \iff \varphi(g) < \varphi(h)$. As remarked in Chapter 10.1 in [CR15], this yields a group action of $\text{Aut}(G)$ on $\text{LO}(G)$ via homeomorphisms.

Proposition 5.23. *Given a group G , there is a homeomorphism*

$$\text{Aut}(G) \rightarrow \text{Homeo}(\text{LO}(G)), \quad \varphi \mapsto F_\varphi,$$

where

$$F_\varphi : < \mapsto <_\varphi.$$

Proof. Clearly $<_{\text{id}} = <$ and for $\varphi, \psi \in \text{Aut}(G)$ we have

$$g <_{\varphi \circ \psi} \iff \varphi(\psi(g)) < \varphi(\psi(h)) \iff g(<_\varphi)_\psi h,$$

hence $\varphi \circ \psi$ is mapped to $F_\varphi \circ F_\psi$.

Regarding continuity of F_φ and its inverse for any choice of $\varphi \in \text{Aut}(G)$, take a basic open set in $\text{LO}(G)$ of the

form

$$V_g = \{< \in \text{LO}(G) \mid 1 < g\}.$$

Then

$$F_\varphi^{-1}(V_g) = \{< \in \text{LO}(G) \mid 1 <_\varphi g\} = V_{\varphi(g)}$$

and

$$(F_\varphi^{-1})^{-1}(V_g) = F_{\varphi^{-1}}^{-1}(V_g) = V_{\varphi^{-1}(g)}.$$

Hence, F_φ is a homeomorphism. $\square$

Remark 5.24. Recall that for $g \in G$ and $< \in \text{LO}(G)$, the conjugated order $<_g$ is given by $h < h' \iff hg^{-1} < h'g^{-1}$. Since G embeds into $\text{Aut}(G)$ via

$$G^{\text{op}} \rightarrow \text{Aut}(G), \quad g \mapsto (h \mapsto ghg^{-1}),$$

where G^{op} is the *opposite group* of G with the same underlying set but with multiplication reversed, we obtain the *conjugation action*

$$G^{\text{op}} \rightarrow \text{Homeo}(\text{LO}(G)), \quad g \mapsto <_g,$$

which expresses itself this time as a right-action. Note that we follow here what seems to be convention in the literature. As any right-action, the conjugation action could easily be converted into a left-action by precomposing with the isomorphism $G \rightarrow G^{\text{op}}, g \mapsto g^{-1}$.

The following example stems from the discussion before Problem 10.2 in [CR15].

Example 5.25. Recall the fundamental group of the Klein bottle

$$K = \langle x, y \mid xyx^{-1} = y^{-1} \rangle.$$

We have seen earlier in Corollary 3.16 that $\text{LO}(K)$ consists of only 4 points, which only differ on how one compares first exponents of the x -component and then exponents of the y -component. Let us check what the conjugation action on $\text{LO}(K)$ does to these orders. First note that for some $< \in \text{LO}(K)$ we have

$$x^m y^k <_y x^{m'} y^{k'} \iff x^m y^{k-1} < x^{m'} y^{k'-1},$$

so the conjugation action by y does not change anything to how the exponents compare.

On the other hand, consider the conjugation action by x . Here we have

$$\begin{aligned} x^m y^k <_x x^{m'} y^{k'} &\iff x^m y^k x^{-1} < x^{m'} y^{k'} x^{-1} \\ &\iff x^{m-1} y^{-k} < x^{m'-1} y^{-k'}. \end{aligned}$$

Hence, the order on the x -component is not affected, but the order on the y -component is reversed, since

$$-k < -k' \iff k > k'.$$

As a result, we obtain that the four elements of $\text{LO}(K)$ are swapped pairwise by the action of the generator x ,

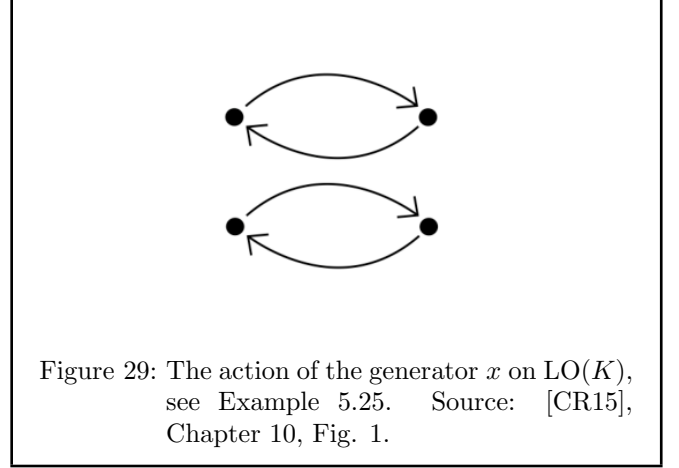


Figure 29: The action of the generator x on $\text{LO}(K)$, see Example 5.25. Source: [CR15], Chapter 10, Fig. 1.

as depicted in Fig. 29. One can also see that there is no left-order that is fixed by under all conjugations, since every orbit contains two points. This is yet another way to see that there are no bi-orders on K .

With the action of $\text{Aut}(G)$ on $\text{LO}(G)$ at hand we can show that the space of left-orders on F_∞ as well as its space of bi-orders have no isolated points. The idea for this comes from Proposition XIV.1.8 in [Deh+08].

Proposition 5.26. *Both $\text{LO}(F_\infty)$ and $\text{BO}(F_\infty)$ have no isolated points.*

Proof. Consider

$$F_\infty = \langle x_1, x_2, \dots \rangle.$$

Choose some left-order $< \in \text{LO}(F_\infty)$. Denote the generating sets for the topology on $\text{LO}(F_\infty)$ as

$$V_g = \{<' \in \text{LO}(F_\infty) \mid 1 <' g\}.$$

Choose some basic open neighborhood U of $<$, which means choosing a finite set of elements $g_1, \dots, g_k \in F_\infty$, such that

$$< \in U = \bigcap_{i=1}^k V_{g_i} = \{<' \in \text{LO}(F_\infty) \mid 1 <' g_i \text{ for all } i\}.$$

Now each g_i involves only finitely many of the generators x_j of F_∞ , hence there exists $n \in \mathbb{N}$ with

$$g_1, \dots, g_k \in \langle x_1, \dots, x_{n-1}, x_{n+1}, \dots \rangle.$$

Consider the automorphism

$$\varphi : F_\infty \xrightarrow{\cong} F_\infty, \quad x_j \mapsto \begin{cases} x_j, & j \neq n, \\ x_j^{-1}, & j = n. \end{cases}$$

This induces a new order $<_\varphi \in \text{LO}(F_\infty)$ given by

$$g <_\varphi h \iff \varphi(g) < \varphi(h).$$

Then by definition we have $1 <_{\varphi} g_i$ for all i , but $< \neq <_{\varphi}$, since $x_n > 1 \iff x_n <_{\varphi} 1$.

Hence every neighborhood of $<$ contains another left-order different from $<$, so $<$ is a limit-point in $\text{LO}(F_{\infty})$. The same arguments work for $\text{BO}(F_{\infty})$. $\square$

Since F_{∞} is countable, we can use Theorem 5.18.

Corollary 5.27. *$\text{LO}(F_{\infty})$ is homeomorphic to the Cantor set.*

The same is true for $\text{BO}(F_{\infty})$, but first we need to show that it is compact. This follows from a more general statement about groups acting by homeomorphisms, which is based on the second part of Problem 10.2 in [CR15]. Before that, we quickly need to show an alternative characterization of the Hausdorff property.

Proposition 5.28. *A space X is Hausdorff if and only if the diagonal Δ is closed in $X \times X$.*

Proof. Suppose Δ is closed in $X \times X$ and take $x \neq y \in X$. Since $X \times X \setminus \Delta$ is open, there is a basic open set $U \times V \subseteq X \times X \setminus \Delta$ containing (x, y) , i.e. $U, V \subseteq X$ are open in X . Also $U \cap V = \emptyset$, since otherwise we would have $U \times V \cap \Delta \neq \emptyset$, leading to a contradiction. But then we found disjoint open neighborhoods for x and y .

On the other hand, suppose that X is Hausdorff. We show that then $X \times X \setminus \Delta$ is open. Let $(x, y) \in X \times X \setminus \Delta$, then there are disjoint open neighborhoods U_x of x and U_y of y in X . But then $U_x \times U_y$ is an open neighborhood of (x, y) , which is contained in $X \times X \setminus \Delta$, since otherwise U_x and U_y would not be disjoint. If we do this for any pair (x, y) , we can write

$$X \times X \setminus \Delta = \bigcup_{x, y \in X \times X \setminus \Delta} U_x \times U_y,$$

which is a union of open sets. $\square$

Proposition 5.29. *Suppose that a group G acts on a Hausdorff space X by homeomorphisms. Then for any $g \in G$, the fixed-point set*

$$X^g := \{x \in X \mid g \cdot x = x\}$$

is closed in X . Moreover, the global fixed point set, i.e. all points in X which are fixed by all $g \in G$, is closed in X .

Proof. For the first part, let $g \in G$ and consider the continuous map

$$h : X \rightarrow X \times X, \quad x \mapsto (g \cdot x, x).$$

Then $X^g = h^{-1}(\Delta)$, where $\Delta \subseteq X \times X$ is the diagonal. Since X is Hausdorff, Δ is closed in $X \times X$ by Proposition 5.28 and thus also X^g is closed in X .

For the second part, observe that the global fixed point set is given by

$$\bigcap_{g \in G} X^g,$$

which is closed as an intersection of closed sets. $\square$

Proposition 5.30. *The space of bi-orders $\text{BO}(G)$ is closed, and thus compact, in $\text{LO}(G)$.*

Proof. This follows directly from Proposition 5.29, since $\text{LO}(G)$ is Hausdorff as a subspace of $\mathcal{P}(G)$ and $\text{BO}(G)$ is the global fixed point set of the conjugation action, which we have seen to be an action by Homeomorphisms of $\text{LO}(G)$. $\square$

Corollary 5.31. *$\text{BO}(F_{\infty})$ is homeomorphic to the Cantor set.*

The conjugation action can also be used for showing that every left-order on the braid group on infinitely many strands is a limit point, as was shown in Proposition XIV.2.10 in [Deh+08]. Here, we consider

$$B_{\infty} = \langle \sigma_1, \sigma_2, \dots \mid \sigma_i \sigma_j = \sigma_j \sigma_i \text{ for } |i - j| > 1, \\ \sigma_i \sigma_j \sigma_i = \sigma_j \sigma_i \sigma_j \text{ for } |i - j| = 1 \rangle.$$

Likewise, denote the braid group on n strands as

$$B_n = \langle \sigma_1, \dots, \sigma_{n-1} \mid \sigma_i \sigma_j = \sigma_j \sigma_i \text{ for } |i - j| > 1, \\ \sigma_i \sigma_j \sigma_i = \sigma_j \sigma_i \sigma_j \text{ for } |i - j| = 1 \rangle.$$

Note that B_n sits inside of B_{∞} via the canonical embedding

$$B_n \rightarrow B_{\infty}, \quad \sigma_i \mapsto \sigma_i.$$

Proposition 5.32. *Each left-order on B_{∞} is a limit of its conjugates. As a consequence, $\text{LO}(B_{\infty})$ has no isolated points and thus is homeomorphic to the Cantor set.*

Proof. Take a left-order $< \in \text{LO}(B_{\infty})$ with positive cone P and a finite subset $S \subseteq P$. Then there exists $n \in \mathbb{N}$, such that $S \subseteq B_n$. Hence, for $i > n$ every braid in S commutes with σ_i . This already implies

$$S = \sigma_i S \sigma_i^{-1} \subseteq \sigma_i P \sigma_i^{-1},$$

where the set on the right is the positive cone for the conjugate of $<$ by σ_i .

Now what is left to show is, that we can indeed find such $i > n$ with $P \neq \sigma_i P \sigma_i^{-1}$. Consider the shift homomorphism

$$\text{sh} : B_{\infty} \rightarrow B_{\infty}, \quad \sigma_i \mapsto \sigma_{i+1}.$$

Then $P \cap \text{sh}^n(B_{\infty})$ and $(\sigma_i P \sigma_i^{-1}) \cap \text{sh}^n(B_{\infty})$ define left-orders on $\text{sh}^n(B_{\infty})$. Suppose $P = \sigma_i P \sigma_i^{-1}$ for all $i > n$, then in particular $P \cap \text{sh}^n(B_{\infty})$ is invariant under conjugation by all elements of $\text{sh}^n(B_{\infty})$, hence $\text{sh}^n(B_{\infty}) \cong B_{\infty}$ would be bi-orderable. But this is not true, as we have seen in Corollary 2.10. $\square$

5.4 Cardinality of Spaces of Orders

An interesting fact is that the space of left-orders for a group is either finite or uncountable, as first shown in [Lin11]. We showcase here a different approach based on Section 2.2.1 in [DNR16], which is using the conjugation action along with topological properties of $\text{LO}(G)$.

Regarding the topological side, we need an additional definition along with a result that can be found as Theorem 2-80 in [HY61].

Definition 5.33. A subset of a space is called *perfect*, if it is closed and has no isolated points.

Theorem 5.34. *No compact Hausdorff space and no complete metric space is both countable and perfect.*

Given a left-orderable group G , define $M \subseteq \text{LO}(G)$ as a *minimal subset* of $\text{LO}(G)$ if it is non-empty, closed and invariant under the conjugation action (see Remark 5.24) and additionally it does not contain a proper subset with these three properties.

Next, consider the set of all limit points $L(M)$ of M . They form a closed subset of $\text{LO}(G)$. Furthermore, since conjugation acts on $\text{LO}(G)$ by homeomorphisms, invariance of M under the action also implies invariance of $L(M)$. Hence, by the minimality assumption, we either have $M = L(M)$ or $L(M) = \emptyset$. In the first case, M has no isolated points and is thus uncountable by Theorem 5.34.

In the second case, M must be finite as it has no limit points. Then for any $< \in M$, the stabilizer

$$G_{<} = \{g \in G \mid <_g = <\}$$

of $<$ under the conjugation action has finite index, since M is finite and invariant under the action. Furthermore, since the condition $< = <_g$ is equivalent to

$$x < y \iff xg^{-1} < yg^{-1},$$

the order $<$ restricted to $G_{<}$ is bi-invariant. Then the following result can be used.

Proposition 5.35. *Let $(G, <)$ be a left-ordered group with a finite-index subgroup $G_0 \subseteq G$, such that $<$ restricted to G_0 is bi-invariant. If $<$ has a neighborhood in $\text{LO}(G)$ containing only countably many left-orders, then $\text{LO}(G)$ is finite.*

We omit the proof, as it uses the theory of Conradion orders, which is beyond the scope of this thesis. For more details, see section 3.2.6 in [DNR16].

Since either M is uncountable or the stabilizer $G_{<}$ has finite index, the result follows.

Theorem 5.36 (Linnell). *The space of left-orders of a group is either finite or uncountable.*

6 Characterizations for Orderability

In this section, we explore two equivalent characterizations for a group G to be left-orderable, which is the same as $\text{LO}(G)$ being a non-empty subset of $\mathcal{P}(G)$. What is interesting here is that the topology on $\mathcal{P}(G)$ plays a key role in some of the proofs. Throughout the section, we mainly follow Chapter 1.6 from [CR15] and sometimes borrow concepts from chapter 2.2.2 in [DNR16].

6.1 Characterization Using the Cayley Graph

The first equivalent characterization uses a “local” version of positive cones in order to deduce statements about the existence of “global” positive cones. Here, by “local” we mean restrictions of positive cones to neighborhoods of 1 in the Cayley graph.

Recall the the notions of length and N -balls in a Cayley graph Γ_A with respect to a generating set A of a group G given in Definition 4.3. Building on that, the following three definitions are based on concepts from Chapter 2.2.2 in [DNR16].

Definition 6.1. Let $S \subseteq G$ be some subset and let

$$B_N = \{g \in G \mid |g| \leq N\}$$

denote the N -ball of G with respect to some generating set.

- i) S is called *total at length N* , if it is *antisymmetric*, i.e. $S \cap S^{-1} = \emptyset$, and $B_N \setminus \{1\} \subseteq S \cup S^{-1}$ holds.
- ii) S is called *stable* if for all $f, g \in S$ we have that

$$|fg| \leq \max\{|f|, |g|\} \implies fg \in S.$$

In this case we call S a *stable basis* for the semigroup $\langle S \rangle^+$.

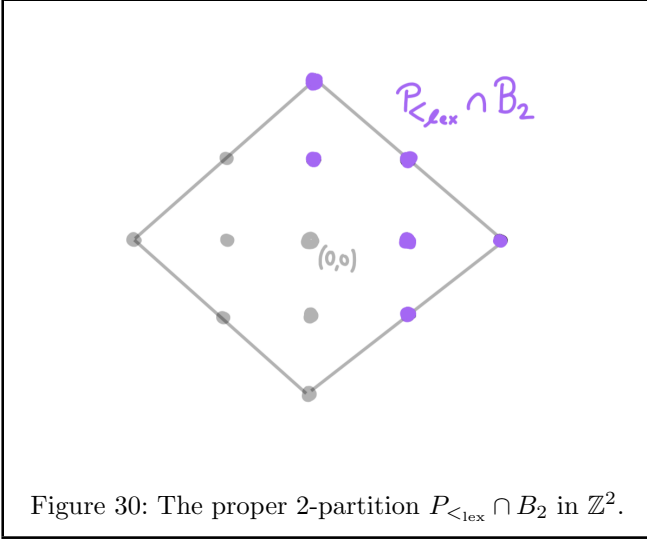
- iii) S is called *N -stable* if for all $f, g \in S$ we have that

$$|fg| \leq N \implies fg \in S.$$

Of course, for $S \subseteq B_N$, N -stable implies stable.

Remark 6.2. The intuition here is that if S is total at length N and N -stable, then it resembles a left-invariant total order when restricted to words of length N . This is reflected in the fact that, for any $N \in \mathbb{N}$, the restriction $P \cap B_N$ of a positive cone $P \in \text{LO}(G)$ to the N -ball B_N is both N -stable and total at length N .

Example 6.3. For a more concrete example of Remark 6.2, take the standard lexicographic order $<_{\text{lex}}$ on $\mathbb{Z}^2$ with its standard Cayley graph, then $P_{<_{\text{lex}}} \cap B_2$ is visualized in Fig. 30 for the case $k = 2$.



This motivates the following definition, which is again based on Chapter 1.6 in [CR15].

Definition 6.4. We call $Q \subseteq B_N$ a *proper N -partition* of B_N if it is N -stable and total at length N .

We continue with some additional (non-)examples of stability.

Example 6.5. Let $F_2 = \langle a, b \rangle$.

- i) The set $\{ab, b^{-1}\}$ is not stable, since $ab \cdot b^{-1} = a \notin S$. However, it becomes stable if we add a to the set.
- ii) Consider the set S_N consisting of all elements in F_2 with length exactly N . This set is not stable for $N > 1$, because it is not antisymmetric and $1 \notin S_N$.

Example 6.5 suggests, that whenever a finite set is not stable, one can add the missing elements step by step until the resulting set is stable. This is a finite process.

Example 6.6. i) Let P_E denote the positive cone from the order defined in Section 4.2.1, then

$$S := (P_E \cap B_2 \setminus \{b\}) \cup \{b^{-1}\}$$

is total at length 2. However, it is not stable, since $a^{-1}b \cdot b^{-1} = a^{-1} \in S$ would violate the semigroup property of P_E . Note that this also means that S is not induced as the intersection of some positive cone with B_2 .

- ii) Consider $T := P_E \cap B_2 \setminus \{ab\} \subseteq B_2$. It is stable, but not 2-stable, since $|ab| \leq 2$ but $ab \notin T$.
- iii) Given any set $S \subseteq F_n$, the intersection $\langle S \rangle^+ \cap B_N$ is always N -stable.

Remark 6.7. Part i) of Example 6.6 is a special case of the fact that S being stable means that in a product fg with $f, g \in S$, either $fg \in S$ or the cancellative part h

must be strictly smaller than half of $|f|$ and half of $|g|$. Indeed, write $f = \tilde{f}h$ and $g = h^{-1}\tilde{g}$ so that $fg = \tilde{f}\tilde{g}$ without cancellations and suppose that $|h| \geq \frac{|f|}{2}$. Then, by using $|f| = |\tilde{f}| + |h|$, we obtain

$$|fg| = |\tilde{f}| + |\tilde{g}| \leq \frac{|f|}{2} + |g| - |h^{-1}| \leq \frac{|f|}{2} + |g| - \frac{|f|}{2} = |g|$$

and thus $fg \in S$ by stability. A similar reasoning works for the case when $|h| \geq \frac{|g|}{2}$. Hence, products of two elements with cancellations spanning at least half of either factor are already included in a stable basis.

The concept of stability will become relevant again later in section 7.3 when dealing with the space of left-orders on F_n , but we will leave it for now and focus on the concept of proper N -partitions.

As already observed in Example 6.2, if G is left-orderable, then there always exists a proper N -partition for any N . By contraposition, this yields Proposition 1.42 from [CR15].

Proposition 6.8. *If A is a generating set of G for which there is some $N \in \mathbb{N}$ such that there exists no proper N -partition of B_N , then G is not left-orderable.*

In fact, the converse is also true, which can be shown using topological arguments applied to $\mathcal{P}(G)$. Before coming to that, we need some results from general topology.

Definition 6.9. A family $\{X_i\}_{i \in I}$ of non-empty subsets $X_i \subseteq X$ of an arbitrary set X set has the *finite intersection property* if every finite subcollection $\{X_j\}_{j \in J}$ of those subsets has non-empty intersection

$$\bigcap_{j \in J} X_j \neq \emptyset.$$

The proof for the next result can be found in [Mun14], Theorem 26.9.

Proposition 6.10. *A space X is compact if and only if any collection $\{X_i\}_{i \in I}$ of non-empty closed subsets $X_i \subseteq X$ with the finite intersection property has non-empty total intersection*

$$\bigcap_{i \in I} X_i \neq \emptyset.$$

From this, one can deduce a special case of Cantor's intersection theorem.

Corollary 6.11. *A decreasing sequence of nested, closed subsets in a compact space has nonempty intersection.*

Proof. Consider a nested sequence

$$C_1 \supseteq C_2 \supseteq C_3 \supseteq \dots$$

of nonempty closed subsets in a compact space X . The sequence automatically has the finite intersection property and thus by Proposition 6.10 we have

$$\bigcap_{k=1}^{\infty} C_k \neq \emptyset.$$

□

Now we arrive at our first characterization for left-orderability. The proof is based on Theorem 1.43 in [CR15].

Proposition 6.12. *Let S be a generating set of G . If for all $k \geq 1$ there is a proper k -partition of B_k , then G is left-orderable.*

Proof. For $k \geq 1$ consider

$$P_k := \{A \subseteq G \mid A \cap B_k \text{ is a proper } k\text{-partition}\} \subseteq \mathcal{P}(G).$$

By assumption, this set is non-empty for all k . Furthermore one can show that every P_k is closed in $\mathcal{P}(G)$: The procedure is exactly as in Example 5.5, but with B_k instead of G as index set of the unions. Then as is the case with $\text{LO}(G)$, each P_k can be written as an intersection of three closed sets in $\mathcal{P}(G)$.

Lastly, those sets form a descending sequence

$$P_1 \supseteq P_2 \supseteq P_3 \supseteq \dots$$

and thus Corollary 6.11 implies

$$\bigcap_{k=1}^{\infty} P_k \neq \emptyset.$$

The claim is now, that an element P contained in this intersection corresponds to a left-order on G . Equivalently, we show that P is a positive cone, i.e. it is a semi-group and induces a partition $G = P \sqcup \{1\} \sqcup P^{-1}$.

Take $g, h \in P$. There exists some $k \geq 1$, such that both of those elements lie in B_k and thus also in $P \cap B_k \subseteq P \cap B_{2k}$. Note at this point, that by definition of P_k we know that $P \cap B_k$ is a proper k -partition for any k . Using this property and the fact that $gh \in B_{2k}$, it follows that $P \cap B_{2k}$. In particular $gh \in P$.

For the partition property, first note that

$$(P \cap B_k) \cup (P \cap B_k)^{-1} = B_k \setminus \{1\}$$

for all k , from which $P \cup P^{-1} = G \setminus \{1\}$ follows. Furthermore $(P \cap B_k) \cap (P \cap B_k)^{-1} = \emptyset$ for all k , which implies $P \cap P^{-1} = \emptyset$. □

6.2 Orderability as a Local Property

The next characterization, which is based on Theorem 1.44 from [CR15], shows that left-orderability of a group

is a local property. Different to Section 6.1, “local” in this context means that it suffices to show the property for any finitely generated subgroup. Again, the proof uses compactness of $\mathcal{P}(G)$ in the form introduced in Proposition 6.10, using that the family of subsets

$$\mathcal{Q}(F) = \{Q \subseteq G \mid Q \cap \langle F \rangle \text{ is a positive cone for } \langle F \rangle\}$$

for a group G and all finite subsets $F \subseteq G$ has the finite intersection property. We will see this shortly in the proof of Theorem 6.15. Now, looking at the total intersection

$$\bigcap_{F \subseteq G \text{ finite}} \mathcal{Q}(F),$$

it contains exactly those subsets $Q \subseteq G$, that resemble a positive cone when restricted to any finitely generated subgroup of G . Perhaps unsurprisingly, this intersection already corresponds to the set of positive cones on G itself. This is the content of Problem 1.46 in [CR15], which we prove next.

Proposition 6.13. *With the notation above, we have*

$$\bigcap_{F \subseteq G \text{ finite}} \mathcal{Q}(F) = \text{LO}(G).$$

Proof. First note that an element on the left is a subset $Q \subseteq G$, such that $Q \cap \langle F \rangle$ is a positive cone for $\langle F \rangle$ for every possible choice of finite $F \subseteq G$.

The inclusion “ $\supseteq$ ” is clear, since a positive cone P for G restricts to a positive cone for any subgroup, in particular for any $\langle F \rangle$.

On the other hand, let $Q \in \bigcap_{F \subseteq G \text{ finite}} \mathcal{Q}(F)$.

To see that it is a semigroup, take $g, h \in Q$. Now, $Q \cap \langle g, h \rangle$ is a positive cone and thus $gh \in Q \cap \langle g, h \rangle \subseteq Q$.

Regarding $G = Q \sqcup \{1\} \sqcup Q^{-1}$, observe that for any $g \in G$ we have

$$\begin{aligned} g \in \langle g \rangle &= (Q \cap \langle g \rangle) \sqcup \{1\} \sqcup (Q \cap \langle g \rangle)^{-1} \\ &\subseteq Q \sqcup \{1\} \sqcup Q^{-1}, \end{aligned}$$

since $Q \cap \langle g \rangle$ is a positive cone. □

In other words, if one can show that the intersection from Proposition 6.13 is non-empty, then the corresponding group is left-orderable (and the other way round).

Before coming to the proof of the main statement of this section, we need another result about the topology of $\mathcal{P}(G)$, which is based on Problem 1.34 from [CR15]. Note that this also holds for the power set of an arbitrary set X .

Proposition 6.14. *For a subset $A \subseteq X$ of an arbitrary set X , the space $\mathcal{P}(X)$ retracts onto the closed subset*

$\mathcal{P}(A)$ via the continuous map

$$r_A : \mathcal{P}(X) \rightarrow \mathcal{P}(A), \quad Q \mapsto Q \cap A.$$

Proof. First we show that $\mathcal{P}(A)$ is closed in $\mathcal{P}(X)$. Observe that Q lies in $\mathcal{P}(X) \setminus \mathcal{P}(A)$ if and only if $Q \not\subseteq A$, i.e. there exists $a \in A$ with $a \notin Q$. The latter means $Q \in U_a^c$ and thus

$$\mathcal{P}(X) \setminus \mathcal{P}(A) = \bigcup_{a \in A} U_a^c$$

is open as a union of open sets. It follows that $\mathcal{P}(A)$ is closed.

Now consider the map r_A as given above. In order to check its continuity, it suffices to check that elements of the subbasis elements on $\mathcal{P}(A)$ are open in $\mathcal{P}(X)$. The subbasis on $\mathcal{P}(A)$ for the subspace topology is given by elements of the form

$$U_x^A = \{B \subseteq A \mid x \in B\}, \quad (U_x^A)^c = \{B \subseteq A \mid x \notin B\},$$

where $x \in A$. Thus, an element Q is in $r_A^{-1}(U_x^A)$ if and only if $Q \cap A$ contains $x \in A$, which is equivalent to $x \in Q$, yielding

$$r_A^{-1}(U_x^A) = U_x.$$

This is a subbasis element in $\mathcal{P}(X)$ and thus open. Similarly, one obtains

$$r_A^{-1}((U_x^A)^c) = U_x^c.$$

□

Theorem 6.15. *A group G is left-orderable if and only if each of its finitely generated subgroups is left-orderable.*

Proof. “ $\implies$ ”: This direction is clear, since any subgroup of a left-orderable group is again left-orderable.

“ $\impliedby$ ”: Let $F \subseteq G$ be a finite subset. By assumption, $\langle F \rangle$ is left-orderable, and thus $\mathcal{Q}(F)$ is non-empty. Recall the retraction map $r_{\langle F \rangle} : \mathcal{P}(G) \rightarrow \mathcal{P}(\langle F \rangle)$ from Proposition 6.14. We have seen earlier, that the set of positive cones $\text{LO}(\langle F \rangle)$ is closed in $\mathcal{P}(\langle F \rangle)$ and thus

$$\mathcal{Q}(F) = r_{\langle F \rangle}^{-1}(\text{LO}(\langle F \rangle))$$

is closed in $\mathcal{P}(G)$.

Now consider the family of closed subsets $\{\mathcal{Q}(F)\}_{F \subseteq G \text{ finite}}$ and take a finite subcollection corresponding to the finite subsets $F_1, \dots, F_n \subseteq G$. Since $\langle F_i \rangle$ is a subgroup of $\langle \bigcup_{j=1}^n F_j \rangle$ for any i , in particular any positive cone for $\langle \bigcup_{j=1}^n F_j \rangle$ restricts to a positive cone for $\langle F_i \rangle$. Furthermore, $\langle \bigcup_{j=1}^n F_j \rangle$ is left-orderable by assumption. Hence we have

$$\emptyset \neq \mathcal{Q}\left(\bigcup_{j=1}^n F_j\right) \subseteq \bigcap_{j=1}^n \mathcal{Q}(F_j),$$

showing that the initially considered family has the finite

intersection property.

Finally, we can use the compactness of $\mathcal{P}(G)$ in conjunction with Propositions 6.10 and 6.13 to conclude

$$\text{LO}(G) = \bigcap_{F \subseteq G \text{ finite}} \mathcal{Q}(F) \neq \emptyset.$$

□

Interestingly, this is a proof with a strong topological flavor, yet the statement of the theorem is a purely algebraic one. It followed mainly from the fact that $\mathcal{P}(G)$ is compact, certain subsets are closed, and specific retraction maps are continuous.

As observed in Theorem 1.47 in [CR15], this implies a strong result about bi-orderability of abelian groups.

Corollary 6.16. *An abelian group is bi-orderable if and only if it is torsion-free.*

Proof. As any left-orderable group is torsion-free by Proposition 2.3, we only have to show that an abelian and torsion-free group G is biorderable, which in this case is the same as left-orderable. By Theorem 6.15 it suffices to show that all of its finitely generated subgroups are left-orderable. Any such subgroup must be torsion-free and thus by the fundamental theorem for finitely generated abelian groups, it is isomorphic to $\mathbb{Z}^n$ for some n . The latter is left-orderable, as can be seen for example with repeated application of Proposition 3.4. □

7 Specific Spaces of Orders

7.1 Braid Groups B_n

In this section, we will see that the Dehornoy order $<_D$ from Section 4.4 is not isolated in $\text{LO}(B_n)$. However, it can be used to construct the Dubrovina-Dubrovin order $<_{DD}$ via iterated convex extensions as introduced in Example 3.10. The latter turns out to have a positive cone that is finitely generated as a semigroup, hence it must be isolated by Proposition 5.21. As a consequence, $\text{LO}(B_n)$ is not homeomorphic to the Cantor set.

Recall from Definition 4.23 that we wrote

$$P_i = \{\beta \in B_n \mid \beta \text{ is } \sigma_i\text{-positive}\}$$

and

$$P_D = P_1 \cup \dots \cup P_{n-1}$$

for the positive cone of $<_D$. Also recall the notion of a conjugated order from Definition 3.1.

First, we need a corollary to Proposition 4.25.

Corollary 7.1. *For any $\beta \in B_n$, the conjugation of $<_D$ by β in $\text{LO}(B_n)$ is discrete with smallest positive element $\beta^{-1}\sigma_{n-1}\beta$.*

Proof. Denote the conjugation of $<_D$ by β as $<_\beta$. Assume there exists an element $\alpha \in B_n$ with

$$1 <_\beta \alpha <_\beta \beta^{-1}\sigma_{n-1}\beta.$$

This is equivalent to

$$\beta^{-1} <_D \alpha \beta^{-1} <_D \beta^{-1}\sigma_{n-1},$$

which in turn is equivalent to

$$1 <_D \beta \alpha \beta^{-1} < \sigma_{n-1}.$$

But this contradicts Proposition 4.25. $\square$

The general result that $<_D$ is not isolated in $\text{LO}(B_n)$ follows from the case $n = 3$. The arguments used here follow Lemma 2.2, which is stated here without proof, as well as Proposition 2.3 and Proposition 2.4 in chapter XIV from [Deh+08].

Lemma 7.2. *For any σ_1 -positive braid $\beta_1 \in P_1$, there exists $\beta \in P_1$, such that $1 <_D \beta \leq_D \beta_1$ and $\beta\sigma_2 \neq \sigma_2\beta$.*

Proposition 7.3. *The Dehornoy order $<_D$ on B_3 is a limit point of its conjugates in $\text{LO}(B_3)$.*

Proof. Denote the positive cone of $<_D$ as P_D . By definition of the Tychonoff topology, the statement is true, if for any finite subset $S \subseteq P_D$ we can find some $\beta \in B_3$, such that

- i) $S \subseteq \beta P_D \beta^{-1}$,
- ii) $P_D \neq \beta P_D \beta^{-1}$.

Let β_1 be the smallest element in $(S \cup \{\sigma_1\}) \cap P_1$. By Lemma 7.2, there exists $\beta \in P_1$, such that

$$1 <_D \beta \leq_D \beta_1$$

and $\beta\sigma_2 \neq \sigma_2\beta$. This will turn out to be the β we are looking for.

Now consider $\gamma \in S$. In particular, γ is σ -positive. If $\gamma = \sigma_2^p$, then $p > 0$ and by the subword-property of $<_D$, also $\beta^{-1}\gamma\beta$ is σ_2 -positive, i.e. $\gamma \in \beta P_D \beta^{-1}$. On the other hand, if γ is not a power of σ_2 , it must be σ_1 -positive. By assumption, we then have

$$\beta \leq_D \beta_1 \leq_D \gamma,$$

which implies $1 \leq_D \beta^{-1}\gamma$. In conjunction with $1 <_D \beta$, this yields

$$1 \leq_D \beta^{-1}\gamma <_D \beta^{-1}\gamma\beta,$$

implying $\gamma \in \beta P_D \beta^{-1}$. For the second property, note that $\beta P_D \beta^{-1}$ is the positive cone of the conjugation of

$<_D$ by β^{-1} , hence by Corollary 7.1 it has a smallest element $\beta\sigma_2\beta^{-1}$. Now, if $\beta P_D \beta^{-1} = P_D$ would hold, we would also have $\beta\sigma_2\beta^{-1} = \sigma_2$, which equates to $\beta\sigma_2 = \sigma_2\beta$. But this contradicts the fact that β does not commute with σ_2 . $\square$

The general case then follows from using that B_3 embeds into B_n as

$$B_3 \cong \langle \sigma_{n-2}, \sigma_{n-1} \rangle \subseteq B_n$$

and the fact that the restriction of the Dehornoy order on B_n onto this subgroup corresponds to the Dehornoy order on B_3 . For details, see Proposition XIV.2.4 in [Deh+08].

Proposition 7.4. *For $n \geq 3$, the Dehornoy order on B_n is a limit point of its conjugates in $\text{LO}(B_n)$.*

As already noted before, in particular this means that $<_D$ is not isolated in $\text{LO}(B_n)$. But, interestingly enough, B_n contains certain $<_D$ -convex subgroups that may be flipped as in Example 3.10 to arrive at an isolated order after $n - 2$ steps. This construction follows arguments from the proof of Theorem 2.2.44 in [DNR16], Problems 10.23 and 10.26 in [CR15] and Theorem 5 in [DD01].

Proposition 7.5. *The subgroup $\langle \sigma_j, \dots, \sigma_{n-1} \rangle$ is $<_D$ -convex in B_n for all $j = 1, \dots, n - 1$.*

Proof. Let $\gamma <_D \beta <_D \gamma'$ with $\gamma, \gamma' \in \langle \sigma_j, \dots, \sigma_{n-1} \rangle$ and $\beta \in B_n$. This means we have

$$1 <_D \gamma^{-1}\beta, \quad 1 <_D \beta^{-1}\gamma'$$

and γ as well as γ' only consist of the generators $\sigma_j, \dots, \sigma_{n-1}$ and their inverses.

Suppose $\beta \notin \langle \sigma_j, \dots, \sigma_{n-1} \rangle$. Then β contains at least one $\sigma_k^{\pm 1}$ with $k < j$. Now the claim is that β is σ_m -positive for some $m \leq k$. First note that $\gamma^{-1}\beta$ must be σ_m -positive for some $m \leq k$, since it is σ -positive and σ_k appears in β . By assumption, σ_m does not appear in γ^{-1} , and also generators with lower indices than m cannot cancel when passing from β to $\gamma^{-1}\beta$. Hence β must be σ_m -positive. But this already implies that $\beta^{-1}\gamma'$ is σ_m -negative, i.e. $\beta^{-1}\gamma' <_D 1$, leading to a contradiction. $\square$

Proposition 7.6. *For $n \geq 3$, the set*

$$P_1 \cup P_2^{-1} \cup \dots \cup P_{n-2}^{(-1)^{n-1}} \cup P_{n-1}^{(-1)^n}$$

is a positive cone in B_n .

Proof. This can be proven inductively. For $n = 3$, we know by Proposition 7.5 that $\langle \sigma_2 \rangle$ is $<_D$ -convex. Thus, we can restrict $<_D$ to this subgroup, consider its reversed order and extend it, i.e. we define $<_3$ on B_3 as the convex extension of $\overline{<_D}|_{\langle \sigma_2 \rangle}$ by $<_D$. This yields the positive cone

$$P_{<_3} = (P_D \setminus \langle \sigma_2 \rangle) \cup (P_D^{-1} \cap \langle \sigma_2 \rangle) = P_1 \cup P_2^{-1}.$$

For the general case, first denote the generators of B_{n-1} as $\tilde{\sigma}_i$, i.e.

$$B_{n-1} = \langle \tilde{\sigma}_1, \dots, \tilde{\sigma}_{n-2} \rangle$$

and likewise write $\tilde{P}_i$ for the $\tilde{\sigma}_i$ -positive braids in B_{n-1} . Then by induction hypothesis,

$$P_{<_{n-1}} = \tilde{P}_1 \cup \tilde{P}_2^{-1} \cup \dots \cup \tilde{P}_{n-2}^{(-1)^{n-1}}$$

is a positive cone for B_{n-1} . Furthermore, B_{n-1} injects into B_n via the “right-shift”

$$\underbrace{\langle \tilde{\sigma}_1, \dots, \tilde{\sigma}_{n-2} \rangle}_{=B_{n-1}} \xrightarrow{\iota} \underbrace{\langle \sigma_1, \dots, \sigma_{n-1} \rangle}_{=B_n}, \quad \tilde{\sigma}_i \mapsto \sigma_{i+1}.$$

Under this identification, we have

$$\begin{array}{c} \tilde{P}_1 \cup \tilde{P}_2^{-1} \cup \dots \cup \tilde{P}_{n-2}^{(-1)^{n-1}} \\ \downarrow \iota \\ (P_2 \cup P_3^{-1} \cup \dots \cup P_{n-1}^{(-1)^{n-1}}) \cap \langle \sigma_2, \dots, \sigma_{n-1} \rangle \end{array}$$

This induced order $<_{n-1}$ on $\langle \sigma_2, \dots, \sigma_{n-1} \rangle$ is reversed by

$$P_{\prec_{n-1}} = (P_2^{-1} \cup P_3 \cup \dots \cup P_{n-1}^{(-1)^n}) \cap \langle \sigma_2, \dots, \sigma_{n-1} \rangle.$$

Then the extension of $\prec_{n-1}$ by $<_D$ yields

$$\begin{aligned} P_{<_n} &= (P_D \setminus \langle \sigma_2, \dots, \sigma_{n-1} \rangle) \cup P_{\prec_{n-1}} \\ &= P_1 \cup P_2^{-1} \cup \dots \cup P_{n-1}^{(-1)^n}. \end{aligned}$$

□

Definition 7.7. We denote the left-order $<_{DD}$ given by the positive cone

$$P_{DD} = P_1 \cup P_2^{-1} \cup \dots \cup P_{n-1}^{(-1)^n}$$

as the *Dubrovina-Dubrovin-order* on B_n .

The next claim is that this positive cone is finitely generated as a semigroup by the elements

$$\beta_i = (\sigma_i \dots \sigma_{n-1})^{(-1)^{i-1}}$$

for $i = 1, \dots, n-1$.

Before continuing, note that β_i is σ_i -positive for i odd and σ_i -negative for i even and a word in $\beta_1, \dots, \beta_{n-1}$ with non-negative exponents is σ_j -positive whenever the lowest appearing index j is odd, and σ_j -negative, when the lowest appearing index j is even. Hence we already have the inclusion

$$\langle \beta_1, \dots, \beta_{n-1} \rangle^+ \subseteq P_{DD}.$$

Proposition 7.8. *We have*

$$P_{DD} = \langle \beta_1, \dots, \beta_{n-1} \rangle^+$$

for $n \geq 3$.

Proof. Again, this can be done by induction. For $n = 3$, consider a braid $\gamma \in P_1 \cup P_2^{-1}$. If γ is σ_2 -negative, then $\gamma = \sigma_2^{-m} = \beta_2^m$ for some $m > 0$, hence $\gamma \in \langle \beta_2 \rangle^+ \subseteq \langle \beta_1, \beta_2 \rangle^+$. If γ is σ_1 -positive, it can be written as

$$\gamma = \sigma_2^{m''} \sigma_1 \sigma_2^{m''} \sigma_1 \dots \sigma_2^{m''} \sigma_1 \sigma_2^{m''+1}$$

for some exponents $m''_j \in \mathbb{Z}$. Using $\sigma_1 = \beta_1 \beta_2$, this can be rewritten as

$$\gamma = \beta_2^{m'_1} \beta_1 \beta_2^{m'_2} \beta_1 \dots \beta_2^{m'_{k'}} \beta_1 \beta_2^{m'_{k'}+1}$$

where the exponents m'_j may still be negative. Now one can use that $\beta_1 = \beta_2 \beta_1^2 \beta_2$, or equivalently $\beta_1^2 = \beta_2^{-1} \beta_1 \beta_2^{-1}$ to get rid of the negative exponents. For example, if the term $\beta_2 \beta_1 \beta_2^{-1}$ appears, we can use

$$\beta_2 \beta_1 \beta_2^{-1} = \beta_2^2 \beta_2^{-1} \beta_1 \beta_2^{-1} = \beta_2^2 \beta_1^2.$$

Using this observation, we can finally write

$$\gamma = \beta_2^{m_1} \beta_1 \beta_2^{m_2} \beta_1 \dots \beta_2^{m_k} \beta_1 \beta_2^{m_{k+1}}$$

with non-negative exponents m_j and thus $\gamma \in \langle \beta_1, \beta_2 \rangle^+$. For the general case, we want to show that

$$P_1 \cup P_2^{-1} \cup \dots \cup P_{n-1}^{(-1)^n} = \langle \beta_1, \dots, \beta_{n-1} \rangle^+.$$

Similar to the proof of Proposition 7.6, we consider B_{n-1} as generated by $\tilde{\sigma}_1, \dots, \tilde{\sigma}_{n-2}$. Likewise we write $\tilde{P}_i$ for the $\tilde{\sigma}_i$ -positive braids in B_{n-1} and $\tilde{\beta}_i = (\tilde{\sigma}_i \dots \tilde{\sigma}_{n-2})^{(-1)^{i-1}}$. By induction hypothesis, we can assume that

$$\tilde{P}_1 \cup \tilde{P}_2^{-1} \cup \dots \cup \tilde{P}_{n-2}^{(-1)^{n-1}} = \langle \tilde{\beta}_1, \dots, \tilde{\beta}_{n-2} \rangle^+.$$

Again, we use the “right-shift” inclusion

$$\underbrace{\langle \tilde{\sigma}_1, \dots, \tilde{\sigma}_{n-2} \rangle}_{=B_{n-1}} \xrightarrow{\iota} \underbrace{\langle \sigma_1, \dots, \sigma_{n-1} \rangle}_{=B_n}, \quad \tilde{\sigma}_i \mapsto \sigma_{i+1}$$

as well as the isomorphism

$$\varphi : B_{n-1} \xrightarrow{\cong} B_{n-1}, \quad \tilde{\sigma}_i \mapsto \tilde{\sigma}_i^{-1}.$$

Then the concatenation $\iota \circ \varphi$ maps $\tilde{P}_i \mapsto P_{i+1}^{-1}$ and $\tilde{\beta}_i \mapsto \beta_{i+1}$. Hence we have

$$\begin{array}{ccc} \tilde{P}_1 \cup \tilde{P}_2^{-1} \cup \dots \cup \tilde{P}_{n-2}^{(-1)^{n-1}} & = & \langle \tilde{\beta}_1, \dots, \tilde{\beta}_{n-2} \rangle^+ \\ \downarrow \iota \circ \varphi & & \downarrow \iota \circ \varphi \\ P_2^{-1} \cup P_3 \cup \dots \cup P_{n-1}^{(-1)^n} & = & \langle \beta_2, \dots, \beta_{n-1} \rangle^+ \end{array}$$

Now take $\gamma \in P_1 \cup P_2^{-1} \cup \dots \cup P_{n-1}^{(-1)^n}$. If $\gamma \in P_2^{-1} \cup P_3 \cup \dots \cup P_{n-1}^{(-1)^n}$, then by the previous observation we already

have $\gamma \in \langle \beta_2, \dots, \beta_{n-1} \rangle^+$. On the other hand, if $\gamma \in P_1$, then γ can be written as

$$\gamma = \tilde{w}_{k_1} \sigma_1 \tilde{w}_{k_2} \sigma_1 \cdots \tilde{w}_{k_{m-1}} \sigma_1 \tilde{w}_{k_m}$$

for some elements $\tilde{w}_j \in \langle \sigma_2, \dots, \sigma_{n-1} \rangle$. Again, using that $\sigma_1 = \beta_1 \beta_2$ we can rewrite this as

$$\gamma = w_{k_1} \beta_1 w_{k_2} \beta_1 \cdots w_{k_{m-1}} \beta_1 w_{k_m}$$

with $w_{k_j} \in \langle \sigma_2, \dots, \sigma_{n-1} \rangle$. Using the induction hypothesis in the form of

$$\begin{aligned} B_{n-1} &= \langle \tilde{\sigma}_1, \dots, \tilde{\sigma}_{n-2} \rangle \\ &= \langle \tilde{\beta}_1, \dots, \tilde{\beta}_{n-2} \rangle^+ \sqcup \{1\} \sqcup \langle \tilde{\beta}_1^{-1}, \dots, \tilde{\beta}_{n-2}^{-1} \rangle^+ \end{aligned}$$

together with the isomorphisms

$$\begin{array}{c} \langle \sigma_2, \dots, \sigma_{n-1} \rangle \\ \uparrow \cong \\ \langle \tilde{\sigma}_1, \dots, \tilde{\sigma}_{n-2} \rangle \\ \parallel \\ \langle \tilde{\beta}_1, \dots, \tilde{\beta}_{n-2} \rangle^+ \sqcup \{1\} \sqcup \langle \tilde{\beta}_1^{-1}, \dots, \tilde{\beta}_{n-2}^{-1} \rangle^+ \\ \downarrow \cong \\ \langle \beta_2, \dots, \beta_{n-1} \rangle^+ \sqcup \{1\} \sqcup \langle \beta_2^{-1}, \dots, \beta_{n-1}^{-1} \rangle^+ \end{array}$$

we can say that each w_{k_j} is either equal to 1, a word in $\beta_2, \dots, \beta_{n-1}$ with only positive exponents or a word in these elements with only negative exponents. In the first two cases, we are done. For the latter case, similar to the case $n = 3$, one can iteratively get rid of negative exponents by using

$$(\beta_2 \beta_3^{-1} \cdots \beta_{n-1}^{(-1)^{n-1}}) \beta_1^{n-1} (\beta_2 \beta_3^{-1} \cdots \beta_{n-1}^{(-1)^{n-1}}) = \beta_1$$

and

$$(\beta_2 \beta_3^{-1} \cdots \beta_{n-1}^{(-1)^{n-1}})^2 = \beta_2^{n-1}.$$

□

Corollary 5.21 then implies that $<_{DD}$ is an isolated order.

Corollary 7.9. *The Dubrovina-Dubrovin-order $<_{DD}$ is isolated in $\text{LO}(B_n)$. Hence $\text{LO}(B_n)$ is not homeomorphic to the Cantor space.*

7.2 Free Abelian Groups $\mathbb{Z}^n$ and Sikora's Theorem

This section is about Sikora's Theorem, which implies that $\text{LO}(\mathbb{Z}^n)$ is homeomorphic to the Cantor set. It was originally proven in [Sik04]. However, we adapt the approach of Theorem 10.5 and Problem 10.6 in [CR15], as

it showcases applications of some of the results regarding convex subgroups from earlier in the thesis.

Theorem 7.10 (Sikora). *For $n \geq 2$, the space of left-orders $\text{LO}(\mathbb{Z}^n)$ has no isolated points.*

We split the proof into two parts, first showing the statement for Archimedean orders, then for non-Archimedean ones.

Proposition 7.11. *An Archimedean left-order $<$ on $\mathbb{Z}^n$ is not isolated in $\text{LO}(\mathbb{Z}^n)$.*

Proof. By Hölder's Theorem, see Theorem 2.22, there is an order-preserving and injective homomorphism

$$f : \mathbb{Z}^n \rightarrow \mathbb{R}.$$

Since f is determined by the generators

$$(1, 0, \dots, 0), \dots, (0, \dots, 0, 1)$$

of $\mathbb{Z}^n$, it can be extended to a linear map

$$\bar{f} : \mathbb{R}^n \rightarrow \mathbb{R}$$

by

$$\bar{f}(x_1, \dots, x_n) = x_1 \cdot f(1, 0, \dots, 0) + \cdots + x_n \cdot f(0, \dots, 0, 1).$$

Using the Riesz representation theorem, $\bar{f}$ is uniquely determined by some vector $v \in \mathbb{R}^n$ via

$$\bar{f} : w \mapsto \langle w, v \rangle.$$

Consider the hyperplane H which is defined by taking v as its normal vector. Then $f(h) = \langle h, v \rangle = 0$ for all $h \in H \cap \mathbb{Z}^n$, showing $H \cap \mathbb{Z}^n = \{0\}$ by injectivity of f . Now take some finite choice of $<$ -positive elements $g_1, \dots, g_k \in \mathbb{Z}^n$. By order-preservation of f we know

$$f(g_i) = \langle v, g_i \rangle > 0$$

for all i , hence they all lie on one side of the hyperplane H . Then a slight perturbation of v yields a different order on $\mathbb{Z}^n$ where all the g_i remain positive, showing that $<$ is not isolated. □

For the non-Archimedean case, some additional preparation is needed. First, we need Lemma 9.7 from [CR15], which we state here without proof.

Lemma 7.12. *If a left-order $<$ on G has the property that for any two positive elements $g, h \in G$ there exists $n > 0$ with $g < hg^n$, and if it has no proper non-trivial convex subgroup, then $<$ is Archimedean.*

The first condition from the lemma is always satisfied for bi-orders by choosing $n = 1$, so in particular it holds for any left-order on $\mathbb{Z}^2$.

By contraposition, the lemma yields that any non-Archimedean left-order $<$ on $\mathbb{Z}^n$ must have a proper non-trivial subgroup C , giving a short exact sequence

$$1 \longrightarrow C \xhookrightarrow{\iota} \mathbb{Z}^n \twoheadrightarrow^p \mathbb{Z}^n/C \longrightarrow 1$$

Since subgroups of free abelian groups are again free abelian, we have $C \cong \mathbb{Z}^p$. Moreover, $\mathbb{Z}^n/C$ is left-orderable by Proposition 3.12, hence it must be torsion-free, giving $\mathbb{Z}^n/C \cong \mathbb{Z}^q$ by the fundamental theorem for finitely generated abelian groups. We obtain that $p, q < n$ and $p + q = n$.

In particular, there exist left-orders $<_C$ on C and $<$ on $\mathbb{Z}^n/C$, such that the order $<$ on G is induced by the short exact sequence, see Corollary 3.15.

We can now prove the missing part of Sikora's theorem.

Proposition 7.13. *If $< \in \text{LO}(\mathbb{Z}^n)$ is non-Archimedean, then it is not isolated.*

Proof. This can be done by induction. For $n = 2$, using the notation from above, we have that $C \cong \mathbb{Z}$, i.e. $C = \langle (a, b) \rangle$ for some element $(a, b) \in \mathbb{Z}^2$. This means, that for $<_C$ there are only two choices, namely the order where (a, b) is positive and one where it is negative. Furthermore, we have that

$$\mathbb{Z}^n/C \cong \mathbb{Z}^2/\langle (a, b) \rangle \cong \mathbb{Z},$$

hence also for $<$ there are only two choices, which amount to declaring one side of $\mathbb{Z}^2 \setminus \langle (a, b) \rangle$ to be positive, the other as negative.

Having a look at how $<$ relates to $<_C$ and $<$ by Corollary 3.15, we see that

$$1 < (n, m) \iff \text{Either } ((n, m) \notin \langle (a, b) \rangle \text{ and } 1 < (n, m)) \\ \text{or } ((n, m) \in \langle (a, b) \rangle \text{ and } 1 <_C (n, m))$$

for $(n, m) \in \mathbb{Z}$. This shows that $<$ is actually a “line order” as constructed in Example 2.14 iv) via the line L spanned by (a, b) in $\mathbb{R}^2$. Similar to the Archimedean case, given $<$ -positive elements $g_1, \dots, g_k \in \mathbb{Z}^2$, one can then slightly perturb the line to obtain a new order, where the elements stay positive, see Fig. 31. Note that this works, since even if some of the positive points lie on L , they all must be on the same connected component of $L \setminus \{(0, 0)\}$. For the general case $n \geq 3$, note that at least one of the indices p, q must be greater than 1. Pick some $<$ -positive elements $g_1, \dots, g_k \in \mathbb{Z}^n$. If $p > 1$, then by induction hypothesis $<_C$ is not isolated, so there exists some order $<'_C$ which is different from $<_C$, but also satisfies $1 <'_C g_i$ for all g_i that lie in C . Then the order induced by $<'_C$ and $<$ via the short exact sequence is different from $<$ but also has the points g_i as positive elements.

On the other hand, if $q > 1$, then again by the induction

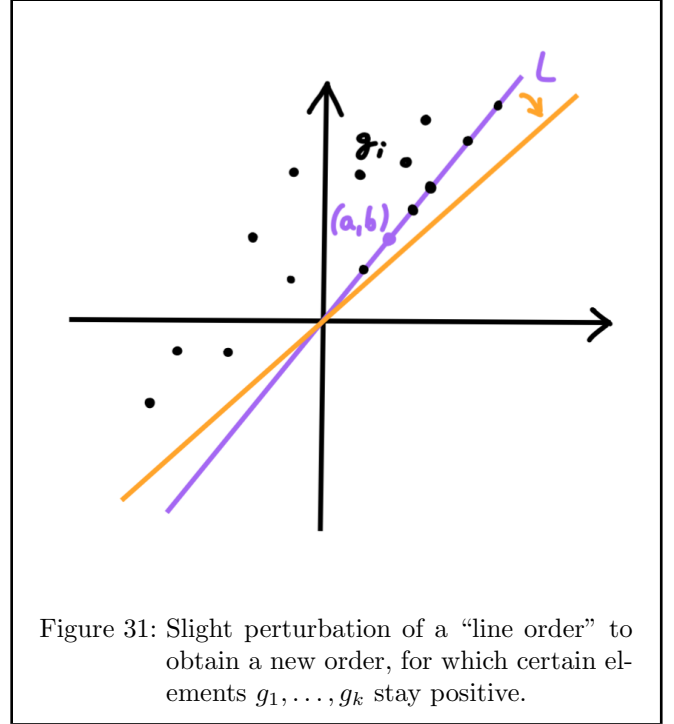


Figure 31: Slight perturbation of a “line order” to obtain a new order, for which certain elements $g_1, \dots, g_k$ stay positive.

hypothesis, there is an order $<'$ on $\mathbb{Z}^n/C$, different from $<$, for which $(0, 0) + C < p(g_i)$ implies $(0, 0) + C <' p(g_i)$. As before, the short exact sequence induces a new order on $\mathbb{Z}^n$ where the g_i stay positive. $\square$

Corollary 7.14. *$\text{LO}(\mathbb{Z}^n) = \text{BO}(\mathbb{Z}^n)$ is homeomorphic to the Cantor set for $n \geq 2$.*

Remark 7.15. Throughout the preceding proofs it was established that every left-order on $\mathbb{Z}^n$ is defined via some hyperplane $H \subseteq \mathbb{R}^n$, where the two connected components of $\mathbb{Z}^n \setminus H$ determine positivity or negativity of points. For points on the hyperplane $\mathbb{Z}^n \cap H$, we either have that the intersection is empty or isomorphic to $\mathbb{Z}^k$ for some $k < n$ and then one continues inductively until arriving at the case $\mathbb{Z}^k \cap H \cong \mathbb{Z}$, where one side of $\mathbb{Z} \setminus \{0\}$ is chosen to be positive.

7.3 Free Groups F_n

In this Section we will see that the space of left-orders on the free group F_n has no isolated points. We follow the geometric/combinatorial approach from Chapter 2.2.2 in [DNR16]. It mainly builds on observations of how words in F_n combine in a product and what happens to the word length.

We have seen in Corollary 5.21 that a finitely generated positive cone is automatically isolated in the space of left-orders. While the converse does not hold in general, see Proposition 2.4 in [Riv12], it does hold for the case of F_n . Then one can proceed to show that there

is no finitely generated positive cone in F_n . The latter actually follows from a more general result about free products of left-orderable groups.

The first step is to show that every isolated positive cone in F_n must be finitely generated. This is Theorem 2.2.33 in [DNR16] and it is split in 5 different Claims, where claims (ii)-(v) serve as preparation for Claim (i), from which the theorem follows.

Theorem 7.16. *If $P \in \text{LO}(F_n)$ is isolated, then it is finitely generated as a semigroup.*

Before looking at the various claims needed to prove this theorem, recall the notions of length, totality and stability from Definitions 4.3 and 6.1. Furthermore, throughout this section, let $B_N := B_N(1)$ denote the ball of radius N with respect to the standard generators of F_n , i.e.

$$B_N := \{w \in F_n \mid |w| \leq N\}.$$

We start with Claim ii), which is about (partial) products of elements from a stable set. The Claim implies that when we consider a product consisting of elements from a stable basis S that is already written in a minimal form with respect to S , the length strictly decreases when we are omitting factors on either end of the product.

A crucial observation for the proof of Claim ii) was already made in Remark 6.7: If S is stable and we have $f, g \in S$, then the cancellative part in the product is either strictly smaller than $\frac{|f|}{2}$ and strictly smaller than $\frac{|g|}{2}$ or we already have $fg \in S$ by stability. The latter is illustrated in Fig. 32. For example, considering a stable set $S \subseteq F_2 = \langle a, b \rangle$ and say we have $f = aba \in S$ and $g = a^{-1}b^{-1}a \in S$, then $fg = a^2$, so more than half of f cancels in the product. Now a^2 must also be an element of S by stability, hence $fg = a^2$ is the minimal way to write the product fg with factors from S .

With this observation in mind, intuitively Claim ii) must hold, since cancellations in products fg that eliminate at least half of one of the factors f or g are already respected in a stable basis S . Thus, if a product is written in a minimal form with respect to S , the partial products must increase in size when considering more and more factors. In a non-stable basis, this must not be true as seen in Example 6.5 i).

Proposition 7.17 (Claim ii)). *If $S \subseteq F_n$ is stable and $w \in \langle S \rangle^+$ is represented by*

$$w = h_1 \cdots h_k$$

with $h_i \in S$ and k minimal, then

$$|h_{i+1} \cdots h_k| < |h_i h_{i+1} \cdots h_k|$$

for all $i = 1, \dots, k-1$.

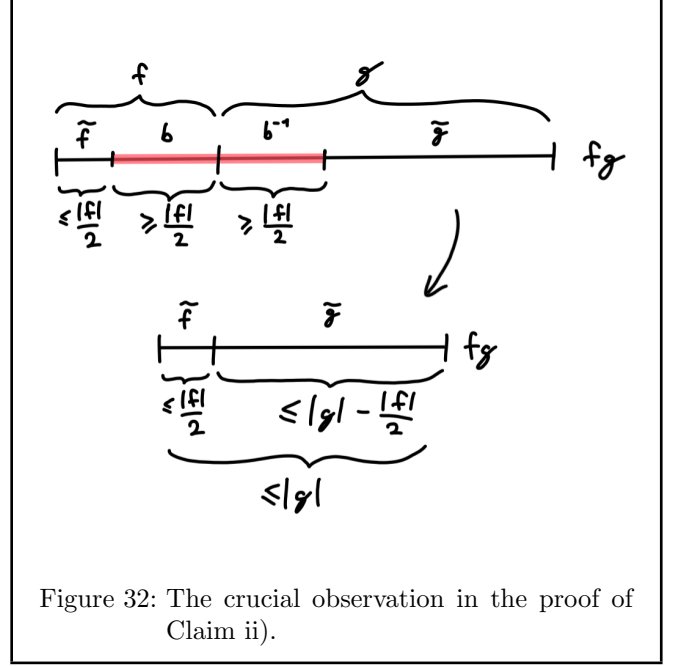


Figure 32: The crucial observation in the proof of Claim ii).

Proof. First write

$$h_i = f_i b_i, \quad h_{i+1} = b_i^{-1} f_{i+1},$$

so that $h_i h_{i+1} = f_i f_{i+1}$ without cancellation, i.e. b_i is exactly the part that cancels in the product. We first show the inequalities

$$|b_i| < \frac{1}{2}|h_i|, \quad |b_i| < \frac{1}{2}|h_{i+1}|$$

Note that

$$|f_i| = |h_i| - |b_i|, \quad |f_{i+1}| = |h_{i+1}| - |b_i|.$$

Suppose we have $|b_i| \geq \frac{1}{2}|h_i|$. Then

$$\begin{aligned} |h_i h_{i+1}| &= |f_i| + |f_{i+1}| \\ &= (|h_i| - |b_i|) + (|h_{i+1}| - |b_i|) \\ &\leq (|h_i| - \frac{|h_i|}{2}) + (|h_{i+1}| - \frac{|h_i|}{2}) \\ &= |h_{i+1}|. \end{aligned}$$

Hence we have $|h_i h_{i+1}| \leq |h_{i+1}|$, implying $h_i h_{i+1} \in S$ by stability. But this contradicts the assumption that k is minimal.

Similarly, the case $|b_i| \geq \frac{1}{2}|h_{i+1}|$ also leads to a contradiction.

As a consequence, the cancellative part can only eliminate strictly less than half of h_i and h_{i+1} , respectively. Continuing with the notation that we write b_i for the cancellative part in the product $h_i h_{i+1}$, we write $h_i = b_{i-1}^{-1} a_i b_i$ without cancellation for each i . With the previ-

ous observation, we can deduce that $a_i \neq 1$ and

$$|b_i| < \frac{|h_i|}{2} = \frac{|b_{i-1}| + |a_i| + |b_i|}{2},$$

implying

$$|b_i| < |b_{i-1}| + |a_i|.$$

With this notation, we obtain

$$h_i h_{i+1} = b_{i-1}^{-1} a_i a_{i+1} b_{i+1}$$

without cancellation, hence inductively

$$h_{i+1} \cdots h_k = b_i^{-1} a_{i+1} \cdots a_k$$

without cancellation. It then follows that

$$\begin{aligned} |h_{i+1} \cdots h_k| &= |b_i| + |a_{i+1}| + \cdots + |a_k| \\ &< |b_{i-1}| + |a_i| + |a_{i+1}| + \cdots + |a_k| \\ &= |h_i \cdots h_k| \end{aligned}$$

□

Corollary 7.18. *Let $S \subseteq F_n$ be stable. If $\langle S \rangle^+$ is not antisymmetric, then $1 \in S$.*

Proof. $\langle S \rangle^+$ not being antisymmetric implies the existence of $g \in \langle S \rangle^+$ with $g^{-1} \in \langle S \rangle^+$. Then $1 = gg^{-1} \in \langle S \rangle^+$, meaning we can write it in the form

$$1 = h_1 \cdots h_k$$

with $h_i \in S$ and k minimal. Claim ii) then implies

$$|h_k| < |h_{k-1} h_k| < \cdots < |h_1 \cdots h_k| = |1| = 0.$$

This already implies $k = 1$ and $1 = h_1 \in S$, as all other cases lead to a contradiction. □

Recall that S being N -stable means if we multiply two elements $f, g \in S$ for which the product fg has length $|fg| \leq N$, then $fg \in S$. The next result Claim iii) then tells us, that for a N -stable subset $S \subseteq B_N$ of the N -ball, the same is also true when considering products with finitely many factors coming from S , i.e. if $h_1 \cdots h_k \in \langle S \rangle^+ \cap B_N$, then $h_1 \cdots h_k \in S$.

Proposition 7.19 (Claim iii)). *For $S \subseteq B_N$ we have that S is N -stable if and only if $S = \langle S \rangle^+ \cap B_N$.*

Proof. It is clear that $S = \langle S \rangle^+ \cap B_N$ implies N -stability of S . On the other hand, assume that S is N -stable and let $g \in \langle S \rangle^+ \cap B_N$. This means, we can write $g = h_1 \cdots h_k$ with $h_i \in S$ and $k \leq N$ minimal. By Claim ii) we have

$$|h_k| < |h_{k-1} h_k| < \cdots < |h_1 \cdots h_k| = |g| \leq N.$$

Since N -stability implies stability for $S \subseteq B_N$, we obtain

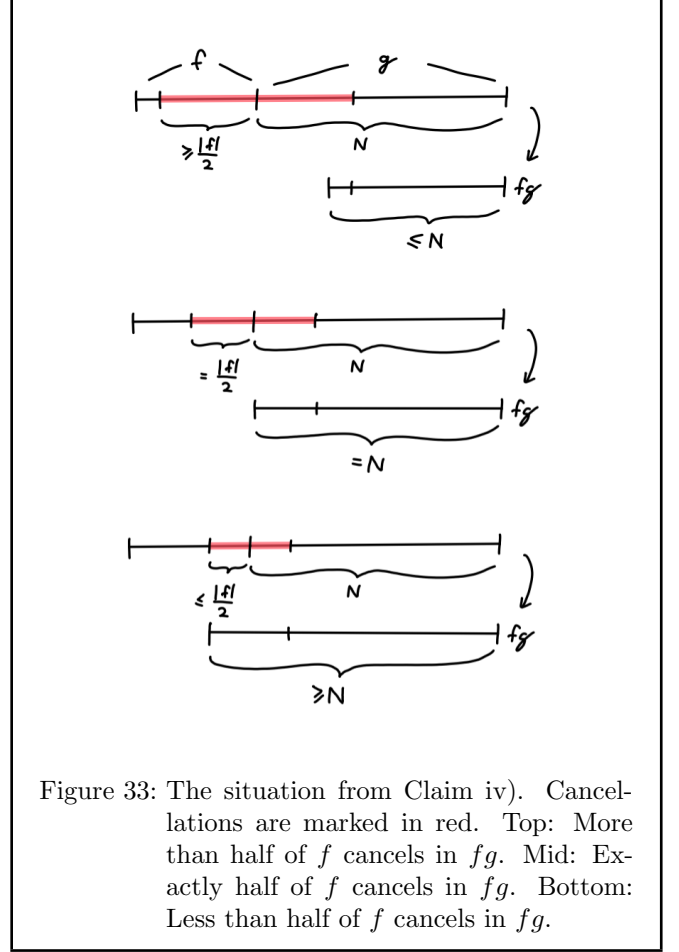


Figure 33: The situation from Claim iv). Cancellations are marked in red. Top: More than half of f cancels in fg . Mid: Exactly half of f cancels in fg . Bottom: Less than half of f cancels in fg .

$h_{k-1} h_k \in S$ and inductively deduce

$$h_{k-2} h_{k-1} h_k, \dots, \underbrace{h_1 \cdots h_k}_{=g} \in S.$$

□

Intuitively, the next Claim means the following: If g already has length N and a product fg should have length N , there must be cancellations happening in the product. Denoting the cancellative part as h , its length $|h|$ must be exactly half of $|f|$. If it would be less, the resulting product would have length greater than N and if it would be bigger, the product would have length smaller than N . The situation is visualized in Fig. 33.

Note that it also suffices to consider the case $|f| \leq N$ and $|g| = N$, since otherwise for $|f| = \tilde{N} > N$, we can take the product $g^{-1} f^{-1}$ instead. Then use $|g^{-1} f^{-1}| = |fg|$, $|g^{-1}| = |g| \leq \tilde{N}$ and $|f^{-1}| = |f| = \tilde{N}$, which returns us to the original case.

Proposition 7.20 (Claim iv)). *Consider reduced words $f, g \in F_n$ with*

$$|fg| = N, \quad |f| \leq N, \quad |g| = N.$$

Then

- a) $|f|$ is even;
- b) Exactly half of f cancels in fg ;
- c) After cancellation, at least the right half of fg must be the same as the right half of g .

Proof. Denote as h the part that cancels in the product fg , i.e. write

$$f = h_1 h, \quad g = h^{-1} h_2, \quad fg = h_1 h_2$$

without cancellation. Then

$$\begin{aligned} |h_1| + |h| &= |f| \leq N, \\ |h_2| + |h| &= |g| = N, \\ |h_1| + |h_2| &= |g| = N. \end{aligned}$$

The last two equalities imply $|h| = |h_1|$, hence $|f| = 2|h_1|$ is even. Moreover, exactly half of f cancels, since $|h| = |f|/2$. Finally, the first two relations combine to

$$|h_1| - |h_2| = |h_1| + |h| - |h_2| - |h| \leq 0,$$

yielding $|h| = |h_1| \leq |h_2|$. In other words, at least the right half of g survives in fg . $\square$

Before continuing, we need some additional notation. Given $S \subseteq B_N$ and $g \in B_N \setminus (S \cup S^{-1})$ define

$$S_1 := S \cup \{g\}$$

and

$$S_{i+1} := S_i \cup \{fg \mid f, g \in S_i, |fg| \leq N, fg \notin S_i\}$$

for $i > 0$. Intuitively, in each step we add all elements that are missing for N -stability. Since B_N is finite, there is some index j with $S_j = S_{j+1}$, i.e. the process terminates at some point. The set S_j is then the smallest subset of B_N that is N -stable and contains S as well as g . By construction we have $\langle S_j \rangle^+ \cap B_N = \langle S, g \rangle^+ \cap B_N$, hence Claim iii) implies $S_j = \langle S, g \rangle^+ \cap B_N$.

Example 7.21. Let $F_2 = \langle a, b \rangle$.

- i) Consider $S = \{a, b\} \subseteq B_2$ and $g = a^{-1}b^{-1}$. Then

$$\begin{aligned} S_1 &= \{a, b, a^{-1}b^{-1}\}, \\ S_2 &= \{a, b, a^{-1}b^{-1}, b^{-1}, a^{-1}, a^2, b^2, ab, ba\}, \\ S_3 &= B_2, \end{aligned}$$

since $a, b, a^{-1}, b^{-1} \in S_2$. See Fig. 34 for a visualization.

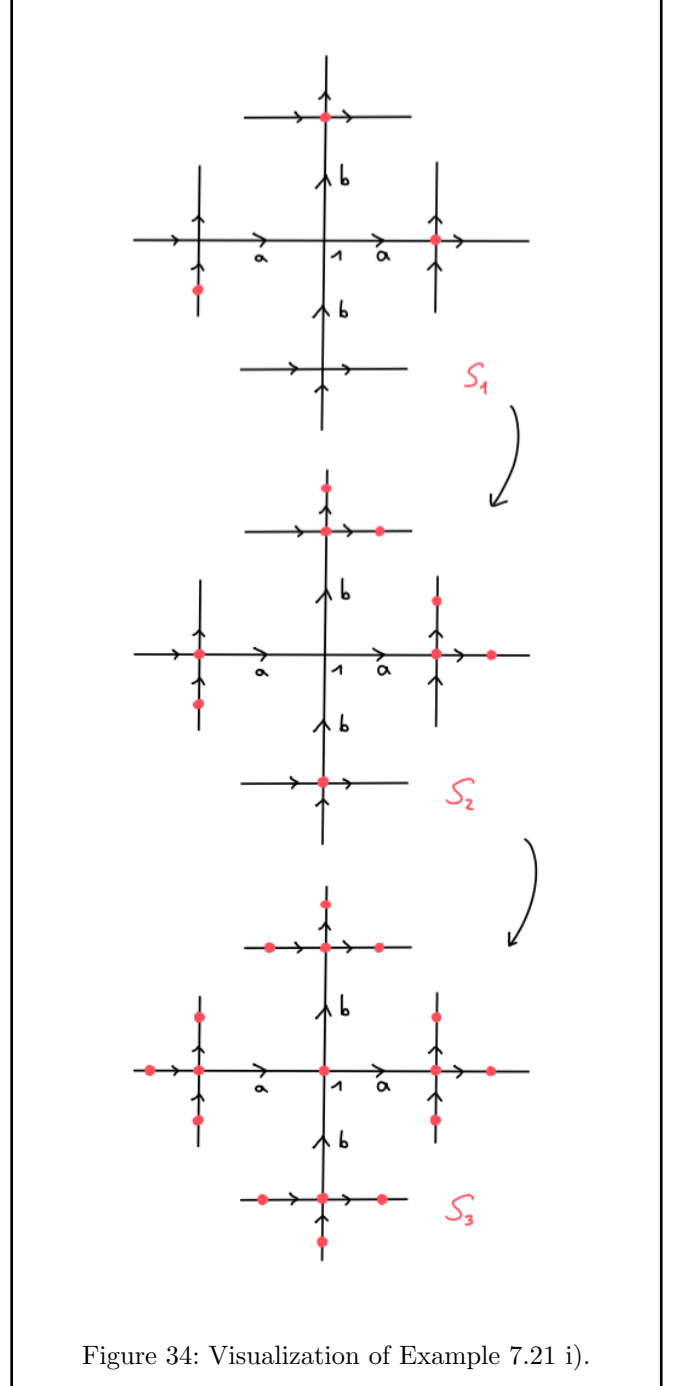


Figure 34: Visualization of Example 7.21 i).

ii) Consider $S = \{a\} \subseteq B_2$ and $g = b^{-1}a^{-1}$. Then

$$\begin{aligned} S_1 &= \{a, b^{-1}a^{-1}\}, \\ S_2 &= \{a, b^{-1}a^{-1}, b^{-1}, a^2\}, \\ S_3 &= \{a, b^{-1}a^{-1}, b^{-1}, a^2, ab^{-1}, b^{-1}a, b^{-2}\}, \\ S_4 &= S_3. \end{aligned}$$

See Fig. 35 for a visualization.

Now we move a step closer to the prerequisites of Claim i). From now on let $S \subseteq B_N$ and assume that it is total at length $N-1$ and $S = \langle S \rangle^+ \cap B_N$. In particular, $1 \notin S$ by totality and S is N -stable by Claim iii). Then take $g \in B_N \setminus (S \cup S^{-1})$ and suppose $\langle S, g \rangle^+$ is not antisymmetric. Then $\langle S_j \rangle^+$ is also not antisymmetric, and since S_j is stable, the Corollary of Claim ii) implies $1 \in S_j$. Since $1 \notin S$, there must be a smallest index k , such that for some $h \in S_k$, both h and h^{-1} are contained in S_k . Note that $|g| = N$, since S is total at length $N-1$.

Before coming to Claim v) from [DNR16], we state two additional remarks about the index k and the elements h, h^{-1} from above.

Remark 7.22. For $1 \leq \ell < k$, every element in $S_\ell \setminus S$ has length N . To see this, suppose there is $f \in S_\ell \setminus S$ with $|f| < N$. Since S is total at length $N-1$, we have $f \in S \cup S^{-1}$. But $f \notin S$, hence $f^{-1} \in S$ and thus also $f, f^{-1} \in S_\ell$, contradicting the minimality of k .

Remark 7.23. Without loss of generality, one can assume that both h and h^{-1} are elements of $S_k \setminus S$. To see this, suppose one of them is contained in S , say $h \in S, h^{-1} \in S_k$. Then we can write $h^{-1} = h_1 h_2$ with $h_1, h_2 \in S_{k-1}$. It must hold that $h_1 \notin S$ or $h_2 \notin S$, otherwise we would have $h^{-1} \in S$, contradicting antisymmetry of S . Assume the former, i.e. $h_1 \in S_{k-1} \setminus S$ (the other case is analogous), then

$$h_1^{-1} = \underbrace{h_2}_{\in S_{k-1}} \cdot \underbrace{h}_{\in S} \in S_k.$$

Also one has $h_1^{-1} \notin S$, otherwise we would have $h_1, h_1^{-1} \in S_{k-1}$, contradicting minimality of k .

Summarizing the arguments, we have that $h_1, h_1^{-1} \in S_k$, but $h_1, h_1^{-1} \notin S$, showing that whenever h or h^{-1} is contained in S , we can just consider h_1 and h_1^{-1} instead.

Proposition 7.24 (Claim v)). *Given the setting from above, every element in $S_{k-1} \setminus S$ as well as h and h^{-1} can be written as a product $h_1 g h_2$, where*

- $h_1, h_2 \in S \cup \{1\}$,
- $|h_1|$ and $|h_2|$ are even,
- Exactly the left half of h_2 cancels in the product $h_1 g h_2$,
- Exactly the right half of h_1 cancels in the product $h_1 g h_2$.

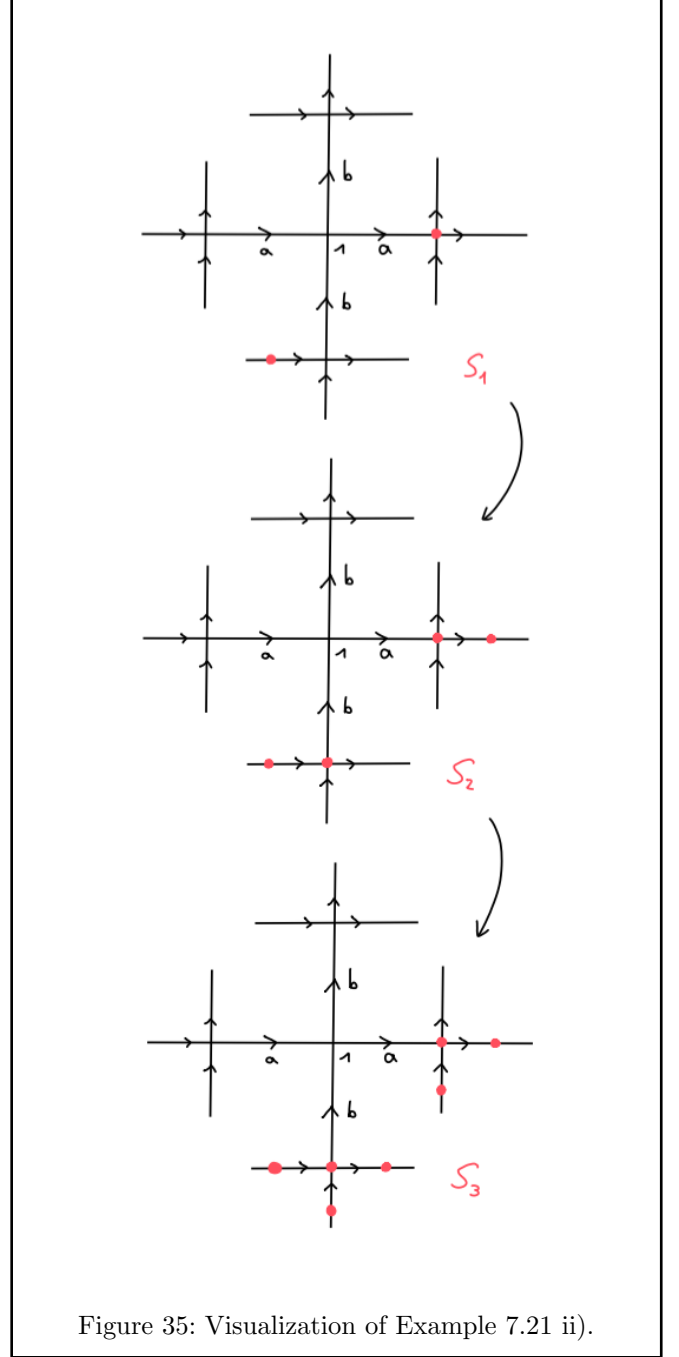


Figure 35: Visualization of Example 7.21 ii).

In particular, we have

$$|h_1gh_2| = |h_1g| = |gh_2| = N.$$

Proof. The proof goes by induction over $i < k$ for elements $f \in S_i \setminus S$. By the first remark from above, such an element has length $|f| = N$.

The base case $i = 1$ is covered by the fact that $f \in S_1 \setminus S$ implies $f = g = 1g1$.

Now assume that the statement holds for elements in $S_\ell \setminus S$ where $\ell < i$ and take $f \in S_i \setminus S$. Then f can be written as a product $f = f_1f_2$ for some $f_1 \in S_m, f_2 \in S_{m'}$ with $m, m' < i$. Then there are three cases to consider. Case a): $f_1, f_2 \notin S$.

By the induction hypothesis, in this case f has the form

$$f = h_1gh_2\bar{h}_1\bar{g}\bar{h}_2$$

with $h_1, h_2, \bar{h}_1, \bar{h}_2$ as in the hypothesis. In particular, we have $|h_1gh_2| = |\bar{h}_1\bar{g}\bar{h}_2| = N$, hence Claim iv) implies that N is even and exactly the right half of h_1gh_2 cancels with the left half of $\bar{h}_1\bar{g}\bar{h}_2$ in the product. Since $|h_1g| = |gh_2|$, the former is half of g followed by h_2 , whereas $|\bar{h}_1\bar{g}| = |g\bar{h}_2|$ implies that the latter is $\bar{h}_1$ followed by half of g . Overall, this yields

$$h_1gh_2\bar{h}_1\bar{g}\bar{h}_2 = h_1g\bar{h}_2$$

and we are done in this case.

Case b): $f_1 \in S, f_2 \notin S$.

In this case, f has the form

$$f = \bar{f}h_1gh_2$$

with $\bar{f} \in S, h_1gh_2 \in S_i \setminus S$, where $|\bar{f}| \leq N$ and h_1, h_2 as in the hypothesis, in particular $|h_1gh_2| = N$. Again, by Claim iv), exactly the right half of $\bar{f}$ cancels in the product.

If $|\bar{f}| < |h_1|$, then the cancellation happens completely inside of $\bar{f}h_1$ and since exactly half of $\bar{f}$ cancels, $|\bar{f}| \leq N$ and $|h_1| \leq N$, we have

$$|\bar{f}h_1| \leq \frac{N}{2} + \frac{N}{2} = N.$$

Thus by stability of S we obtain $\bar{f}h_1 \in S$.

On the other hand, if $|h_1| \leq |\bar{f}|$, then at least the left half of h_1 cancels in $\bar{f}h_1$, yielding

$$|\bar{f}h_1| \leq |\bar{f}| \leq N$$

and thus again $\bar{f}h_1 \in S$ by stability.

Additionally, in both cases we have

$$|\bar{f}h_1gh_2| = N, \quad |\bar{f}h_1| \leq N, \quad |gh_2| = N,$$

hence by Claim iv) the length $|\bar{f}h_1|$ is even and half of it cancels in the product, so we are done.

Case c): $f_1 \notin S, f_2 \in S$. This case is similar to Case b).

After having covered the induction for elements in $S_{k-1} \setminus S$, the final induction step also works for h and h^{-1} . $\square$

Finally, we arrive at the proof of Claim i). The proof builds mainly on Claim v).

Proposition 7.25 (Claim i)). *If $S \subseteq B_N$ is total at length $N - 1$ and satisfies $S = \langle S \rangle^+ \cap B_N$, then for all $g \in B_N \setminus (S \cup S^{-1})$, the semigroup $\langle S, g \rangle^+$ remains anti-symmetric.*

Proof. The proof is done by contradiction. Assuming that $\langle S, g \rangle^+$ is not antisymmetric yields h and h^{-1} as in the discussion preceding this Claim.

We know that h and h^{-1} can be written as

$$h = h_1gh_2, \quad h^{-1} = \bar{h}_1h\bar{h}_2$$

with $h_1, h_2, \bar{h}_1, \bar{h}_2$ as given in Claim v).

Case a): N is even.

In this case, g can be written as $g = g_1g_2$ with

$$|g_1| = |g_2| = \frac{|g|}{2} = \frac{N}{2}.$$

Then we have

$$1 = h \cdot h^{-1} = h_1g_1g_2h_2 \cdot \bar{h}_1g_1g_2\bar{h}_2.$$

Because h and h^{-1} are inverse to each other, in particular the left half of h^{-1} cancels with the right half of h . By Claim v), the former is equal to $\bar{h}_1g_1$ while the latter is the same as g_2h_2 , hence

$$1 = h_1g_1g_2\bar{h}_2.$$

But this equality yields $g = h_1^{-1}\bar{h}_2^{-1} \in S^{-1}$, which contradicts the assumption that $g \in B_N \setminus (S \cup S^{-1})$.

Case b): N is odd.

In this case, g can be written as

$$g = g_1 \cdot x_\ell \cdot g_2$$

with no cancellation, where x_ℓ is some generator of F_n . Also we have

$$|g_1| = |g_2| = \frac{|g| - 1}{2} = \frac{N - 1}{2}.$$

With similar arguments as in the preceding case, writing

$$1 = h \cdot h^{-1} = h_1g_1x_\ell \underbrace{g_2h_2 \cdot \bar{h}_1g_1}_{=1} x_\ell g_2\bar{h}_2$$

yields

$$1 = h_1g_1x_\ell^2g_2\bar{h}_2.$$

But with the initial assumptions about x_ℓ , there are no cancellations in the product on the right-hand side, hence it cannot be equal to 1, leading to a contradiction. $\square$

Now given $S \subseteq B_N$ that satisfies the conditions of Claim i), i.e. it is total at length $N - 1$ and N -stable, choose $g \in B_N \setminus (S \cup S^{-1})$, provided that such a g exists. Then the Claim implies that $\langle S, g \rangle^+$ is antisymmetric. In particular, $\langle S, g \rangle^+ \cap B_N$ stays total at length $N - 1$ and it is N -stable by definition, hence Claim i) can be applied again, but this time to $\langle S, g \rangle^+ \cap B_N$ instead of S . This can be repeated until

$$\langle S, g, g_1, \dots, g_m \rangle^+ \cap B_N$$

is total at length N , where the g_i are elements, such that $g_i, g_i^{-1} \notin \langle S, g, g_1, \dots, g_{i-1} \rangle^+ \cap B_N$. But then $\langle S, g, g_1, \dots, g_m \rangle^+ \cap B_{N+1}$ is total at length N and $N + 1$ -stable, hence the process can be repeated again. This can be done for all balls $B_{\tilde{N}}$ with $\tilde{N} \geq N$, and thus inductively defines a positive cone, where in particular g and all elements of S are positive.

On the other hand, the same process works for g^{-1} instead of g , but this time yielding a positive cone where g^{-1} and all elements of S are positive. As a result, we obtain two different left-orders, both for which all elements of S are positive.

Example 7.26. We illustrate this construction with a concrete example. The process is illustrated in Fig. 36, so one can follow along. Consider $F_2 = \langle a, b \rangle$ and

$$S = \{a, b, ab, ba, a^2, b^2\} \subseteq B_2.$$

It is 2-stable, total at length 1, but not at length 2. Choosing $g = b^{-1}a \in B_2 \setminus (S \cup S^{-1})$, we obtain

$$\langle S, g \rangle^+ \cap B_2 = \{a, b, ab, ba, a^2, b^2, b^{-1}a\},$$

which remains antisymmetric, but is still not total at length 2. Hence we repeat the process, choosing $g_1 = ab^{-1}$ and arrive at

$$\langle S, g, g_1 \rangle^+ \cap B_2 = \{a, b, ab, ba, a^2, b^2, b^{-1}a, ab^{-1}\},$$

which is total at length 2.

Here we have to pass to B_3 and we notice that

$$\begin{aligned} \langle S, g, g_1 \rangle^+ \cap B_3 = \{ & a, b, ab, ba, a^2, b^2, b^{-1}a, ab^{-1}, a^3, b^3, \\ & ab^{-1}a, a^2b^{-1}, a^2b, aba, ab^2, b^{-1}a^2, \\ & b^{-1}ab, b^2a, ba^2, bab^{-1}, bab, ba^2 \} \end{aligned}$$

is not total at length 3. This allows us to choose an element $\tilde{g}$ of

$$B_3 \setminus (\langle S, g, g_1 \rangle^+ \cup \langle S^{-1}, g^{-1}, g_1^{-1} \rangle^+)$$

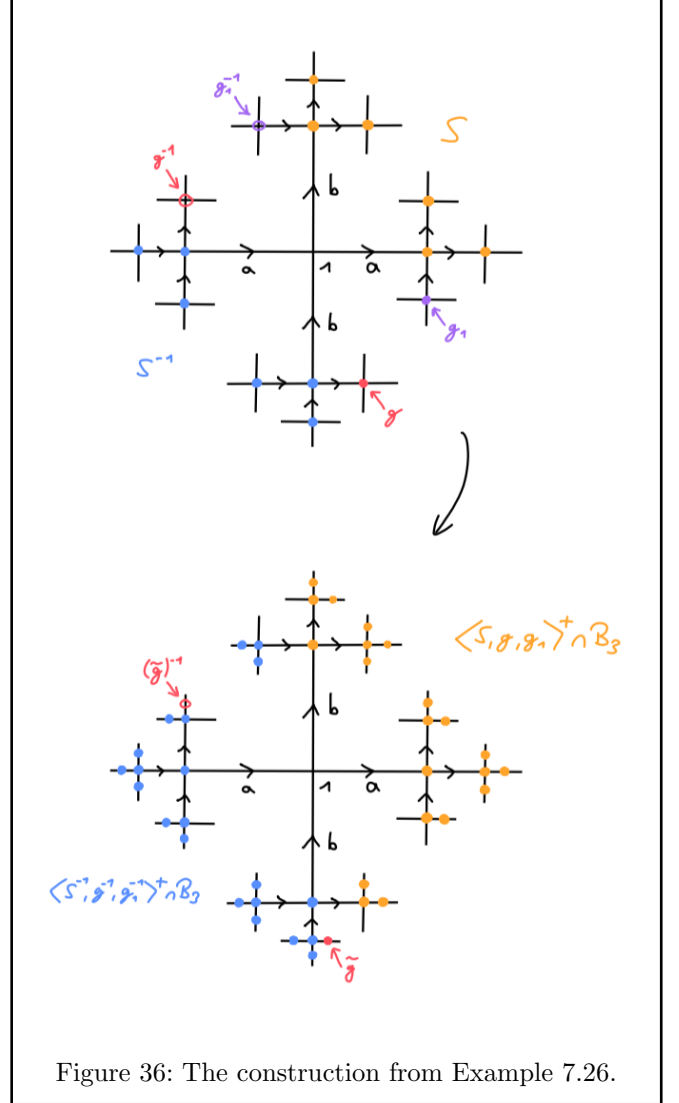


Figure 36: The construction from Example 7.26.

and repeat the process.

With this observation in mind, we can finally move to the proof of Theorem 7.16.

Proof of Theorem 7.16. Let $P \in \text{LO}(F_n)$ be isolated, i.e. there are $f_1, \dots, f_k \in F_n$ such that P is the unique positive cone containing these elements. In particular, there is an index $\tilde{N} > 1$, such that all the elements f_i are contained in $B_{\tilde{N}}$.

Assume that P is not finitely generated as a semigroup. First note that there must exist $N > \tilde{N}$, such that

$$S_N := \langle P \cap B_{\tilde{N}} \rangle^+ \cap B_N$$

is not total at length N . To see this, assume the contrary, i.e. S_N is total at length N for all $N > \tilde{N}$. But this means that $\langle P \cap B_{\tilde{N}} \rangle^+$ is a finitely generated positive cone, for which all the elements f_i are positive. By the initial assumption, P is unique over all positive cones satisfying this condition, implying $P = \langle P \cap B_{\tilde{N}} \rangle^+$ is finitely generated, leading to a contradiction.

In particular, for such a N , the set $B_N \setminus (S_N \cup S_N^{-1})$ is nonempty. Additionally, by taking the minimal such N , we can assume that S_N is total at length $N - 1$. Furthermore it is N -stable by construction. This means, it satisfies the conditions of Claim i), hence we can use the procedure discussed above to obtain elements $g, g_1, \dots, g_m$ such that

$$\langle S_N, g, g_1, \dots, g_m \rangle^+ \cap B_N$$

is total at length N .

Now similar to before, one can show that there must exist $N' > N$ such that $\langle S_N, g, g_1, \dots, g_m \rangle^+ \cap B_{N'}$ is not total at length N' , since otherwise $\langle S_N, g, g_1, \dots, g_m \rangle^+$ would be a finitely generated positive cone for which the elements f_i are positive, implying the contradictory statement $P = \langle S_N, g, g_1, \dots, g_m \rangle^+$. Hence we can continue the construction as above to arrive at a positive cone, for which all the f_i as well as g are positive.

Repeating the construction for g^{-1} instead of g , we obtained two different positive cones, both for which all elements $f_1, \dots, f_k$ are positive. This contradicts the assumption that P is isolated. $\square$

The next step is to show that there is no finitely generated positive cone in F_n . At this point, we deviate from the approach in [DNR16] and instead use Corollary 2 from [Hv17], which states that no order on a free product $A * B$ of two non-trivial, finitely generated, left-orderable groups A, B has a finitely generated positive cone.

Theorem 7.27. *Let A and B be two non-trivial, finitely generated and left-orderable groups. Then there is no left-order on the free product $A * B$ with a positive cone that is finitely generated as a semigroup.*

Combined with Theorems 7.16 and 5.18, and using that

$$F_n \cong (\dots (\mathbb{Z} * \mathbb{Z}) * \mathbb{Z}) * \dots * \mathbb{Z},$$

we obtain the desired result.

Corollary 7.28. *There is no finitely generated positive cone in F_n . Hence, there is no isolated left-order on F_n and $\text{LO}(F_n)$ is homeomorphic to the Cantor set.*

8 Concluding Remarks and Further Directions

Let us review some of the highlights of this thesis.

First, recall from Proposition 3.4 that a short exact sequence

$$1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{p} H \longrightarrow 1$$

where N and H are left-ordered, induces a left-order on G in a lexicographic way.

As a special case, this allows us to reconstruct an order on G if we are given left-orders on a normal subgroup N and on the quotient G/N , since then

$$1 \longrightarrow N \hookrightarrow G \twoheadrightarrow G/N \longrightarrow 1$$

is a short exact sequence.

On the contrary, while it does not work in general, we have seen that starting from the middle term is possible when N is relatively convex. Suppose we start with a left-order $<$ on G and a homomorphism $\varphi : G \rightarrow H$ such that $\ker \varphi$ is $<$ -convex. Then by Proposition 3.12 this induces a left-order $\prec$ on the image $\varphi(G)$ such that φ is order-preserving. Moreover, Corollary 3.15 showed that in this case, the order $<$ on G is equal to the left-order induced by the exact sequence

$$1 \longrightarrow (\ker \varphi, <) \hookrightarrow (G, <) \xrightarrow{\varphi} (\varphi(G), \prec) \longrightarrow 1$$

where $\ker \varphi$ is left-ordered by the restriction of $<$. As a special case, this argument works for the sequence

$$1 \longrightarrow N \hookrightarrow G \twoheadrightarrow G/N \longrightarrow 1$$

if N is relatively convex, i.e. convex for some left-order on G . This places a heavy restriction on orders of that kind, in the most extreme case we have seen in Corollary 3.16 that it shrinks the space of left-orders on the fundamental group of the Klein bottle to just 4 possible left-orders.

This construction also helped for proving Sikora's Theorem 7.10, where we could use that a non-Archimedean order $<$ on $\mathbb{Z}^n$ has a proper non-trivial subgroup. This implied that $<$ must be induced by an exact sequence of the form

$$1 \longrightarrow \mathbb{Z}^p \longrightarrow \mathbb{Z}^n \longrightarrow \mathbb{Z}^q \longrightarrow 1$$

with $p, q < n$ and $p + q = n$. This has allowed an inductive approach to show that $<$ is not isolated in $\text{LO}(\mathbb{Z}^n)$.

The Archimedean case, where no proper non-trivial convex subgroup exists, could be handled more easily with the help of Hölder's Theorem 2.22. In both cases, however, we have seen that the order in question is determined by a hyperplane. Returning to one of our initial family of examples, namely the “line orders” on $\mathbb{Z}^2$ from Example 2.14 iii) and iv), the Archimedean case corresponds to irrational slopes, and the non-Archimedean case to rational ones.

Let us return to the case where we have a $<$ -convex subgroup $C \subseteq G$. As shown Proposition 3.14, we could choose a new order $<'$ on C and consider the convex extension $<_{\text{ext}}$ of $<'$ by $<$ on G induced by

$$1 \longrightarrow C \hookrightarrow G \twoheadrightarrow G/C \longrightarrow 1$$

where G/C is ordered as in Proposition 3.12. Following all the definitions, the order $<_{\text{ext}}$ coincides with $<$ on $G \setminus C$ and with $<'$ on C . Furthermore, C stays $<_{\text{ext}}$ -convex. As a special case, we considered “flipping” a convex subgroup $C \subseteq G$, which means reversing the order on C while keeping it the same on $G \setminus C$.

Perhaps surprisingly, this flipping process helped us in analyzing the space of orders $\text{LO}(B_n)$ of the braid group B_n in Section 7.1. Let $\sigma_1, \dots, \sigma_{n-1}$ be the generators of B_n and denote the Dehornoy order as $<_D$, then B_n contains $n - 2$ proper $<_D$ -convex subgroups

$$\langle \sigma_2, \dots, \sigma_{n-1} \rangle \supseteq \langle \sigma_3, \dots, \sigma_{n-1} \rangle \supseteq \dots \supseteq \langle \sigma_{n-1} \rangle.$$

While $<_D$ is a limit point of its conjugates and thus not isolated in $\text{LO}(B_n)$, iteratively flipping those convex subgroups resulted in the isolated Dubrovina-Dubrovin order $<_{DD}$. Moreover, by Proposition 7.8 the positive cone P_{DD} of the latter is finitely generated as the semigroup

$$P_{DD} = \langle \beta_1, \dots, \beta_{n-1} \rangle^+$$

for certain braids $\beta_1, \dots, \beta_{n-1}$. As a consequence, the $<_{DD}$ -positive braids are exactly those elements that can be expressed as a word in $\beta_1, \dots, \beta_n$ using only positive exponents, thus giving a somewhat explicit description of the order.

As a final mention of group extensions, the Artin-Magnus order on the pure braid group PB_n introduced in Section 4.3 served as an excellent application of Proposition 3.4. Structural properties of PB_n revealed that it can be split into $n - 1$ factors as an iterated semidirect product

$$PB_n = U_n \rtimes (U_{n-1} \rtimes (\dots \rtimes (U_4 \rtimes (U_3 \rtimes U_2) \dots)),$$

where each of the factors U_i is homeomorphic to a finitely generated free group. Hence, each U_i is already ordered by using the Magnus order introduced in section 4.2.2. This allowed us to iteratively piece together a left-order on the whole of P_n using a chain of exact sequences as in Example 3.5 iv).

Regarding the space of orders $\text{LO}(G)$ for a group G , the correspondence between left-orders and positive cones allowed us to equip it with the Tychonoff topology in a natural way, making it into a totally disconnected and compact Hausdorff-space. For countable G the space is also second-countable. Thus, the reformulation of Brouwer's Theorem 5.18 directed our main interest to identifying isolated points in $\text{LO}(G)$. It turned out that the latter admit an explicit algebraic description by Lemma 5.20, which was vital for the remainder of the thesis.

Furthermore, the conjugation action and more generally the action of $\text{Aut}(G)$ on $\text{LO}(G)$ by homeomorphisms introduced in Proposition 5.23 proved a useful tool for analyzing spaces of orders. On one hand it was often used to construct converging sequences in $\text{LO}(G)$, see Propositions 5.26, 5.32 and 7.3. On the other hand, its property to act by homeomorphisms were important in Section 5.4 and for showing that the space of bi-orders $\text{BO}(G)$ is a closed subset of $\text{LO}(G)$ through Proposition 5.29.

In Section 6.1 we explored connections of $\text{LO}(G)$ with Cayley graphs of G . A key notion was that of a proper N -partition of the N -ball B_N in the Cayley graph, which resembles a local version of a positive cone (see Remark 6.2). The main result Proposition 6.12 of this Section showed that the whole of G can be left-ordered if each of the individual N -balls can be left-ordered by a proper N -partition. Interestingly, the proof mainly followed from topological arguments.

The concept of proper N -partitions made another appearance in Section 7.3, where we have seen that the space of left-orders $\text{LO}(F_n)$ on the finitely generated free group F_n has no isolated points. In fact, this is a special case of the more general fact that there are no isolated orders on the free product $G * H$ of two non-trivial, finitely generated left-orderable groups $G * H$, see Theorem A in [Riv12]. The proof from the reference, however, relies on the notion of a dynamical realization of a left-order, which resembles an embedding of $G * H$ into $\text{Homeo}_+(\mathbb{R})$ in an order-preserving way. Roughly speaking, the dynamical realization of a given left-order $<$ on $G * H$ can be perturbed slightly to obtain a different left-order that remains close to $<$ in $\text{LO}(G * H)$.

Note that the authors of [Mul24] and [DM23] used

dynamical realizations of bi-orders to show that also the space of bi-orders $\text{BO}(G * H)$ for two non-trivial bi-orderable groups G and H has no isolated points, making it homeomorphic to the Cantor set. This result from 2024 answers Conjecture 2.2 from [Sik04].

The downside to this dynamical approach is that it requires a lot of additional machinery. On the contrary, the proof from [DNR16] that we showcased in Section 7.3 uses mainly elementary combinatorial observations on how words in F_n combine in products and what this implies for lengths of the words involved. As a result, the proof is highly constructive and the individual steps are straightforward to follow. However, the overarching intuition remains somewhat obscure, and it is not immediately clear which of these results generalize to free products or other non-free groups.

Let us quickly recall the main idea of the proof for Theorem 7.27 that follows from Proposition 7.25, which we refer to as Claim i). Note that because of Proposition 7.19, which we refer to as Claim iii), the condition $S = \langle S \rangle^+ \cap B_N$ in Claim i) can be replaced with S being N -stable.

Let $B_N \subseteq F_n$ be the N -ball in F_n with respect to the standard generators. Suppose we have a set $S \subseteq B_N$ that is total at length $N - 1$ and N -stable, but not total at length N . Then in particular there exists $g \in B_N \setminus (S \cup S^{-1})$ and Claim i) implies that $\langle S, g \rangle^+$ is antisymmetric. Hence, $\langle S, g \rangle^+ \cap B_N$ stays total at length $N - 1$ and it is also N -stable. Since B_N is finite, Claim i) can be applied repeatedly until we arrive at the proper N -partition

$$\langle S, g, g_1, \dots, g_m \rangle^+ \cap B_N$$

of B_N . In other words, we successfully “left-ordered” B_N in such a way, that all elements of S and g are positive. In particular, the set

$$\tilde{S}_N := \langle S, g, g_1, \dots, g_m \rangle^+ \cap B_{N+1}$$

is total at length N and $N + 1$ -stable. If there is no index $M \geq N$, such that $\tilde{S}_M$ is total at length M but not total at length $M + 1$, then $\langle S, g, g_1, \dots, g_m \rangle^+$ defines a finitely generated positive cone on F_n . Otherwise, for such an M , we can repeat the construction starting with $\tilde{S}_M$, which inductively defines a positive cone on F_n for which S and g are positive.

Now consider a positive cone $P \in \text{LO}(F_n)$ that is isolated and not finitely generated as a semigroup. By Lemma 5.20 there is a finite set $T \subseteq P$, such that P is the unique positive cone containing T . In particular, there is a radius $\tilde{N} \in \mathbb{N}$ with $T \subseteq P \cap B_{\tilde{N}}$. Similar to before, there must exist $N > \tilde{N}$ such that $\langle P \cap B_{\tilde{N}} \rangle^+ \cap B_N$ is total at length $N - 1$ but not at length N . Otherwise $\langle P \cap B_{\tilde{N}} \rangle^+$ is a positive cone containing T , implying P is finitely

generated by uniqueness. At this point, by choosing

$$S := \langle P \cap B_{\tilde{N}} \rangle^+ \cap B_N,$$

we can do the construction above once for $g \in B_N \setminus (S \cup S^{-1})$ and once for g^{-1} . In this way, we arrive at two different positive cones, which both contain T , contradicting uniqueness of P .

Looking closely, we note that, aside from Claim i) and Claim iii), every other argument in the proof holds for any finitely generated group G . So the question of generalization reduces to those two Claims. Furthermore, the statement of Claim iii) that $S = \langle S \rangle^+ \cap B_N$ implies N -stability for $S \subseteq B_N$ also holds for any group, while the opposite direction follows immediately from Proposition 7.17, which we refer to as Claim ii).

It might be that Claim ii), as well as Claim i), hold in a more general setting, since the statements only involve the concepts of length, stability and a certain minimality requirement. However, the proof of Claim ii) for the case of F_n is very dependant on exact cancellations and it is not so clear what happens when relations are involved.

The proof Claim i) also relies heavily on Propositions 7.20 and 7.24, which we refer to as Claim iv) and Claim v). The latter two most certainly do not generalize to anything other than free groups, since their statements are very specific to F_n .

Unfortunately, at the time of writing this thesis, we were not able to identify generalizations of the presented Claims. Promising directions for further investigation include finitely generated groups that are in some sense “close to being free”, as well as free products $G * H$ of non-trivial, finitely generated left-orderable groups G, H .

After this retrospect, let us discuss some possible further directions, which we sadly could not consider in the thesis due to time constraints.

As already remarked in [DNR16] after Question 2.2.38, for the case where $\text{LO}(G)$ is homeomorphic to the Cantor set, there still remain open questions regarding more geometric informations about this space. At this point it should be noted that $\text{LO}(G)$ is metrizable for countable G , see Proposition 1.3 in [Sik04]. Hence one can ask for example about the Hausdorff-dimension of $\text{LO}(G)$, depending on the chosen metric.

For the case where $\text{LO}(G)$ is not finite and also not homeomorphic to the Cantor set $\mathcal{C}$, there are still (infinitely many) copies of $\mathcal{C}$ hidden inside of $\text{LO}(G)$. This is due to the fact that the subspace $L(\text{LO}(G))$ of limit points in $\text{LO}(G)$ stays compact, totally disconnected, Hausdorff and second-countable and has, by definition, no isolated points. Questions remain, however, in which form those copies of $\mathcal{C}$ may appear. For example, regard-

ing the braid group B_n with $n \geq 3$, in Section XIV.2.3 from [Deh+08] the authors constructed a copy of $\mathcal{C}$ inside $\text{LO}(G)$ by taking the Dehornoy-order $<_D$ with its positive cone P_D from Section 4.4 and considering its set of conjugates

$$Z_n = \{\beta^{-1}P_D\beta \mid \beta \in B_n\}.$$

Similar to Proposition 7.3, each point of Z_n is a limit point, so in particular the closure $\overline{Z}_n$ of Z_n in $\text{LO}(B_n)$ satisfies the requirements for Theorem 5.18.

Throughout the thesis and in a lot of references we used, only total left-invariant orders are considered, but the theory could also be done for partial left-orders. In this setting, for example the positive cone $P_<$ of some partial left-order $<$ on a group G does not satisfy the property $G = P \cup \{1\} \cup P^{-1}$ anymore. It would certainly be interesting to see if specific results still hold, or if they require some modification. In particular, which properties remain and which get lost when passing from the space of left-orders $\text{LO}(G)$ to the bigger space $\text{PLO}(G)$ of partial left-orders?

Furthermore, there are algorithmic aspects to consider. For example, in Section 6 we saw that a group G is left-orderable if and only if a proper k -partition of the k -ball B_k exists for all $k \in \mathbb{N}$. As remarked in [CR15] after Theorem 1.43, if G is finitely generated, then it is a finite process to check whether a proper k -partition on B_k exists for any given k . For an algorithm to check orderability of G through existence of proper k -partitions for all k , however, there are two major hinderances. On one hand, the word problem has to be decidable algorithmically, and on the other hand, if G is left-orderable, then the algorithm never stops. Hence such an algorithm could only be used to disprove orderability, although one does not know beforehand, if and when the algorithm will stop.

Finally, while spaces of orders form an interesting topic that provides connections between group theory and topology, questions remain to which mathematical areas beyond dynamical systems these results can actually be applied. Since the study of spaces of orders seems to be a relatively young field, there is hope that some interesting applications will emerge. (Of course, as mathematicians we are not strictly bound to concerns of applicability, but it is still nice to know.)

References

- [CR15] Adam Clay and Dale Rolfsen. Ordered groups and topology. 2015. arXiv: 1511.05088 [math.GT].
- [DD01] T. V. Dubrovina and N. I. Dubrovin. “On braid groups”. In: Sbornik: Mathematics 192.5 (2001), pp. 693–703. DOI: 10.1070/SM2001v192n05ABEH000564.
- [Deh+08] Patrick Dehornoy et al. Ordering Braids. Vol. 148. Mathematical Surveys and Monographs. American Mathematical Society, 2008.
- [DM23] Serhii Dovhyi and Kyrylo Muliarchyk. “On the topology of the space of bi-orderings of a free group on two generators”. In: Groups, Geometry, and Dynamics 17.2 (2023). DOI: 10.4171/GGD/712.
- [DNR16] B. Deroin, A. Navas, and C. Rivas. Groups, Orders, and Dynamics. 2016. arXiv: 1408.5805v2 [math.GR].
- [Fra12] Michael Francis. Two Topological Uniqueness Theorems for Spaces of Real Numbers. 2012. arXiv: 1210.1008 [math.GN].
- [Hv17] Susan Hermiller and Zoran Šunić. “No positive cone in a free product is regular”. In: International Journal of Algebra and Computation 27.08 (2017), pp. 1113–1120.
- [HY61] John G. Hocking and Gail S. Young. “Topology”. In: Addison-Wesley Series in Mathematics (1961).
- [KR03] Djun Maximilian Kim and Dale Rolfsen. “An Ordering for Groups of Pure Braids and Fibre-Type Hyperplane Arrangements”. In: Canadian Journal of Mathematics 55.4 (2003), 822–838. DOI: 10.4153/CJM-2003-034-2.
- [KT08] C. Kassel and V. Turaev. Braid Groups. 1st ed. Graduate Texts in Mathematics. Springer New York, 2008.
- [Lin11] Peter A. Linnell. “The space of left orders of a group is either finite or uncountable”. In: Bulletin of the London Mathematical Society 43.1 (2011), 200–202. DOI: 10.1112/blms/bdq099.
- [MKS04] W. Magnus, A. Karrass, and D. Solitar. Combinatorial Group Theory. 2nd Revised. Dover Books on Mathematics. Dover Publications, 2004.
- [Mul24] Kyrylo Muliarchyk. Free products of bi-orderable groups. 2024. arXiv: 2403.14779 [math.GR].
- [Mun14] James Munkres. Topology. 2nd ed. Pearson Education Unlimited, 2014.
- [Riv12] Cristóbal Rivas. “Left-orderings on free products of groups”. In: Journal of Algebra 350.1 (2012), pp. 318–329. DOI: 10.1016/j.jalgebra.2011.10.036.
- [Sik04] Adam S. Sikora. “Topology on the spaces of orderings of groups”. In: Bulletin of the London Mathematical Society 36.1 (2004), pp. 519–526. DOI: 10.1112/S0024609303003060.
- [use12] user642796. The Cantor ternary set is totally disconnected. <https://math.stackexchange.com/questions/253535/the-cantor-ternary-set-is-totally-disconnected>. [Online; accessed 21-August-2025]. 2012.
- [Š13] Zoran Šunić. “Explicit left orders on free groups extending the lexicographic order on free monoids”. In: Comptes Rendus. Mathématique 351.13-14 (2013), pp. 507–511. DOI: 10.1016/j.crma.2013.07.001.